

INFLUENCE OF BUILT ENVIRONMENT ON ELECTRICITY CONSUMPTION IN DHAKA CITY

by

Rumkey Chowdhury (1915024)

**A thesis submitted for the degree of
BACHELOR OF URBAN AND REGIONAL PLANNING, BUET**



**Department of Urban and Regional Planning
Bangladesh University of Engineering and Technology**

Dhaka – 1000, Bangladesh

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**I dedicate this thesis
to my family**

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ABSTRACT

This study examines the impact of socioeconomic, urban, and environmental factors on electricity consumption in two distinct neighborhoods, Dhanmondi and Lalbagh in Dhaka. It integrates trend analysis utilizing a 7-year period data from 2017 to 2023 of four uses, spatial distribution, and regression models of five land uses based on consumption data in 2023. The study reveals how urban layouts, and socio-economic conditions affect energy usage across multiple sectors. Total electricity consumption in both Lalbagh and Dhanmondi increased across residential, commercial, industrial, and institutional sectors over the study period. Dhanmondi exhibited higher electricity consumption, particularly in residential and commercial sectors due to its affluent population and active commerce, while Lalbagh maintained lower and steadier usage, reflecting its residential-industrial nature. In 2020, both neighborhoods saw a significant drop in electricity use due to COVID-19, with a robust recovery in Dhanmondi's commercial and industrial sectors highlighting its economic resilience. Lalbagh's electricity consumption is lower in the north and center but higher near southeastern industrial hubs due to its residential and mixed-use nature with moderate land use diversity. In contrast, Dhanmondi's planned urban development, featuring residential, commercial, and institutional zones, leads to high electricity consumption across its northern and central areas. Lalbagh displays a variety of building structures, including permanent (pucca), semi-permanent (semi-pucca), and temporary (katcha), reflecting its diverse development phases and socioeconomic spectrum. In contrast, Dhanmondi primarily consists of permanent (pucca) buildings, indicative of its systematic planning, affluent demographics, and contemporary infrastructure. Moreover, the interplay of all three factors (use, location and structure type) creates specific scenarios where consumption is influenced in ways that cannot be predicted by looking at each factor individually. Electricity consumption in Lalbagh is influenced mainly by environmental and occupancy factors, with additional effects from building characteristics such as floor number, size etc. Conversely, Dhanmondi's consumption is shaped by structural attributes like building volume and density, significantly impacting residential, institutional, and industrial sectors. The analysis reveals Dhanmondi's consumption patterns are non-linear, unlike Lalbagh's more linear trends. While industrial consumption is effectively explained in Dhanmondi and less so in Lalbagh, other sectors, like institutional and commercial, are not as well accounted for in both neighborhoods. This discrepancy highlights the need for localized energy management strategies tailored to the specific context and demand.

LIST OF ABBREVIATIONS

NDVI: Normalized Difference Vegetation Index

MNDWI: Modified Normalized Difference Water Index

DPDC: Dhaka Power Distribution Company

MLR : Multiple Linear Regression

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CHAPTER 1 : INTRODUCTION

1.1 Background of the Study

Access to electricity is essential for modern development. It helps economies grow, improves people's lives, and supports the growth of sustainable cities. Bangladesh has made remarkable progress in expanding electricity access, with a significant increase in per capita consumption from 110 kWh to 378 kWh during the same period (Siddique, 2023). These achievements reflect the government's strategic efforts to integrate more areas into the national grid. However, rising per capita electricity consumption alone does not guarantee holistic development, as disparities in consumption patterns persist between urban and rural areas (Siddique, 2023). For instance, Dhaka, the capital city, accounts for 40–45% of the country's total electricity generation, underscoring the disproportionate energy demand in urban centers (Siddique, 2023). This disparity is further influenced by variations in the built environment, which plays a critical role in shaping energy consumption patterns (Lim & Kang, 2022).

Dhaka, one of the fastest-growing megacities in the world, is home to over 16 million people within an area of 415.16 km² (Istiaque & Khan, 2018). The city exhibits significant diversity in land use, socioeconomic conditions, and the physical characteristics of its built environment, all of which influence energy consumption patterns (Dewan & Yamaguchi, 2009; Wang et al., 2024). Studies have shown that factors such as land use patterns, building density, green spaces, and socio-economic conditions significantly impact energy demand (Ali & Nitivattananon, 2012; Woo & Cho, 2018; Mohammadi et al., 2021). Despite these findings, there remains a lack of research on how the built environment influences electricity consumption in Bangladesh, particularly within urban areas like Dhaka. Furthermore, spatial differences in electricity consumption patterns within cities remain underexplored, especially in low- and middle-income neighbourhoods.

The need for such research is heightened by Bangladesh's commitment to reducing greenhouse gas (GHG) emissions under the Paris Agreement, which it ratified on 21 September 2016. The country aims to reduce GHG emissions in the power, industry, and transport sectors by 5% below business-as-usual levels by 2030, relying solely on domestic resources (GoB, 2015). As the largest contributor to GHG emissions, the power sector requires urgent attention, and identifying the factors influencing electricity consumption is a critical step toward achieving this goal (Debnath & Mourshed, 2022). Understanding the relationship between the built

environment and electricity consumption can inform policies aimed at promoting energy efficiency and sustainable urban development.

There is limited research exploring how the built environment, encompassing buildings, infrastructure, and urban design, shapes electricity consumption in urban areas like Dhaka. This study seeks to fill this knowledge gap by focusing on two distinct areas of the city: Dhanmondi, planned residential and commercial neighbourhoods, and Lalbagh, a historic area with denser, older urban development. These areas represent contrasting socio-economic and built environment characteristics, making them ideal for comparative analysis. By analyzing trends in electricity consumption over the years in these locations and examining the influence of built environment factors, such as building types, density, and infrastructure layout, the research aims to uncover critical insights into the relationship between urban design and energy use.

1.2 Objective of the Study

The study aims to achieve two primary objectives:

- (1) To identify trends in electricity consumption over the years in selected areas of Dhaka
- (2) To examine the relationship of built environment factors with electricity consumption.

By addressing these objectives, the research aims to provide actionable insights for policymakers and urban planners to design energy-efficient strategies tailored to Bangladesh's unique urban context.

1.3 Rationale of the Study

There is a pressing need to examine the relationship between the built environment and electricity consumption in Bangladesh amid rapid urbanization and increasing energy demands. As Dhaka continues to grow as a densely populated megacity, disparities in electricity usage between urban and rural areas highlight the urgency of understanding consumption patterns within urban settings. Despite the recognized influence of factors such as building design, urban layout, and infrastructure on energy use, there is a significant gap in research specific to Bangladesh's urban context. This study addresses this gap by investigating electricity consumption trends and the role of built environment characteristics in two contrasting areas—Dhanmondi and Lalbagh—offering a comparative perspective. The findings provide valuable insights for developing energy-efficient urban planning strategies, reducing inequities in energy access, and supporting Bangladesh's commitments to sustainable development and greenhouse gas reduction. By bridging this knowledge gap, the research aims to contribute

actionable solutions to enhance energy sustainability and inform policy decisions in a rapidly evolving urban landscape.

1.4 Scope and Limitation of the Study

The scope of this research encompasses an analysis of electricity consumption trends per consumer across residential, commercial, institutional, industrial and mixed-use usage categories. Utilizing historical data, the study aims to identify consumption patterns and develop predictive models based on a set of built environment variables. The research employed quantitative methods, such as statistical analysis, to investigate the relationships between these variables and electricity usage. An integrating analysis of the influence of both natural and built environment, provided valuable insights for urban planners to regulate the built environment, offer environmentalists a platform to emphasize natural concerns, and empower policymakers to formulate comprehensive strategies for managing electricity demand. However, further examinations may be needed in future to reach an integrated decision.

This research is constrained by several limitations that impact its scope and findings. Firstly, data for high-income areas within Dhaka city were unavailable, limiting the study to middle-income and low-income areas only and excluding insights into electricity consumption patterns in wealthier neighbourhoods. Secondly, additional variables critical to understanding electricity usage—such as solar exposure of buildings, building orientation, the extent of electric device usage, and residents' income levels—could not be incorporated due to a lack of accessible data. These factors are significant in capturing the socio-economic and environmental influences on consumption. Furthermore, conducting a primary survey across the large study area was deemed too time-consuming, leading to reliance on secondary data. Consequently, the findings may not fully represent the broader dynamics of electricity consumption, and these limitations should be considered when interpreting the results.

CHAPTER 2 : LITERATURE REVIEW

Built Environment: The built environment refers to the human-made structures and spaces that provide the setting for human activity. This includes buildings, parks, transportation systems, and other infrastructure that shape the physical landscape of urban and rural areas. The built environment significantly influences various aspects of life, including social interactions, economic activities, and environmental impacts. It plays a crucial role in determining energy consumption patterns, particularly in urban areas where population density and land use diversity are pronounced (IPCC, 2014). Understanding the built environment is essential for developing effective energy management strategies, especially in rapidly urbanizing regions.

Although cities occupy only 3% of the Earth's land, they consume between 67% and 76% of global energy (IPCC, 2014). Research globally highlights that variables such as land use, building characteristics, and spatial configurations significantly affect energy efficiency, with implications for urban planning and policy design (Chen et al., 2017; Wang et al., 2023). For instance, studies in Iran, China, and the United States demonstrate that urban density, vegetation cover, and building age directly correlate with energy intensity, while socio-economic factors like occupancy levels further modulate consumption patterns (Ali et al., 2020; Anderson et al., 2015; Debnath et al., 2019). This review synthesizes key findings on built environment variables and their relationship with electricity consumption, providing a foundation for analyzing their interdependencies in urban contexts.

Normalized Difference Vegetation Index (NDVI): The Normalized Difference Vegetation Index (NDVI) quantifies vegetation coverage and health by analyzing satellite reflectance in visible and near-infrared spectra. Positive NDVI values indicate vegetation density, while negative values suggest barren surfaces. Studies highlight that higher NDVI correlates with reduced energy consumption in urban areas, as vegetation mitigates microclimates by lowering ambient temperatures, thereby decreasing cooling demands (Nie et al., 2021; Yuan et al., 2019). For instance, Chen et al. (2022) noted that urban greenery indirectly enhances energy efficiency by shading buildings and reducing heat island effects, reinforcing NDVI's role in sustainable urban planning.

Modified Normalized Difference Water Index (MNDWI): MNDWI measures water body distribution using remote sensing data, emphasizing surface water dynamics in urban and peri-urban regions. Positive values indicate the presence of open water bodies (e.g., lakes, rivers, ponds), and negative values suggest non-water areas, such as urban surfaces, vegetation, or barren land. Proximity to water bodies, as captured by MNDWI, influences localized cooling effects, which can reduce energy consumption for air conditioning (Lee et al., 2023). Research by Movahedi and Derrible (2021) highlights the utility of MNDWI in assessing hydrological patterns and their indirect impact on energy demand, particularly in managing thermal comfort through evaporative cooling and heat dissipation.

Distance from Nearby Permeable Areas: The distance from buildings to permeable areas (PD) (e.g., parks, water bodies) significantly affects local thermal environments. Structures farther from these areas often experience higher ambient temperatures, increasing reliance on cooling systems and electricity use (Cho et al., 2019; Pisello et al., 2016). Wang et al. (2023) emphasize that urban greening strategies, such as reducing PD, can lower energy demands by enhancing shading and evaporative cooling, highlighting PD's critical role in energy-efficient urban design.

Floor Number: The total number of floors in a building influences energy use patterns, particularly for heating, ventilation, and air-conditioning (HVAC) systems. Taller buildings often exhibit higher energy consumption due to the "stack effect," which exacerbates heat loss in winter and heat gain in summer, necessitating intensified HVAC operations (Cho et al., 2019; Mutani et al., 2015). Floor number's impact on vertical energy dynamics underscores its relevance in modelling urban energy demand and optimizing building design.

Building Density: Building density (BD), measured as the number of structures within a radius, shapes energy consumption by intensifying urban heat island effects. High BD elevates ambient temperatures, increasing cooling loads, but also enables shared energy systems and compact designs that promote efficiency (Hong et al., 2020; Wang et al., 2023). This duality positions BD as a critical variable for balancing energy demands and sustainable urban development.

Building Area and Building Volume: Building area (BA) and volume (BV) directly correlate with energy requirements, as larger structures demand more energy for lighting, heating, and cooling. BA reflects ground coverage, while BV incorporates vertical expansion, both influencing thermal mass and insulation needs (Bastida-Molina et al., 2024). Ghadi and

Baniyounes (2018) note that BV's impact on thermal dynamics necessitates careful consideration in energy performance assessments, particularly for high-rise structures.

Shape Factor: The shape factor (SF), defined as the perimeter-to-area ratio, indicates a building's geometric efficiency in minimizing energy loss. Compact designs (lower shape factor) reduce exposed surfaces, limiting heat exchange with the environment and lowering HVAC demands (Mutani et al., 2015). The role of the shape factor in optimizing energy efficiency underscores its importance in architectural planning and retrofitting strategies.

Customer No: Customer no representing building occupancy levels significantly influences energy consumption patterns. Higher occupancy correlates with increased appliance use and variable behavioral practices, such as lighting and temperature preferences (Cho et al., 2019; Barmmparesos et al., 2019). Newsham et al. (2009) stress that CN's variability necessitates occupant-centric energy management systems to align consumption with user needs and efficiency goals.

Building Age: Building age measured from construction to the study year (2023), reflects historical construction practices and material efficiency. Older buildings often lack insulation and modern energy systems, leading to higher heating/cooling demands and emissions (Nie et al., 2021; Pickering et al., 2017). Retrofitting aging infrastructure emerges as a key strategy to mitigate energy inefficiencies linked to building age.

Structure Type: Building structure types, specifically katcha, pucca, and semi-pucca, significantly influence the electricity consumption of residential buildings. Katcha houses, typically constructed from temporary or low-quality materials, tend to have poor insulation properties, leading to higher demand for cooling and heating systems during extreme weather, thus increasing electricity consumption (Amber et al., 2019). In contrast, pucca constructions are made from durable, high-quality materials that feature better thermal insulation and energy efficiency, reducing dependence on electrical energy for heating or cooling and thereby lowering overall electricity usage (Brounen et al., 2012). Semi-pucca houses, which incorporate elements of both katcha and pucca designs, present variable electricity consumption patterns influenced by the proportion of permanent fixtures and insulation present. As a result, the energy performance of these different structure types can be effectively quantified, showing that well-designed structures, such as pucca houses, substantially mitigate electricity consumption compared to their katcha counterparts (Wahlström & Hårsman, 2015).

The built environment significantly influences electricity consumption patterns through various factors, including urban form, socio-economic conditions, and technological integration. As cities continue to grow, particularly in developing countries like Bangladesh, further research is needed to explore the specific impacts of the built environment on energy consumption. This understanding will be vital for developing effective strategies to manage energy demand and promote sustainable urban development.

Multilinear Regression Analysis: Multilinear regression analysis is a statistical method used to model the relationship between two or more independent variables and a dependent variable. By fitting a linear equation to observed data, it allows researchers to understand how the independent variables collectively influence the outcome of the dependent variable (Amber et al., 2019). This analysis is particularly useful for predicting outcomes and assessing the strength of predictors. In the context of energy consumption, multilinear regression can help identify which design parameters and external factors most significantly impact energy performance in buildings (Wahlström & Hårsman, 2015)

CHAPTER 3: METHODOLOGY

Methodology refers to the process or steps followed to complete a study. This process varies based on the subject matter and the context of the particular study. It serves as a guideline that, if followed properly, will lead to the same results outlined in the study. In this sense, the reliability of a study significantly depends on its methodology.

3.1 Study Area Selection

The study aimed to analyze electricity consumption patterns across different spatial units with distinct socio-economic characteristics. The Detailed Area Plan (DAP) 2022-2035 of Dhaka provides data on building structures, which helped identify areas representing varying socio-economic groups. Dhanmondi (including Dhanmondi and Jigatola) was chosen to represent middle-income areas, as Dhanmondi is known for its planned residential zones occupied by middle and upper-middle-income households. Lalbagh (encompassing Lalbagh and Kamrangirchar) was selected to represent low-income areas, with Kamrangirchar being one of the most densely populated and underprivileged neighbourhoods in Dhaka. The study intended to include high-income areas as well, but due to the unavailability of electricity consumption data for such regions, they could not be incorporated. Thus, the focus was narrowed to low-income (Lalbagh and Kamrangirchar) and middle-income (Dhanmondi and Jigatola) areas to understand their respective electricity consumption patterns.

3.2 Data Collection

3.2.1 Variables Considered for the Study: The study used a combination of data sources to comprehensively understand the factors of the study area that influence electricity consumption. A total of ten variables shown in Figure 3.1, have been used to model electricity consumption patterns for different types of building use. The study focused on building-level data and the surrounding environment. Variables that have been used are floor no, building area(sqm), NDVI, MNDWI, building density (250m), building volume (m³), distance from the nearest permeable land of each building(m), customer no, building age(year) and perimeter/area ratio. Models of electricity consumption for five distinctive uses (residential, commercial, institutional, industrial and mixed-use) have been prepared based on consumption data in 2023. Trend analysis has been done for four types of building uses (residential, commercial, institutional and industrial) based on consumption data from 2017 to 2023.

Variables	Definition
Normalized Difference Vegetation Index (NDVI)	The amount of vegetation and plant growth status. Negative values indicate no vegetation; positive values indicate vegetation coverage and an increase in coverage.
Average Modified Normalized Difference Water Index (MNDWI)	The amount and distribution of water bodies
Distance from nearby permeable area (D)	Distance of each building from its nearest permeable land such as water body, open space
Floor No (F)	Total number of floors in a building
Building Density(250m)	Number of buildings within 250m buffer around each building
Building Area(sqm)	The ground area of each building
Building Volume (cubic meter)	Multiplication of building area and total building height
Shape Factor	The perimeter of a building is divided by the area of the building's ground area.
Customer No	Number of consumers under a building
Building Age	Duration of building's sanction date to electricity connection
Structure Type	Three categories of structural types are considered. They are pucca, semi pucca and katcha

Figure 3.1: Definition of built environment indices

Source: Author, 2024

3.3.2 Secondary Data Collection: The whole study is based on secondary data (shown in Figure A2 of Appendix A). NDVI and MNDWI data are collected from Landsat 8 (USGS). Dhaka Power Distribution Company Limited (DPDC) provide electricity connections at Lalbagh and Dhanmondi. Building-level data (floor no, building area, perimeter, sanction date and consumer number) has been extracted from a GIS file, which contained building footprints and consumer data. ‘Mixed-use’ is not defined in tariff categorization by DPDC. So, the building is mentioned as “Mixed-use” if different types of uses are present under a building. DPDC has provided the structure type data. Tariff data was given at the consumer level and further converted into building-level. Tariff categorization is illustrated in Figure A1 of the Appendix A.

3.3 Data Analysis

3.3.1 Trend Analysis: Historical data from 2017 to 2023, covering electricity consumption for different types of users, has been analyzed to understand consumption patterns. The data spans seven years and includes the total electricity consumption of all consumers by use category. The electricity consumption per consumer by use category for each year has been graphed using Origin Pro software. Trend analysis was conducted for four categories of consumers: residential, commercial, industrial, and institutional. The graph illustrates how the patterns of electricity consumption per consumer have evolved over the years for each user type. Historical data from 2017 to 2023, covering electricity consumption for different types of users, has also been analyzed to understand consumption patterns based on the total consumption of all consumers by use category. The electricity consumption of all consumers for different uses for each year has been graphed using Origin Pro software. Trend analysis was also conducted for four categories of consumers: residential, commercial, industrial, and institutional in this graph. The graph illustrates how the patterns of total electricity consumption of all consumers have evolved over the years for each user type.

3.3.2 Shannon Diversity Index : The Shannon Diversity Index is an essential metric utilized to evaluate land use diversity by quantifying the diversity and evenness of land cover types within a given area. Originally developed for ecological studies, the Shannon-Wiener Index (H') is commonly employed in the context of land use analysis to assess the relative diversity of various land cover classes. This index captures both richness (the number of different types of land use) and evenness (the distribution of land cover classes) within a defined landscape,

allowing researchers to infer insights regarding ecological health and land management practice. (Navalho et al.,2017).

The calculation involves finding the proportion of each land use type and using the formula

$H' = - \sum_{i=1}^k p_i \ln p_i$, where p_i is the proportion.

The land use diversity index was calculated in both Dhanmondi and Lalbagh, by measuring the area for each type and finding the total area at the initial. Then, proportion was found by the proportion p_i of each type by dividing its area by the total. For each type, the proportion was multiplied by the natural logarithm of p_i (using $\ln$). Finally, all values were added up and then multiplied by -1. A higher H' value indicates greater diversity, which is useful for ecological and urban planning.

3.3.3 Welch T-Test: The Welch t-test is a statistical hypothesis test used to determine whether there is a significant difference between the means of two independent groups. It is particularly useful when the two groups have unequal variances and/or unequal sample sizes. The Welch t-test adjusts the degrees of freedom used in the estimation of the p-value, thus providing a more reliable result when variance homogeneity is not assumed (Bajagain et.al.,2020). The Welch t-test was conducted in the study to determine whether there is a statistically significant difference in mean electricity consumption per square meter (m^2) between Lalbagh and Dhanmondi for different building uses. This test was chosen because the assumption of homogeneity of variances (equal variances between the two groups) was violated, as confirmed by Levene's test for equality of variances. The Welch t-test is specifically designed to handle cases where the variances of the two groups are unequal, making it more appropriate than the standard independent t-test in such scenarios. In this study, the sample sizes were sufficiently large (greater than 30), which rejected the need for a normality test, as the Central Limit Theorem ensures that the sampling distribution approximates normality with large sample sizes. Before running the Welch t-test, outliers in the data were identified using the boxplot method. These outliers were addressed by replacing them with the upper and lower fence values of the boxplot, ensuring that extreme values did not disproportionately influence the results. Since the p-value is much smaller than the chosen level of significance ($\alpha = 1\%$), the null hypothesis, which states that there is no difference in mean electricity consumption per m^2 between the two areas, was rejected. Instead, the alternative hypothesis was supported, indicating that there is indeed a statistically significant difference in electricity consumption patterns between Lalbagh and Dhanmondi for different uses.

3.3.4 ART Three-Way ANOVA Test: The ART (Analysis of Relationships between Treatment) Three-Way ANOVA test is a statistical method used to investigate interactions among three independent categorical variables and their effect on a continuous dependent variable. This approach is particularly powerful for evaluating complex experimental designs where the influence of multiple factors and their interactions on a response variable are examined simultaneously (Sundari et al., 2013). The ART Three-Way ANOVA test can be effectively utilized in cases of heterogeneity of variance, allowing for robust statistical analysis even when variances across groups are unequal. This flexibility enables researchers to explore interactions among multiple factors without the stringent requirement for equal variances that traditional ANOVA methods impose (Liu & Zhang, 2012). The ART (Aligned Rank Transform) Three-Way ANOVA was selected to analyze electricity consumption patterns in Dhaka because Levene's test ($p < 0.001$) confirmed unequal variances across groups, violating assumptions of traditional ANOVA. Electricity consumption data often includes outliers and skewed distributions (e.g., high usage in commercial vs. residential buildings). The study required testing three factors (location, structure type, use) and their interactions, which standard non-parametric tests cannot handle. Adjusted data to isolate the effect of each factor (e.g., removing the influence of "location" when analyzing "structure type"). Performed a three-way ANOVA on ranked data to test main effects (location, structure, use), two-way interactions (e.g., structure $\times$ location) and three-way interaction (structure $\times$ location $\times$ use). Factors Considered for the test are location (Dhanmondi and Lalbagh), structure type (pucca, semi-pucca, katcha) and use (residential, commercial, industrial, institutional and mixed-use). Here, the dependent variable was electricity consumption per area. Since the p-value is much smaller than the chosen level of significance ($\alpha = 1\%$), the null hypothesis was rejected and the alternative hypothesis was accepted.

3.3.5 Models of Electricity Consumption: Models are prepared using Origin Pro for statistical analysis. ArcGIS software is used to extract essential data from raw information. Satellite images from Landsat 8 are used and further processed through ArcGIS to extract the value of NDVI and MNDWI. Building footprints provided by DPDC and land use extracted from DAP (2022- 2035) of the study area are used to find out the value of the distance from the nearest permeable area of each building in ArcGIS. A 250m buffer area is created around each building, and building density is analyzed through ArcGIS. DPDC provided an Excel file including consumer-level data. Each consumer has a specific building ID. A formula is used to count the

repetition of the consumer number under a building. Structure Type is categorical data. Here, numerical values were assigned to each category of structure type. So, encoding technique (binary number 0 or 1) was used to transform the categories into numerical values. Building age is calculated from the duration gap between the sanction date of the building and 2024 (considering the last year) in Excel. Models are prepared to understand the relationship between the built environment with electricity consumption for five types of uses. Electricity consumption data for 2023 (given by DPDC) is used as a dependent variable in the models. They are residential, commercial, industrial, institutional and mixed-use. Co-linearity among the input parameters is common in regression analysis. A Pearson correlation test was conducted to ascertain collinearity between the inputs. Input variables with Pearson correlation coefficients exceeding 0.8 should be excluded to build a statistically stable model (Kim, 2019). However, the omission of variables due to multicollinearity may undermine the practical feasibility of the model (Gregorich et al., 2021). Thus, to acknowledge the practical complexities of electricity consumption, input variables with Pearson correlation coefficient values of 0.85 or higher were considered moderately correlated. Stepwise multi-linear regression analysis is used to create an equation with significant variables for modelling electricity consumption. At first, MLR models were prepared with all variables. Then, the final MLR models were chosen, having good F values, lower RMSE values, and significant p-values (<0.005) at a 95% confidence level after conducting several attempts. Tables 3.1 and 3.2 represent the sample size used for MLR models.

Table 3.1: Sample Size Used for MLR Models in Lalbagh

Source: DPDC

Location		Use Types	Sample Size
Lalbagh		Residential	14239
		Commercial	1595
		Industrial	928
		Institutional	135
		Mixed	5851

Table 3.2: Sample Size Used for MLR Models in Dhanmondi

Source: DPDC

Location		Use Types	Sample Size
Dhanmondi		Residential	7674
		Commercial	2110
		Industrial	40
		Institutional	131
		Mixed	69

CHAPTER 4 : LANDUSE PATTERN, STRUCTURE TYPE AND ELECTRICITY CONSUMPTION IN THE STUDY AREAS

4.1 Location of the Study Area

This chapter examines the relationship between land use patterns and electricity consumption in the two studied neighbourhoods. The study focuses on identifying the characteristics of energy use of settlements with different socio-economic features. Two neighbourhoods of Dhaka, Dhanmondi (which includes localities of Dhanmondi and Jigatola) and Lalbagh (encompassing Lalbagh and Kamrangirchar), were chosen as the study areas as they provide diverse socio-economic and land use contexts. While Dhanmondi is mainly mixed and commercial land use on the side of the road, Lalbagh is dominated by residential and industrial land use. Mainly, high- and middle-income people live in Dhanmondi, while lower-middle-income and low-income groups dominate Lalbagh. Dhanmondi has a total area of 1614 acres, and Lalbagh has a total of 1539 acres, shown in Figure 4.1 (DPDC, nd)

Dhanmondi is located in the western part of Dhaka. Dhanmondi is a planned residential and commercial area characterized by its organized layout, green spaces, and relatively higher socio-economic status compared to other parts of the city. This neighborhood has seen significant urban development and infrastructure improvements, contributing to its role as a hub for educational institutions, healthcare facilities, and commercial activities over the years. The built environment in Dhanmondi, with its modern buildings and amenities, influences electricity consumption patterns, reflecting the lifestyle and energy needs of its residents (Chowdhury et al., 2019).

In contrast, Lalbagh, situated in the older part of Dhaka, represents a historically and culturally significant area with densely packed low-rise structures and limited green spaces (Dewan & Yamaguchi, 2009). Narrow streets, traditional architecture, and a mix of residential and commercial spaces mark this neighborhood. Lalbagh's socio-economic conditions are more varied, with a significant portion of the population living in informal settlements. The built environment in Lalbagh presents unique challenges, including limited access to modern infrastructure and services, which can affect energy consumption patterns (Sikder et al., 2018).

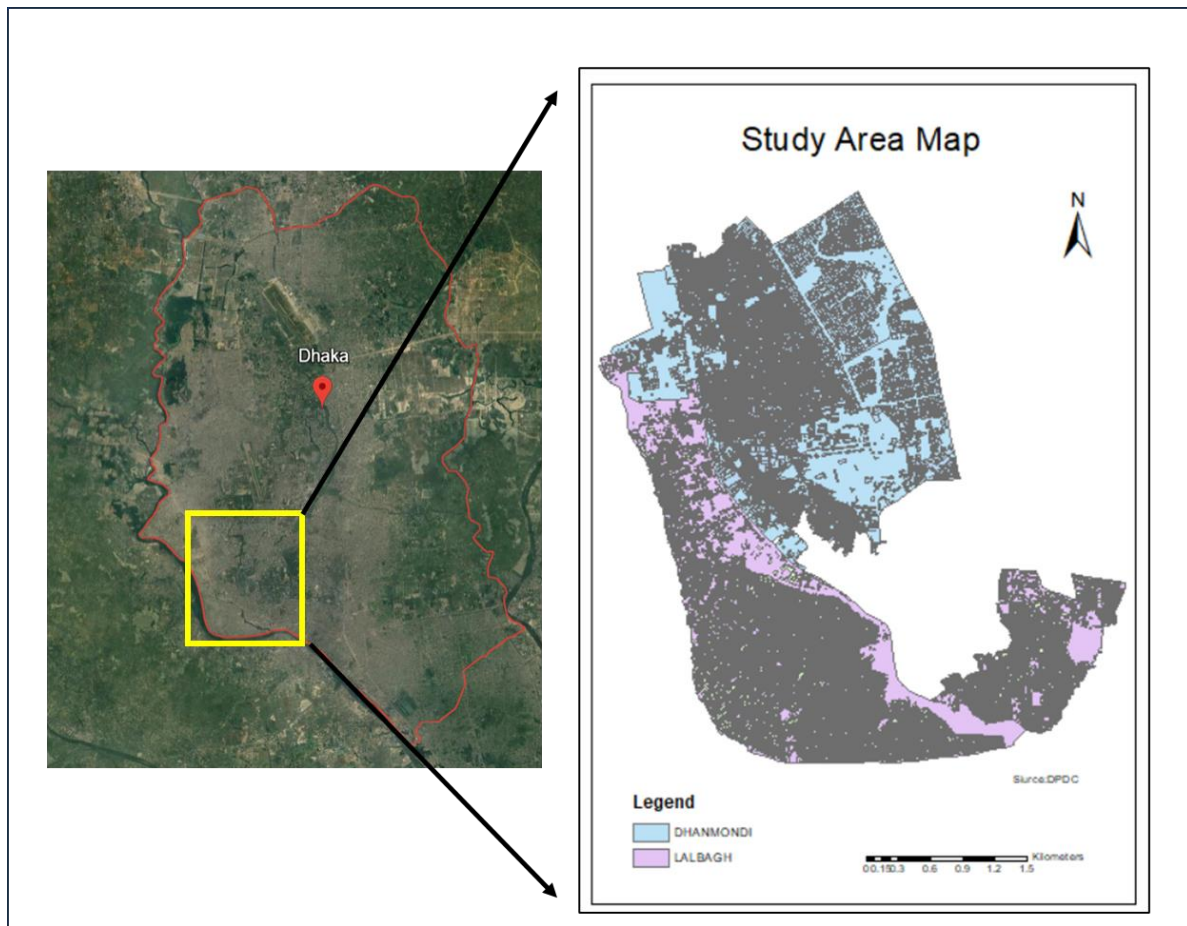


Figure 4.1: Location of Study Area

Source: Google Earth and DPDC

4.2 Land Use of the Study Area

Lalbagh

The land use map of Lalbagh shown in Figure 4.2, portrays a densely developed and multifaceted area with a diverse range of land use categories. Residential zones, marked in yellow, dominate the landscape, particularly in the northern and central regions, indicating a heavily populated core. Mixed-use areas, shown in orange, are also prominent, especially in the central and southern parts, blending residential, commercial, and other functions. Agricultural land, in light yellow, is limited to small patches along the northern and eastern fringes, while commercial areas, in red, cluster near the center and along major thoroughfares. Industrial zones, depicted in dark red, are less widespread and primarily situated in the south. Open spaces in green are scarce and scattered, suggesting minimal recreational or undeveloped land, and water bodies, in blue flow through the center, dividing the region. Additionally, smaller pockets of administrative (light green), community (purple), educational and health,

transport (gray), restricted (light orange), and vacant (light beige) uses enhance the area's functional diversity.

It is clear that Lalbagh's land use is predominantly residential, accounting for 37.52% of the total area, which highlights a significant focus on housing and suggests a high population density (shown in Table C1 of Appendix C). The second-largest category, Mixed Use, spans 28.40% area, reflecting a substantial blend of residential, commercial, and other functions, a key feature of urban complexity. Transport and Communication occupy 110.148 acres (7.15%), indicating well-developed infrastructure that supports connectivity across the region. Water bodies, at 104.01 acres (6.75%), add ecological and aesthetic value, though their limited extent shows constrained natural features. Vacant land is scarce, covering 82.94 acres (5.39%), suggesting high land utilization and potential pressure for future development. Smaller categories further diversify Lalbagh's landscape. These minor categories collectively enhance the functional diversity of Lalbagh, supporting a multifaceted urban environment.

The Shannon Diversity Index (H') for Lalbagh's land use is 1.79, a value that reflects a moderate to high level of diversity across its land use categories (shown in Table C1). This index measures both the number of distinct land use types (15 in total) and the evenness of their distribution across the total area of 1540.10 acres. With 15 categories, the maximum H' would be $\ln(15) \approx 2.70$, which occurs if all categories were equally represented (each at 6.66% of the total area). A higher H' value indicates greater diversity, and Lalbagh's score suggests a varied urban landscape, driven by the presence of numerous categories, from dominant ones like residential (37.52%) and mixed-use (28.40%) to smaller ones like agriculture (0.03%) and administrative (0.02%). However, the significant dominance of residential and mixed-use areas, which together cover over 65% of the land, may temper the index's value compared to a hypothetical scenario where land is more evenly distributed across all categories. The contributions of smaller categories, such as transport and communication (7.15%), water Body (6.75%), and industrial (4.89%), enrich the diversity by adding functional variety, supporting ecological, infrastructural, and economic roles. Thus, an H' of 1.790343 interprets. So, the value of H' represents that Lalbagh as a region with a robust mix of land uses, balancing residential focus with diverse urban functions, though not perfectly even in distribution.

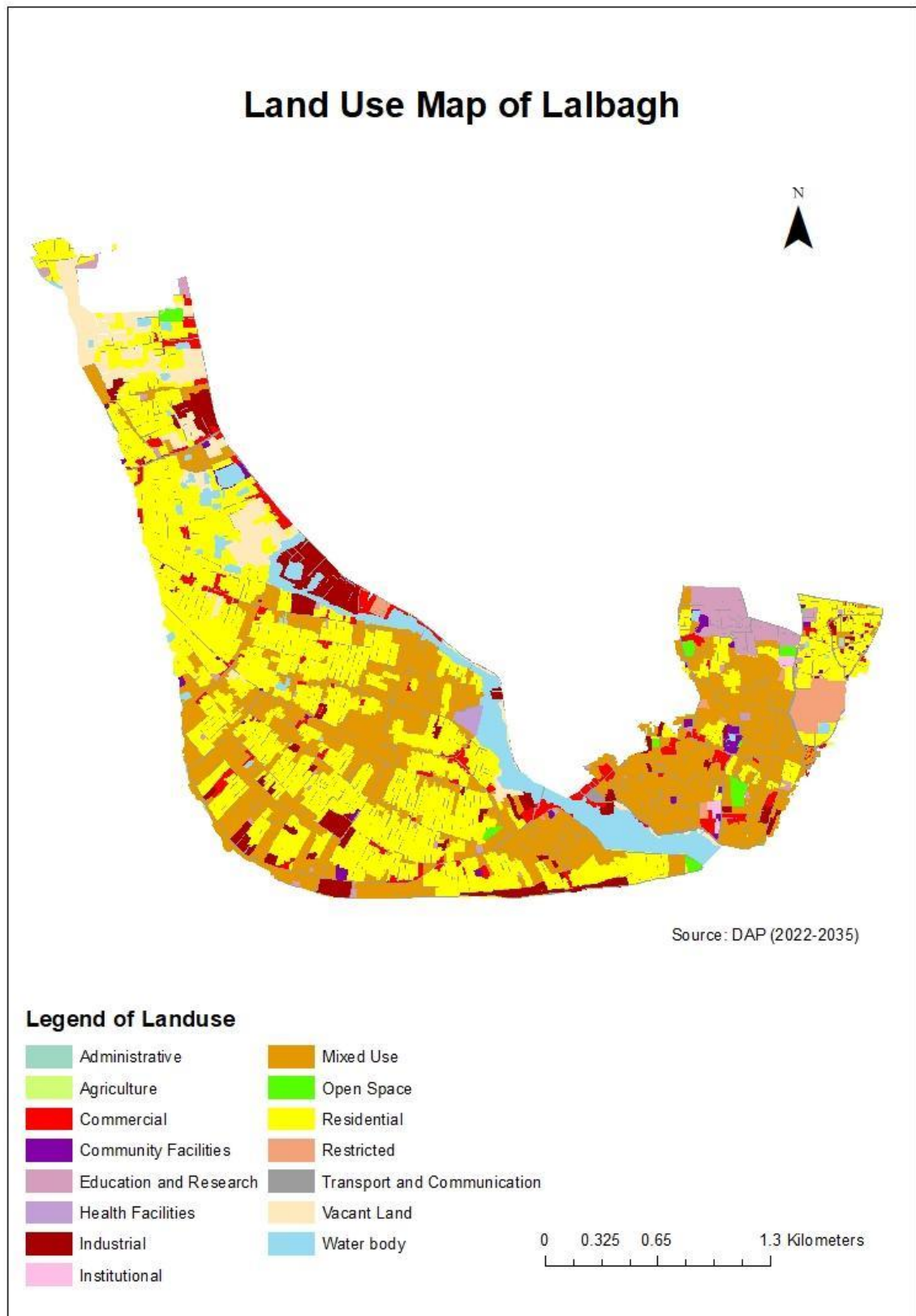


Figure 4.2: Landuse Map of the Lalbagh

Source: DAP (2022-2035)

Dhanmondi

The land use map of Dhanmondi shown in Figure 4.3 portrays a densely developed urban area characterized by a diverse mix of land use categories. Residential zones, marked in yellow, dominate the central and northern regions, establishing Dhanmondi as a primarily residential neighborhood with a high concentration of housing. Commercial areas, indicated in red, are strategically scattered along major roads and concentrated in the eastern and southern parts, reflecting vibrant business and shopping districts. Industrial zones, shown in maroon, are confined to the central-southern area, indicating limited yet notable industrial activity. Community facilities depicted in purple, are dispersed with significant clusters in the northern and central sections, while mixed-use zones, in orange, blend residential and commercial uses in the western and northern parts. Open spaces, marked in green, are scarce but present in the northern and eastern regions, offering limited recreational areas. Educational and research institutions, in brown, are situated in the western and southern areas, and transport and communication zones, in gray, align with major thoroughfares in the western and central parts, supporting urban connectivity. Restricted areas which is “Pilkhana” (peach), vacant land (light beige), and water bodies (blue) are minimal, appearing mainly in the southern, western, and central regions.

Table C2 of Appendix C, provides a detailed breakdown of land use categories in Dhanmondi, listing each category along with its corresponding total area. Residential is the largest category, occupying 37.03% (596.67 acres) of the total area, making it the dominant land use in Dhanmondi. Transport and Communication follows at 11.28% indicating significant land dedicated to roads, highways, or transit facilities. Restricted 10.55 and mixed use 10.47% also cover substantial portions, suggesting areas with specific zoning restrictions or a blend of residential, commercial, and other uses. Smaller categories include institutional, community facilities, and open space, showing limited but present public or undeveloped areas. Other notable categories are education and research (7.55%), industrial (5.16%), and water body (4.90%), contributing to the variety of uses.

The Shannon Diversity Index (H') for Dhanmondi's land use is 2.03, a value that reflects a moderate to high level of diversity across its land use categories (shown in Table C2). This index measures both the number of distinct land use types (12 in total) and the evenness of their distribution across the total area of 1611.12091 acres. With 12 categories, the maximum H' would be $\ln(12) \approx 2.4849$, which occurs if all categories were equally represented (each at

8.33% of the total area). A higher H' value indicates greater diversity, and Dhanmondi's score suggests a varied urban landscape, driven by the presence of numerous categories, from dominant ones like residential (37.03%) and transportation and communication (11.28 %) to smaller ones like institutional (0.075%), community facilities (2.56%), and open space (2.65%). However, the significant dominance of residential and mixed-use areas, which together cover over 47.5% of the land, may temper the index's value compared to a hypothetical scenario where land is more evenly distributed across all categories. The contribution of categories such as education and research (7.55%), industrial (5.16%), and water body (4.90%) enrich the diversity by adding functional variety, supporting ecological, infrastructural, and economic roles. So the value of H' represents that Dhanmondi is a region with a robust mix of land uses, balancing residential focus with diverse urban functions, though not perfectly even in distribution.

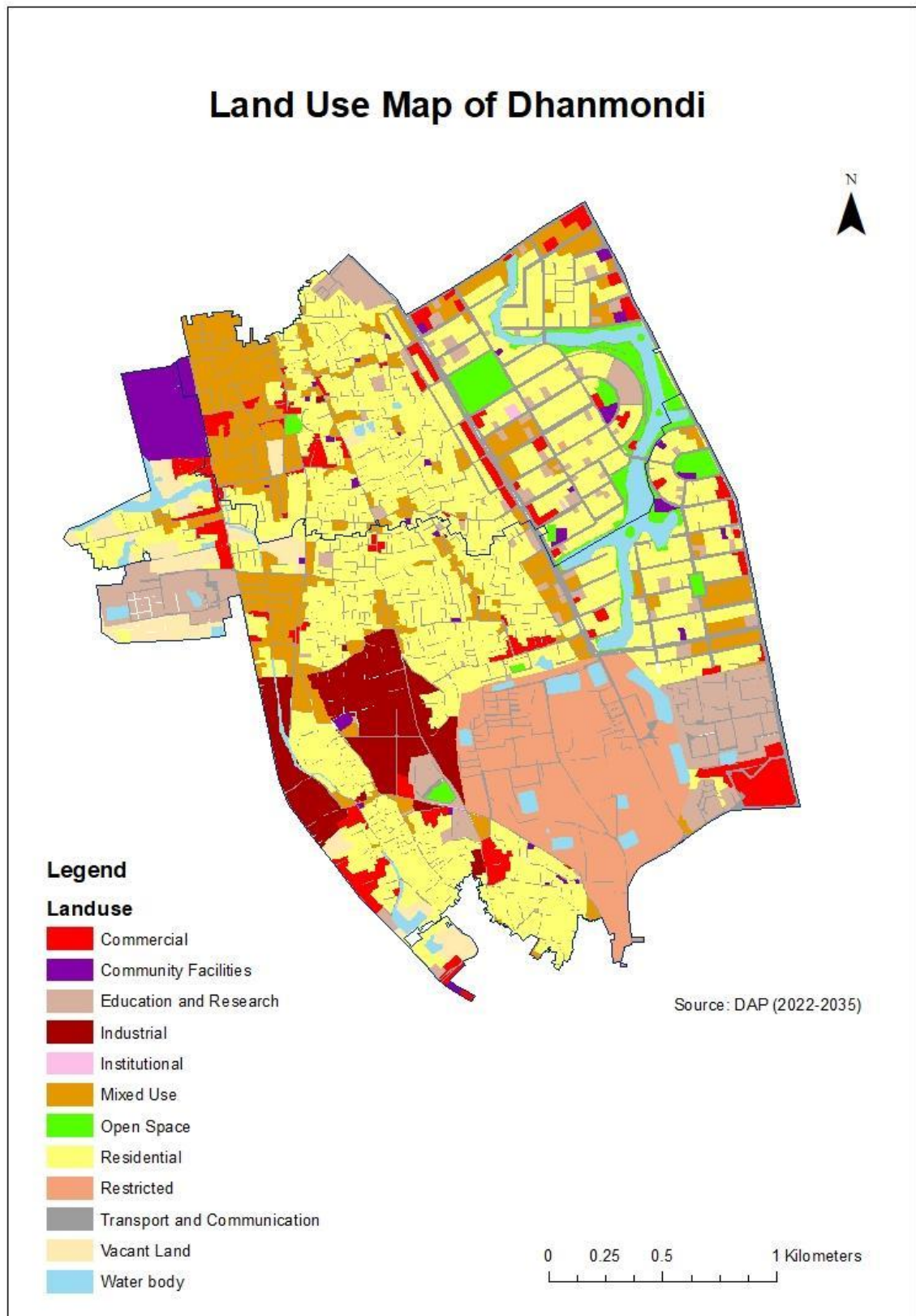


Figure 4.3: Landuse Map of the Dhanmondi

Source: DAP (2022-2035)

4.3 Structure Typology in the Study Area

This portion represents the structure typology (katcha, pucca, semi pucca and under construction) in the study area. The below figures are prepared based on DPDC data. The DPDC data is last updated on July 2019.

Lalbagh

The “Building Structure Typology of Lalbagh “shown in Figure 4.4 illustrates a dense concentration of buildings, with Pucca structures (green) being the most widespread, particularly dominating the central and southern parts of the area. This distribution suggests these regions form the urban core, characterized by permanent, well-constructed buildings. Katcha structures (red) are less frequent and primarily located in the northern and eastern peripheries, indicating areas that may be less developed or host temporary settlements. Semi-Pucca structures (yellow) are moderately distributed, often interspersed with Pucca buildings, reflecting transitional zones or areas with mixed construction quality. Under Construction structures (brown) are sparse and scattered across the map, pointing to limited ongoing building activity throughout Lalbagh.

Table 4.1 represents a quantitative breakdown of structure types in Lalbagh, showing pucca structures as the most prevalent, comprising 52.78% (12,219 buildings) of the total. This dominance underscores the area's overall permanence and development level. Semi-pucca structures account for 36.67% (8,490 buildings), highlighting a significant presence of semi-permanent constructions, likely serving varied socio-economic needs. Katcha structures represent 10.50% (2,431 buildings), indicating a smaller but notable portion of temporary or less durable buildings. Structures under construction are minimal, at 0.05% (11 buildings), suggesting that Lalbagh is largely built-out with little current development or renovation activity.

Table 4.1: Structure Typology in Lalbagh

Source: DPDC

Structure_Type	Total No of Structure	Percentage (%)
Pucca	12219	52.77957756
Semi-Pucca	8490	36.67228197
Katcha	2431	10.50062632
Under Construct	11	0.047514146

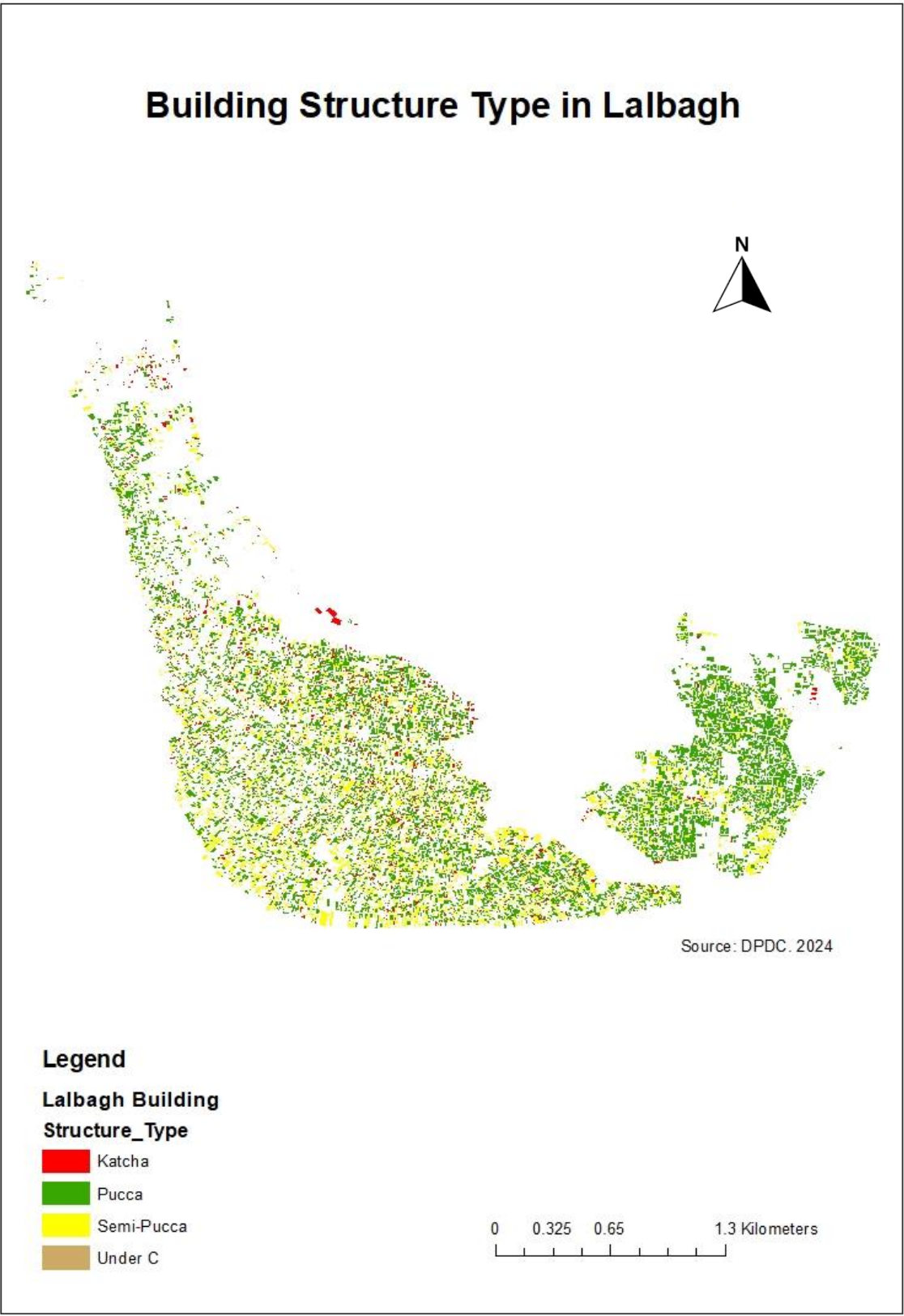


Figure 4.4: Building Structure Typology in Lalbagh

Source: DPDC

Dhanmondi

Figure 4.5 visually represents the distribution of various building structure types within the Dhanmondi area. The legend identifies four categories: Katcha (temporary or mud structures) in red, pucca (permanent or concrete structures) in green, semi-pucca (partially permanent structures) in yellow, and Under Construction in beige. Green dominates the map, especially in the central and eastern regions, indicating that pucca structures are the most prevalent, suggesting a well-developed urban area with durable construction. Red clusters, representing katcha structures, are scattered primarily in the western and southern parts, pointing to pockets of less permanent development. Yellow areas, representing Semi-pucca structures, are dispersed throughout, with noticeable concentrations in the southern and eastern parts. Beige areas, marking buildings under construction, are sparse and scattered across the area.

Table 4.2 lists four categories: semi-pucca, pucca, katcha, and under construction. The data shows 6,254 semi-pucca structures, accounting for 33.41% of the total; 10,079 pucca structures, comprising 53.84%; 2,286 katcha structures, making up 12.21%; and 102 under construction structures, representing just 0.54%. The percentages sum to 100%, offering a complete breakdown of the area's structural composition. This reveals that pucca structures are the most common, aligning with the map's green dominance, and constitute over half of all buildings, reinforcing the interpretation of a stable, developed region. semi-pucca structures are also significant, while katcha structures are less prevalent, and the minimal presence of under construction buildings suggests little current development activity.

Table 4.2: Structure Typology in Dhanmondi

Source: DPDC

Structure_Type	Total No of Structure	Percentage (%)
Semi-Pucca	6254	33.40633513
Pucca	10079	53.83793601
Katcha	2286	12.21088617
Under Constructi	102	0.54484269

Building Structure Type in Dhanmondi

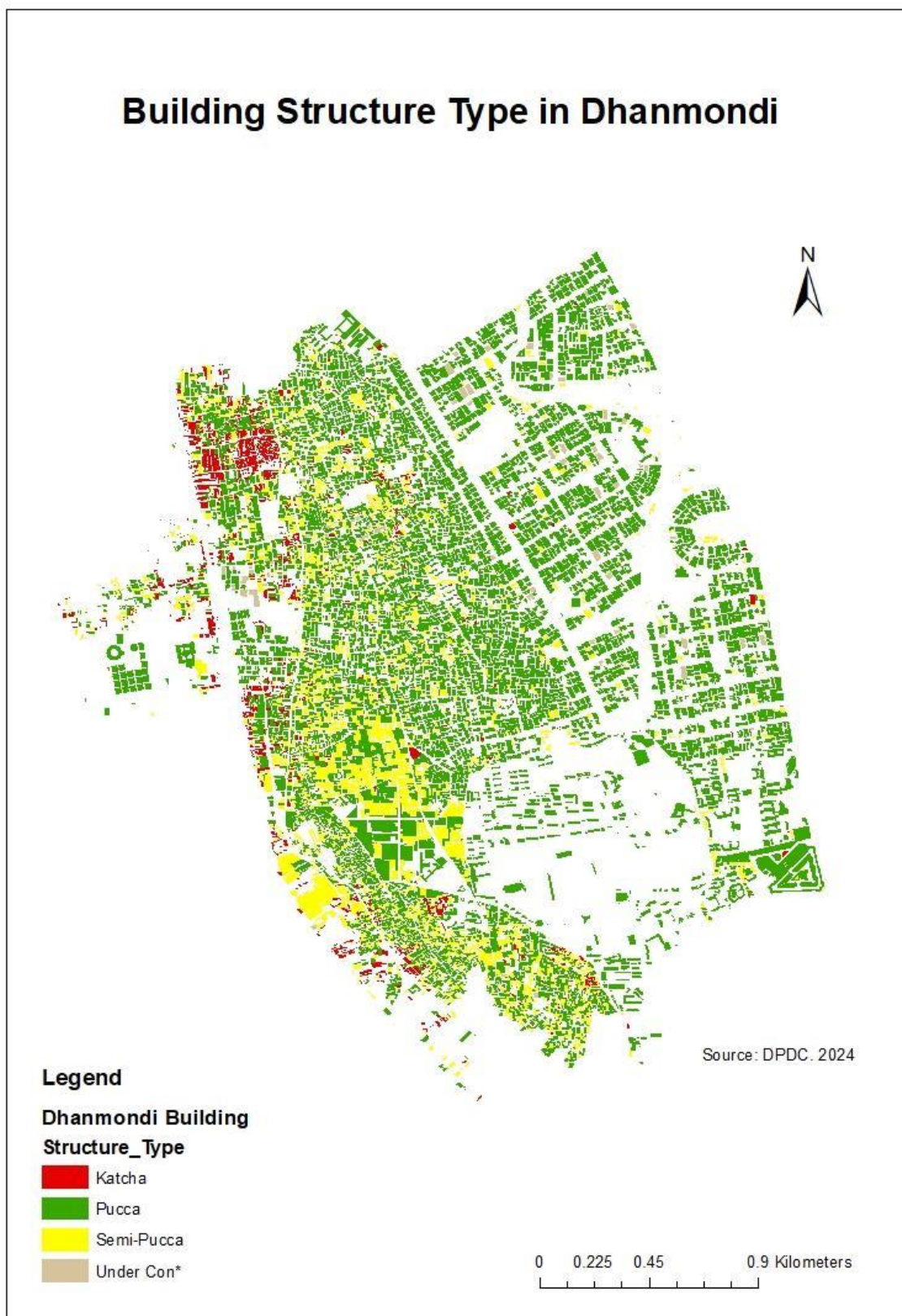


Figure 4.5: Building Structure Typology in Dhanmondi

Source: DPDC

Effect of Building Structure, Location, and Use on Electricity Consumption

To investigate how electricity consumption is influenced by building structure type, location, and use, a three-way ANOVA was employed using the aligned ranks transformation (ART) method. Table 4.3 shows the sample size of the dataset. This approach was chosen due to its robustness in handling non-normal data and heterogeneity of variances, which are present in the data set. Levene test shows that there is no homogeneity in the dataset as $Df = 29$, $F = 87.143$, $p = 2.2e-16 < 0.001$

From Table 4.4, the interpretation is described below.

Main Effects:

1. **Building Structure (StructureT):** The type of building structure significantly impacts electricity consumption. Pucca, semi-pucca, and katcha buildings have different consumption patterns. For example, pucca buildings might consume more electricity than katcha buildings.
2. **Location:** Buildings in Dhanmondi and Lalbagh show significantly different electricity usage.
3. **Use:** Different types of building use (residential, commercial, industrial, institutional, mixed-use) result in distinct electricity consumption levels. For instance, commercial buildings likely consume more electricity compared to residential.

Two-Way Interactions:

1. **StructureT $\times$ Location:** The effect of building structure on electricity consumption varies depending on the location. For example, consumption of pucca and katcha buildings are not equal in Dhanmondi and Lalbagh. This suggests that certain structural types may perform differently based on the specific conditions of each area.
2. **StructureT $\times$ Use:** The impact of building structure depends on how the building is used. For instance, consumption of pucca and katcha buildings is not equal in commercial buildings and residential ones. This indicates that the way a building is used can amplify or reduce the effect of its structure on electricity consumption.
3. **Location $\times$ Use:** The effect of building use on electricity consumption also depends on the location. For example, residential buildings in Dhanmondi consume more electricity

per area compared to those in Lalbagh. This implies that the interaction between location and building use creates unique consumption patterns.

Three-Way Interaction:

- **StructureT × Location × Use:** The combined effect of building structure, location, and use on electricity consumption is complex and not simply additive. Specific combinations exhibit unique consumption patterns. For example, pucca buildings in Dhanmondi used for commercial purposes is different compared to electricity consumption compared to other combinations. This highlights that the interplay of all three factors creates specific scenarios where consumption is influenced in ways that cannot be predicted by looking at each factor individually.

Table 4.3: Sample Size of ART ANOVA Test

	Dhanmondi			Lalbagh		
	katcha	Pucca	Semi-Pucc	katcha	Pucca	Semi-Pucca
Residential	309	5393	1568	1618	7369	5247
Commercial	97	1452	456	271	600	723
Industrial	4	27	6	84	279	563
Institutional	5	85	30	6	111	19
Mixed	3	50	17	419	3715	1712

Table 4.4: Results of the ART ANOVA revealing significant main effects and interactions,

	Df	Df.res	F value	Pr(>F)
StructureT	2	32197	163.343	< 2.22e-16 ***
location	1	32197	773.009	< 2.22e-16 ***
Use	4	32197	455.104	< 2.22e-16 ***
StructureT:location	2	32197	203.115	< 2.22e-16 ***
StructureT:Use	8	32197	273.13	< 2.22e-16 ***
location:Use	4	32197	95.319	< 2.22e-16 ***
StructureT:location:Use	8	32197	94.219	< 2.22e-16 ***
Signif. codes: 0 '***' 0.001 ' ' 0.01 ' ' 0.05 ' ' 0.1 ' ' 0.5 ' ' 1				

4.4 Electricity Consumption Pattern in the Study Area

Figures 4.6 and 4.12 illustrate electricity usage across Lalbagh and Dhanmondi. It defines five consumption ranges: light green (1,000– 15,000kWh), green (15,000 – 60,000 kWh), yellow (60,000 – 1,00,000 kWh), orange (1,00,000 – 2,00,000 kWh), and red (2,00,000 – 38,70,000 kWh).

Lalbagh

Figure 4.6, depicting electricity consumption in Lalbagh, illustrates the spatial distribution of energy usage across the region, measured in kilowatt-hours (kWh). The data is categorized into five distinct consumption ranges, each represented by a specific colour, enabling precise analysis of the consumption distribution. The lowest consumption range, indicated by light green, spans from 1,000,000 to 15,000,000 kWh, and is widely distributed across the northern, western, and central areas of the map, indicating that a significant portion of these regions falls within this category of minimal energy usage. The next range, represented by darker green, covers consumption from 15,000,000.001 to 60,000,000,000 kWh, and is also extensively present in the northern, western, and central areas, suggesting a broad prevalence of moderate energy use in these directions. Higher consumption levels are less widespread but distinctly located in specific directions. The yellow category, representing consumption from 60,000,000.001 to 100,000,000,000 kWh, appears concentrated in the southeastern and eastern areas of the map, indicating a focus of higher energy use in these directions. The orange category, covering consumption from 100,000,000.001 to 200,000,000,000 kWh, is similarly clustered in the southeastern and eastern regions, showing an increase in energy demand in these specific directions. The highest consumption range, depicted in red and spanning from 200,000,000.001 to 387,000,000,000 kWh, is the least common but is distinctly present in isolated areas, primarily in the southeastern direction, highlighting pockets of extremely high energy usage in this part of Lalbagh. Overall, the distribution of electricity consumption in Lalbagh shows a gradient pattern where lower consumption ranges (light green and dark green) dominate the northern, western, and central areas, while higher consumption ranges (yellow, orange, and red) are confined to the southeastern and eastern directions. This pattern reflects a varied but structured distribution of energy use across Lalbagh, with the majority of the region exhibiting lower to moderate consumption and a smaller number of areas in the southeast and east accounting for significantly higher energy demands.

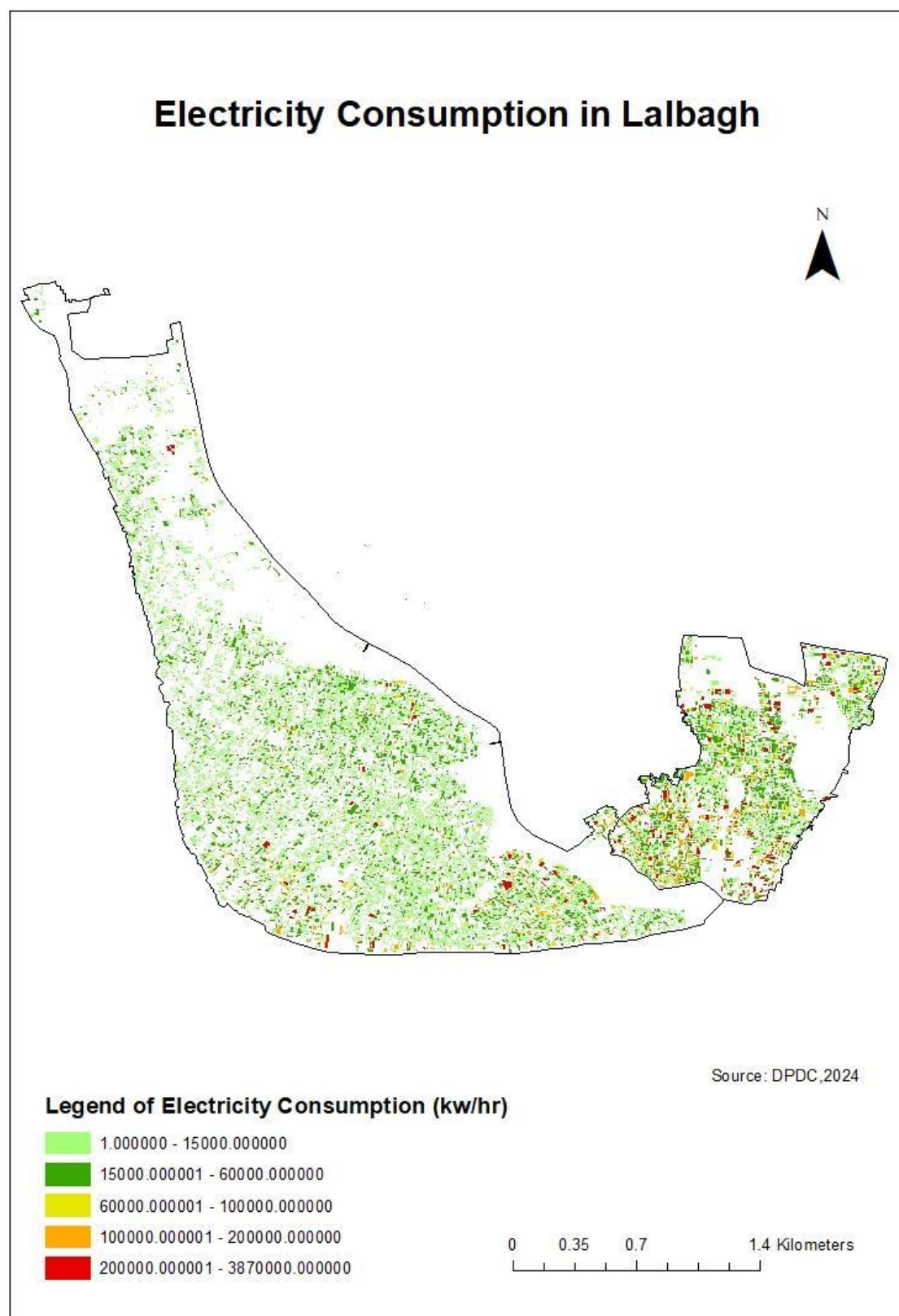


Figure 4.6: Electricity Consumption in Lalbagh

Source: DPDC, 2023

Residential Electricity Consumption:

The box plot shown in Figure 4.7 visualizes the distribution of total electricity consumption (measured in kilowatt-hours, kWh) for the 14,239 residential buildings in Lalbagh. The box plot reveals that the majority of Lalbagh's residential buildings have electricity consumption values clustered between 2,616.5 kWh and 9,475 kWh, with a median consumption of 4,851 kWh. The plot includes several individual dots above the upper whisker, representing outliers. However, the distribution is extremely right-skewed as outliers are scattered at approximately 200,000 kWh, 300,000 kWh, 400,000 kWh, 500,000 kWh, 600,000 kWh, and 700,000 kWh, indicating extremely high electricity consumption values among a small number of residential buildings.

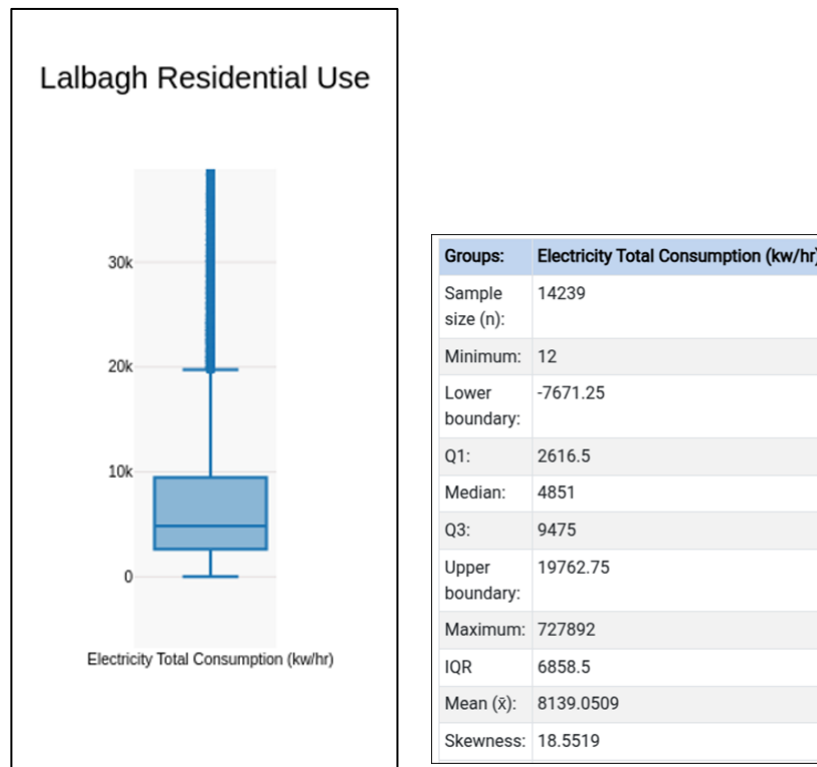


Figure 4.7: Boxplot of Residential Total Electricity Consumption in Lalbagh
Source: DPDC

Commercial Electricity Consumption:

The box plot shown in Figure 4.8 visualizes the distribution of total electricity consumption (measured in kilowatt-hours, kWh) for the 1,595 commercial buildings in Lalbagh. The box plot reveals that the majority of Lalbagh's commercial buildings have electricity consumption values clustered between 1,469 kWh and 7,900.5 kWh, with a median consumption of 3,461 kWh. The plot includes several individual dots above the upper whisker, representing outliers. However, the distribution is extremely right-skewed as outliers are scattered at approximately 100,000 kWh, 150,000 kWh, 200,000 kWh, and 250,000 kWh, indicating extremely high electricity consumption values among a small number of commercial buildings.

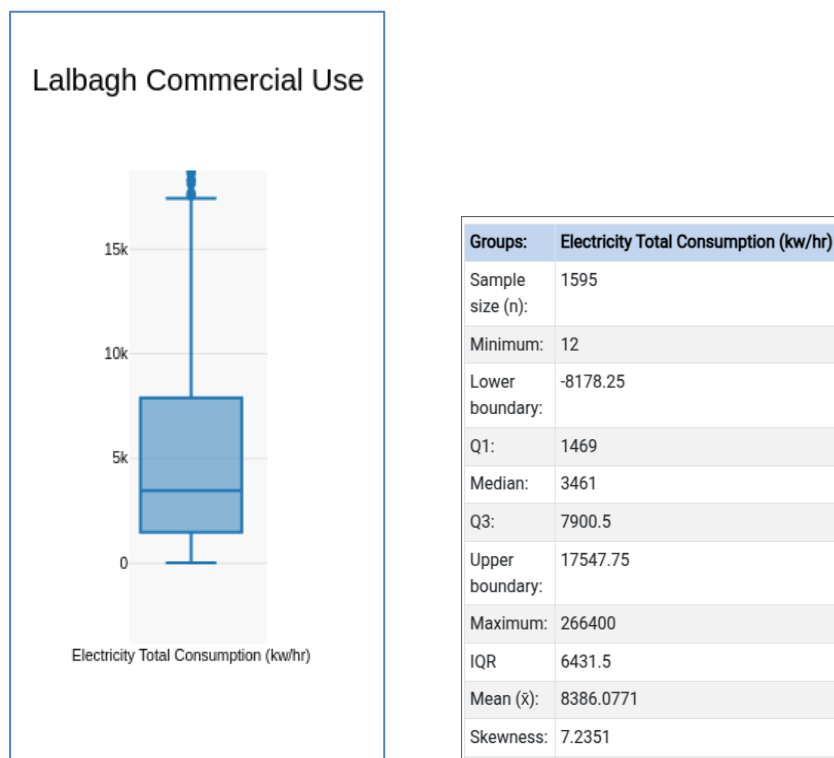


Figure 4.8: Boxplot of Commercial Total Electricity Consumption in Lalbagh
Source: DPDC

Industrial Electricity Consumption:

The box plot shown in Figure 4.9 visualizes the distribution of total electricity consumption (measured in kilowatt-hours, kWh) for the 928 industrial buildings in Lalbagh. The box plot reveals that the majority of Lalbagh's industrial buildings have electricity consumption values clustered between 13871 kWh and 105922.5 kWh, with a median consumption of 40,810 kWh. The plot includes several individual dots above the upper whisker, representing outliers. However, the distribution is extremely right-skewed as outliers are scattered at approximately 600,000 kWh, 900,000 kWh, and 1.2 million kWh, indicating extremely high electricity consumption values among a small number of industrial buildings.

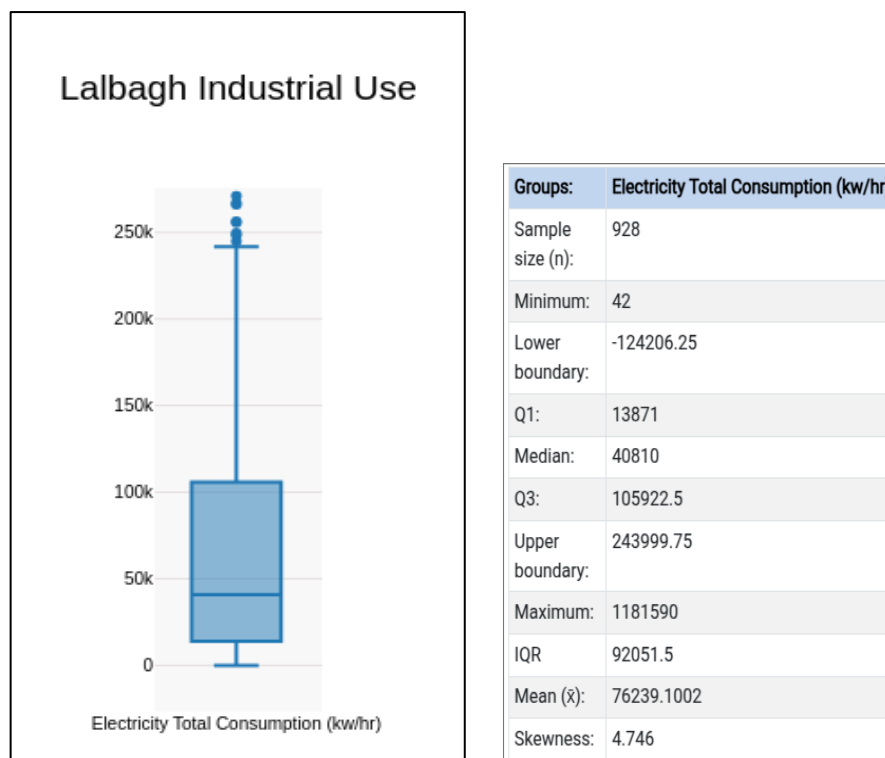


Figure 4.9: Boxplot of Industrial Total Electricity Consumption in Lalbagh
Source: DPDC

Institutional Electricity Consumption:

The box plot shown in Figure 4.10 visualizes the distribution of total electricity consumption (measured in kilowatt-hours, kWh) for 135 institutional buildings in Lalbagh. The box plot reveals that the majority of Lalbagh's institutional buildings have electricity consumption values clustered between 5663.75 kWh and 17929.25 kWh, with a median consumption of 9337 kWh. The plot includes several individual dots above the upper whisker, representing outliers. However, the distribution is extremely right-skewed as outliers are scattered at approximately 50,000 kWh, 100,000 kWh, and 150,000 kWh, indicating extremely high electricity consumption values among a small number of institutional buildings.

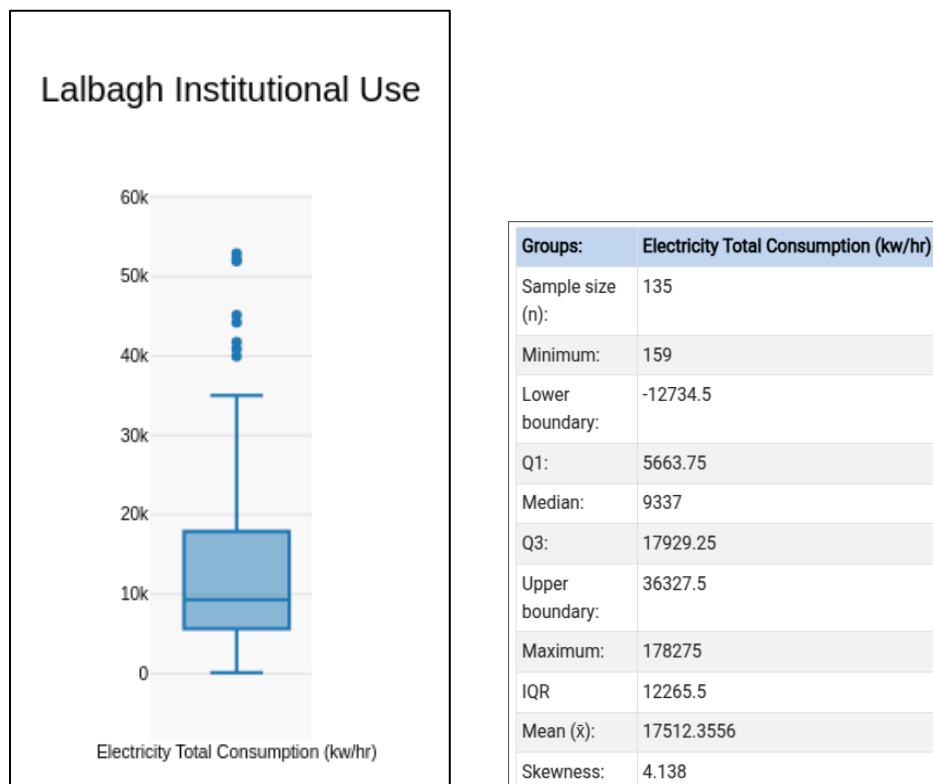


Figure 4.10: Boxplot of Institutional Total Electricity Consumption in Lalbagh
Source: DPDC

Mixed Electricity Consumption:

The box plot shown in Figure 4.11 visualizes the distribution of total electricity consumption (measured in kilowatt-hours, kWh) for 5,581 mixed-use buildings in Lalbagh. The box plot reveals that the majority of Lalbagh's mixed-use buildings have electricity consumption values clustered between 7,898.5 kWh and 359,66.25 kWh, with a median consumption of 16259 kWh. The plot includes several individual dots above the upper whisker, representing outliers. However, the distribution is extremely right-skewed as outliers are scattered at approximately 1 million kWh, 2 million kWh, and 3 million kWh, indicating extremely high electricity consumption values among a small number of mixed-use buildings.

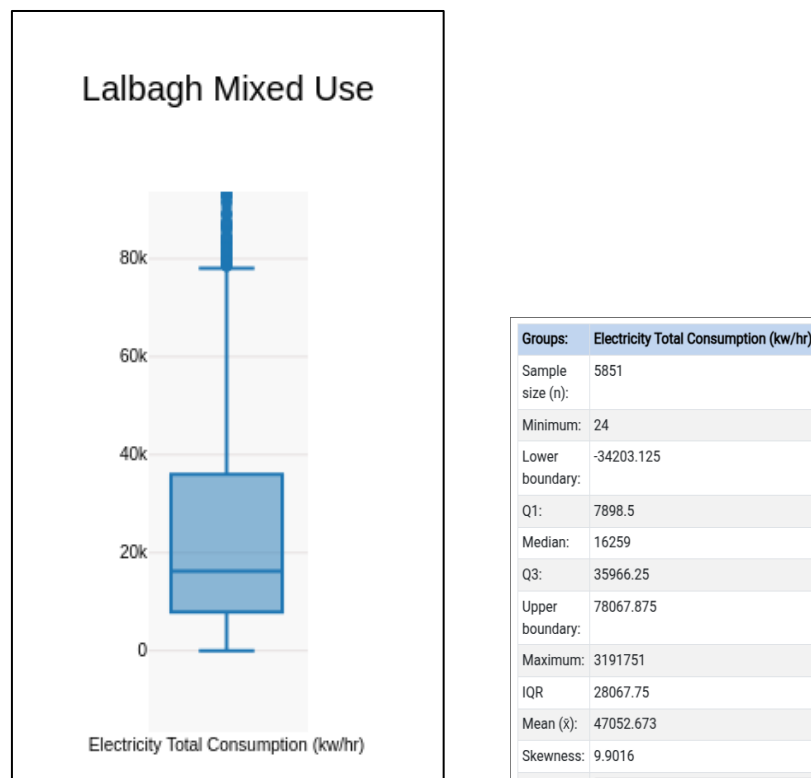


Figure 4.11: Boxplot of Mixed Total Electricity Consumption in Lalbagh
Source: DPDC

Dhanmondi

Figure 4.12, depicting electricity consumption in Dhanmondi, illustrates the spatial distribution of energy usage across the region, measured in kilowatt-hours (kWh). The data is categorized into five distinct consumption ranges, each represented by a specific color. The lowest consumption range, indicated by light green, spans from 1,000,000 to 15,000,000 kWh, and is sparsely distributed across the southwestern and southern areas of the area, indicating that these regions have minimal energy usage. The next range, represented by darker green, covers consumption from 15,000,000.001 to 60,000,000,000 kWh and is present in the southwestern and southern areas, as well as scattered patches in the central region, suggesting a moderate prevalence of energy use in these directions. Higher consumption levels are more concentrated but distinctly located in specific directions. The yellow category, representing consumption from 60,000,000.001 to 100,000,000,000 kWh, appears primarily in the central and northeastern areas of the map, indicating a focus on higher energy use in these directions. The orange category, covering consumption from 100,000,000.001 to 200,000,000,000 kWh, is clustered in the central, northeastern, and eastern regions, showing an increase in energy demand in these specific directions. The highest consumption range, depicted in red and spanning from 200,000,000.001 to 387,000,000,000 kWh, is the least common but is distinctly present in isolated areas, primarily in the central and northeastern directions, highlighting pockets of extremely high energy usage in these parts of Dhanmondi. Overall, the distribution of electricity consumption in Dhanmondi shows a gradient pattern where lower consumption ranges (light green and dark green) dominate the southwestern and southern areas, while higher consumption ranges (yellow, orange, and red) are confined to the central, northeastern, and eastern directions. This pattern reflects a varied but structured distribution of energy use across Dhanmondi, with the majority of the region in the southwest and south exhibiting lower to moderate consumption and a smaller number of areas in the central, northeast, and east accounting for significantly higher energy demands.

Dhanmondi is characterized by a high concentration of orange and red squares, particularly in its northern and central regions, indicating very high electricity consumption. In contrast, Lalbagh predominantly features light green and green squares, suggesting lower to moderate usage. Dhanmondi's dense clustering of high-consumption squares contrasts sharply with Lalbagh's more uniform low to moderate usage, hinting at differing urban densities or activity levels between two areas.

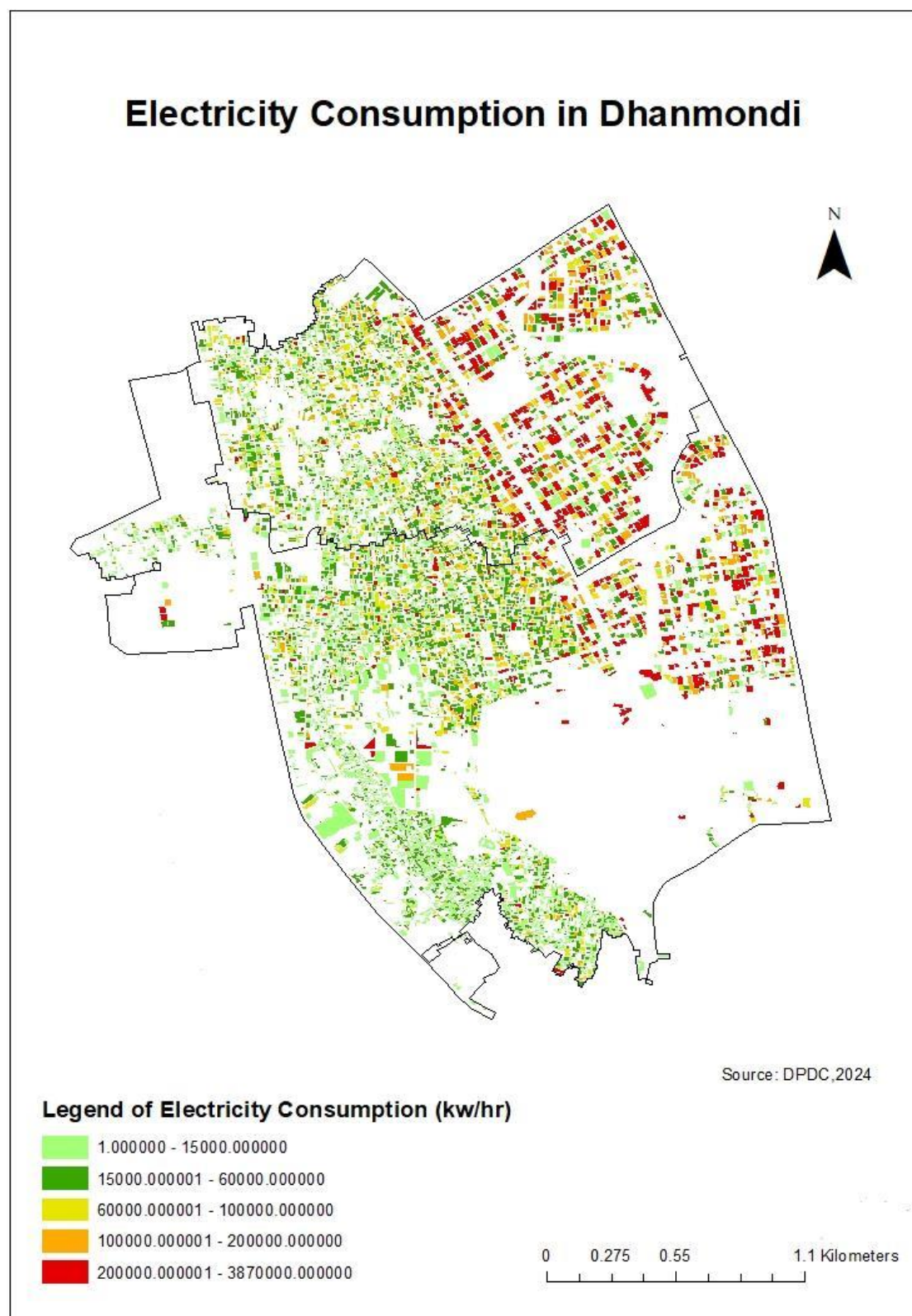


Figure 4.12: Electricity Consumption in Dhanmondi

Source: DPDC, 2023

Residential Electricity Consumption:

The box plot in Figure 4.13 illustrates the electricity consumption for 7,674 residential buildings in Dhanmondi, ranging from 0 to 4 million kWh. The majority of buildings have consumption values between 6,322.82 kWh and 64,212 kWh, with a median of 26,024.755 kWh. Outliers are scattered at approximately 1 million kWh, 1.5 million kWh, 2 million kWh, 2.5 million kWh, 3 million kWh, 3.5 million kWh, and 4 million kWh. The mean consumption of 692,91.0219 kWh is significantly higher than the median, indicating an extremely right-skewed distribution with a skewness value of 10.2559.

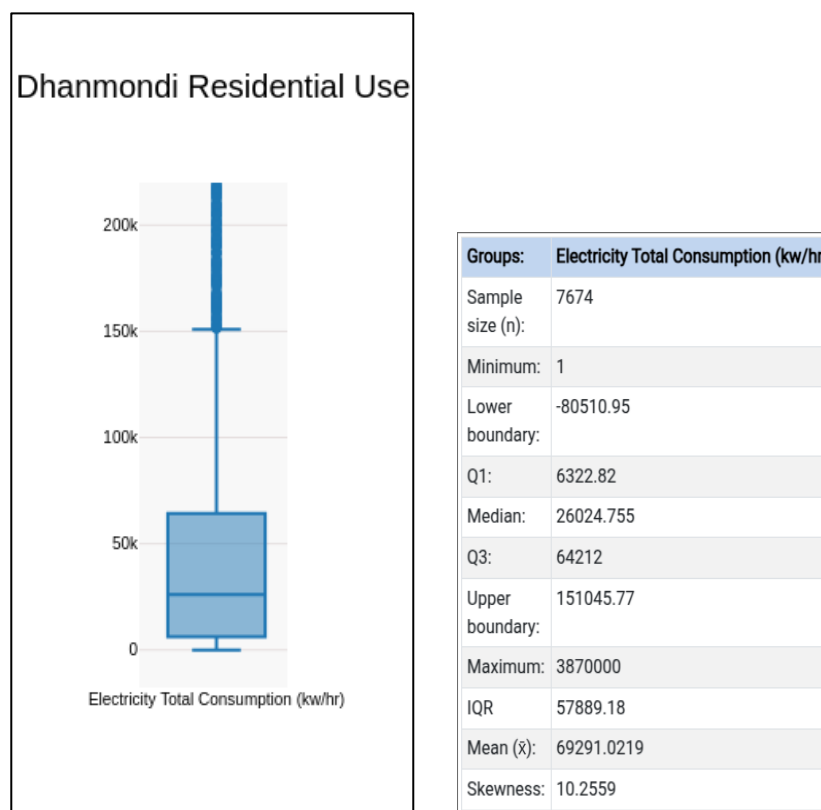


Figure 4.13: Boxplot of Residential Total Electricity Consumption in Dhanmondi
Source: DPDC

Commercial Electricity Consumption:

The box plot Figure 4.14 visualizes the distribution of electricity consumption for 2,110 commercial buildings in Dhanmondi, ranging from 0 to 3 million kWh. The majority of buildings have consumption values between 5,986 kWh and 574,75.24 kWh, with a median of 22484.285 kWh. The presence of outliers at approximately 1 million kWh, 1.5 million kWh, 2 million kWh, 2.5 million kWh, and 3 million kWh indicate extremely high consumption among a few buildings. The mean consumption is unusually high at 647,015,203 kWh, likely due to a data error, while the skewness value of 9.8079 confirms a strongly right-skewed distribution.

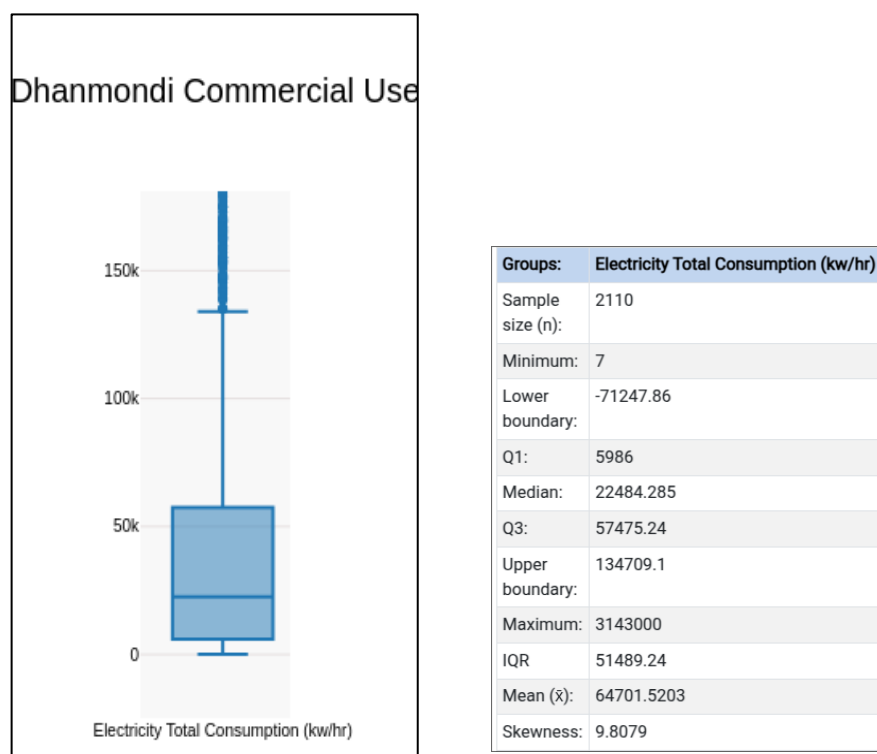


Figure 4.14: Boxplot of Commercial Total Electricity Consumption in Dhanmondi
Source: DPDC

Industrial Electricity Consumption:

The box plot Figure 4.15 illustrates the electricity consumption for 40 industrial buildings in Dhanmondi, spanning from 0 to 500,000 kWh. Most buildings consume between 4513 kWh and 61390.115 kWh, with a median of 22355.2 kWh. Outliers are present at around 200,000 kWh, 300,000 kWh, 400,000 kWh, and 500,000 kWh, indicating high consumption by a small number of buildings. The mean consumption of 58340.0397 kWh is higher than the median, reflecting a right-skewed distribution with a skewness value of 3.0554.

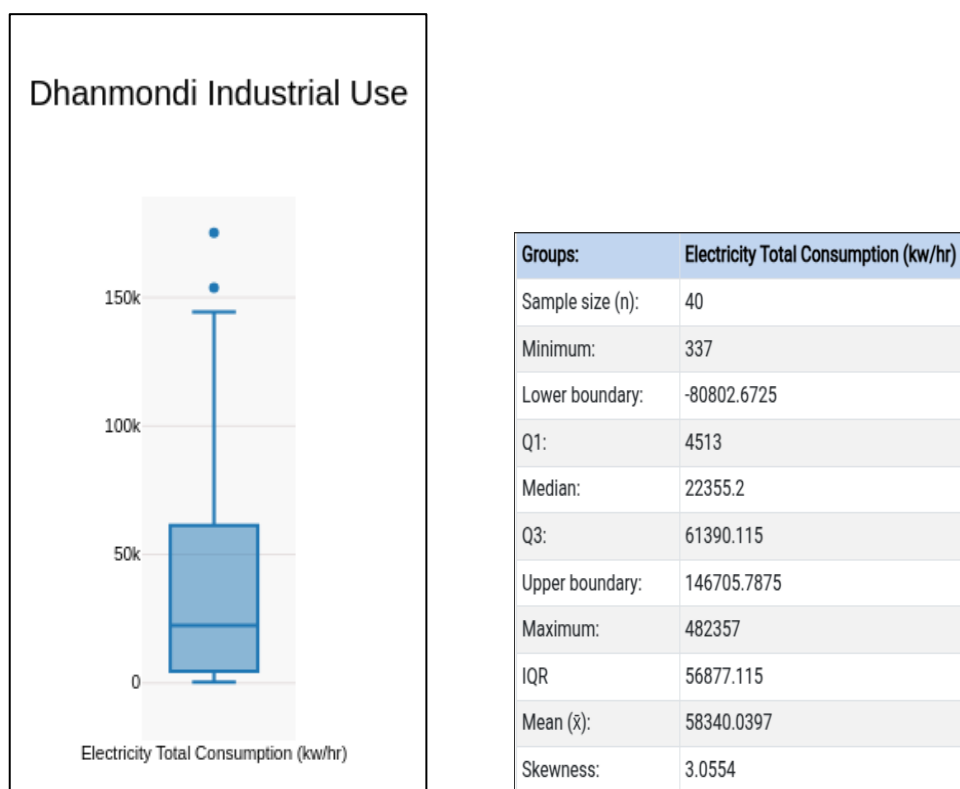


Figure 4.15: Boxplot of Industrial Total Electricity Consumption in Dhanmondi
Source: DPDC, 2024

Institutional Electricity Consumption:

The box plot Figure 4.16 shows the electricity consumption for 131 institutional buildings in Dhanmondi, ranging from 0 to 500,000 kWh. The majority of buildings have consumption values between 6126.925 kWh and 60783.25 kWh, with a median of 26659 kWh. Outliers are scattered at approximately 200,000 kWh, 300,000 kWh, 400,000 kWh, and 500,000 kWh. The mean consumption of 55144.1043 kWh is slightly higher than the median, indicating a right-skewed distribution with a skewness value of 3.216.

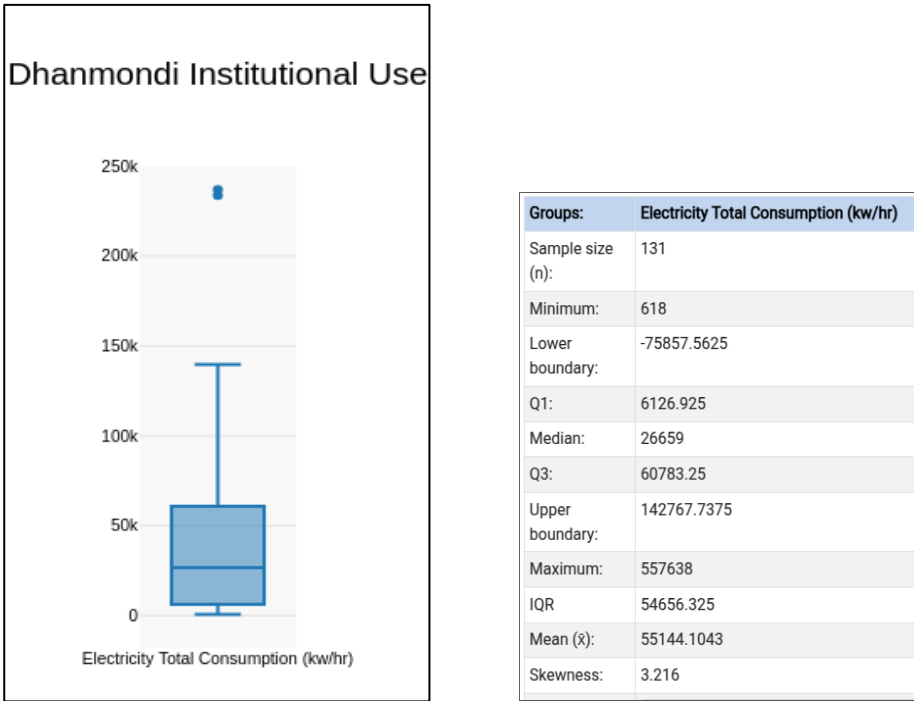


Figure 4.16: Boxplot of Institutional Total Electricity Consumption in Dhanmondi
Source: DPDC

Mixed Electricity Consumption:

The box plot Figure 4.17 visualizes the electricity consumption for 69 mixed-use buildings in Dhanmondi, ranging from 0 to 700,000 kWh. Most buildings consume between 9,821.5 kWh and 46,461.4425 kWh, with a median of 21,824 kWh. Outliers are present at around 200,000 kWh, 300,000 kWh, 400,000 kWh, 500,000 kWh, 600,000 kWh, and 700,000 kWh. The mean consumption of 49,041.1462 kWh is higher than the median, reflecting a right-skewed distribution with a skewness value of 5.2442.

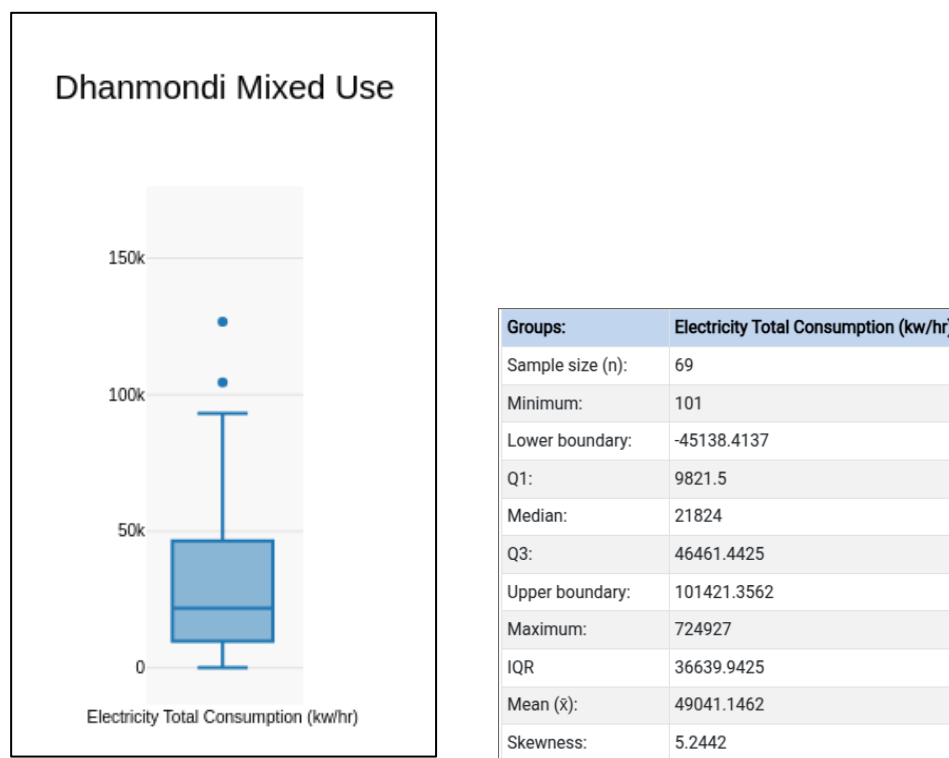


Figure 4.17: Boxplot of Mixed Total Electricity Consumption in Dhanmondi
Source: DPDC

Comparison of Electricity Consumption Per Area

Two sample T test is done to find out whether electricity consumption per area for five different purposes differs between Lalbagh and Dhanmondi.

The box plot for Dhanmondi shown in Figure 4.18 (a) represents electricity consumption per area of 7,674 residential buildings in Dhanmondi, with Q1 at 31.3818, median at 81.6259, and Q3 at 224.3326. The IQR is 192.9508, and the mean is 412.8211, indicating a highly right-skewed distribution (skewness 24.3248). Most buildings consume between 31.38 and 224.33 kWh/hr per m², but high outliers suggest some buildings have significantly higher usage, possibly due to large or atypical residences.

The box plot for Lalbagh in Figure 4.18 (b) shows electricity consumption (kWh/hr per m²) for 14,239 residential buildings in Lalbagh, with Q1 at 30.8668, median at 57.6001, and Q3 at 117.6665. The IQR is 86.7998, and the mean is 104.2529, indicating a right-skewed distribution (skewness 34.3471). Most buildings consume between 30.87 and 117.67 kWh/hr per m², but high outliers suggest some buildings have significantly higher usage, likely due to large or atypical residences.

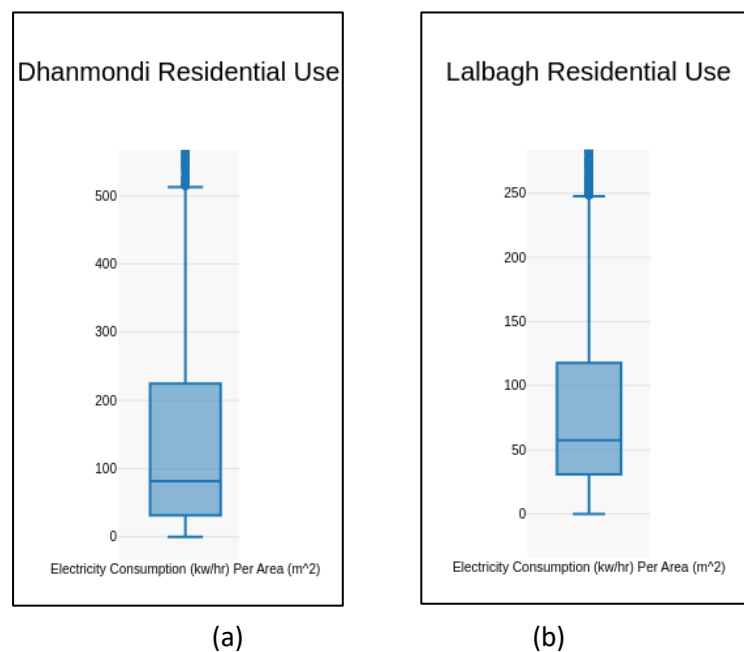


Figure 4.18: Boxplot of Residential Use Electricity Consumption Per Area in Dhanmondi and Lalbagh
Source: DPDC

This study found that electricity consumption per m² for residential use in Dhanmondi is statistically significantly higher compared to Lalbagh.

Location	Mean	t value	df value	p value
Dhanmondi	451.4566	12.854	7758.6	< 2.2e-16
Lalbagh	229.0671			≅ 0 <0.01

The box plot for Dhanmondi shown in Figure 4.19 (a) shows electricity consumption (kWh/hr per m²) for 2,110 commercial buildings in Dhanmondi, with Q1 at 32.1217, median at 80.6778, and Q3 at 229.7155. The IQR is 197.5938, and the mean is 370.604, indicating a right-skewed distribution (skewness 13.2782). Most buildings consume between 32.12 and 229.72 kWh/hr per m², but high outliers suggest some buildings have significantly higher usage, possibly due to large or energy-intensive facilities.

The box plot for Lalbagh shown in Figure 4.19 (b) shows electricity consumption (kWh/hr per m²) for 1,595 commercial buildings in Lalbagh, with Q1 at 20.3366, median at 54.13, and Q3 at 142.3243. The IQR is 121.9876, and the mean is 189.595, indicating a right-skewed distribution (skewness 10.5813). Most buildings consume between 20.34 and 142.32 kWh/hr per m², but high outliers suggest some buildings have significantly higher usage, likely due to large or energy-intensive commercial spaces.

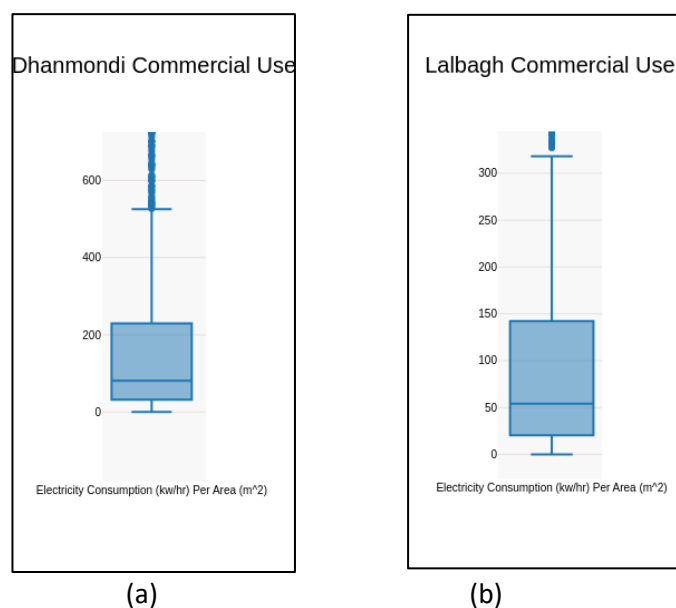


Figure 4.19: Boxplot of Commercial Use Electricity Consumption Per Area in Dhanmondi and Lalbagh
Source: DPDC

This study found that electricity consumption per m² for commercial use in Dhanmondi is statistically significantly higher compared to Lalbagh.

Location	Mean	t value	df value	p value
Dhanmondi	464.0599	40.224	3515.1	2.2e-16
Lalbagh	290.6921			$\cong 0 < 0.01$

The box plot for Dhanmondi shown in Figure 4.20 (a) shows electricity consumption (kWh/hr per m²) for 40 industrial buildings in Dhanmondi, with Q1 at 37.8887, median at 86.2076, and Q3 at 271.7265. The IQR is 233.8379, and the mean is 247.7934, indicating a right-skewed distribution (skewness 2.7478). Most buildings consume between 37.89 and 271.73 kWh/hr per m², but high outliers suggest some buildings have significantly higher usage, likely due to energy-intensive industrial operations.

The box plot for Lalbagh shown in Figure 4.20 (b) shows electricity consumption (kWh/hr per m²) for 928 industrial buildings in Lalbagh, with Q1 at 128.2623, median at 461.158, and Q3 at 1,277.7681. The IQR is 1,149.5058, and the mean is 1,143.7803, indicating a right-skewed distribution (skewness 5.183). Most buildings consume between 128.26 and 1,277.77 kWh/hr per m², but high outliers suggest some buildings have significantly higher usage, likely due to energy-intensive industrial processes.

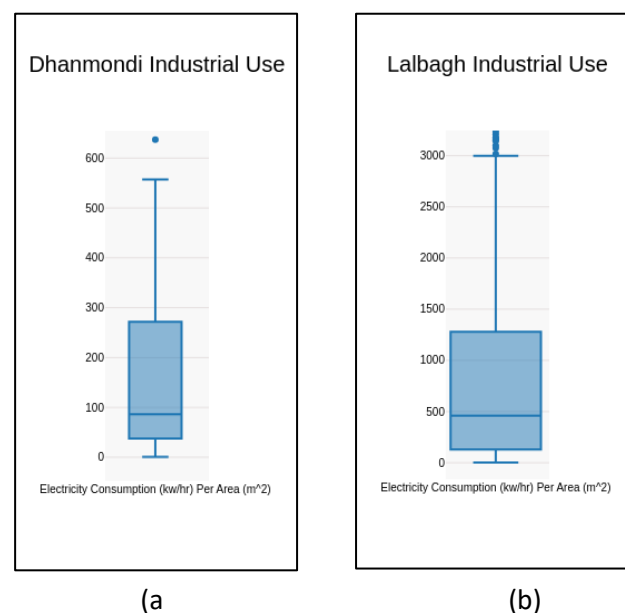


Figure 4.20: Boxplot of Industrial Use I Electricity Consumption Per Area in Dhanmondi and Lalbagh
Source: DPDC

This study found that electricity consumption per m² for industrial use in Dhanmondi is statistically significantly lower compared to Lalbagh

Location	Mean	t value	df value	p value
Dhanmondi	549.405	16.78	966	< 2.2e-16
Lalbagh	2741.895			$\cong 0 < 0.01$

The box plot for Dhanmondi shown in Figure 4.21 (a) shows electricity consumption (kWh/hr per m²) for 131 institutional buildings in Dhanmondi, with Q1 at 30.3022, median at 88.8888, and Q3 at 320.7803. The IQR is 290.478, and the mean is 532.4497, indicating a right-skewed distribution (skewness 7.2527). Most buildings consume between 30.30 and 320.78 kWh/hr per m², but high outliers suggest some buildings have significantly higher usage, possibly due to large or energy-intensive institutions.

The box plot for Lalbagh shown in Figure 4.21 (b) shows electricity consumption (kWh/hr per m²) for 135 institutional buildings in Lalbagh, with Q1 at 16.5986, median at 30.2243, and Q3 at 62.7628. The IQR is 46.1641, and the mean is 145.9553, indicating a right-skewed distribution (skewness 7.4723). Most buildings consume between 16.60 and 62.76 kWh/hr per m², but high outliers suggest some buildings have significantly higher usage, likely due to large or energy-intensive institutions.

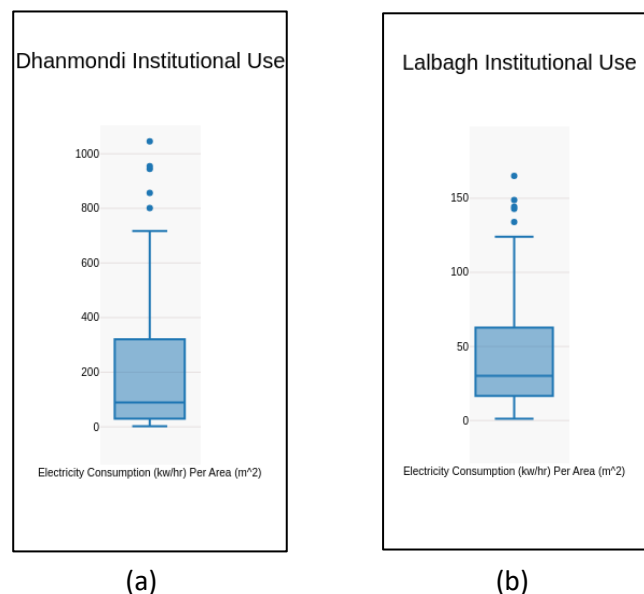


Figure 4.21: Boxplot of Institutional Use Electricity Consumption Per Area in Dhanmondi and Lalbagh
Source: DPDC

This study found that electricity consumption per m² for institutional use in Dhanmondi is statistically significantly higher compared to Lalbagh

Location	Mean	t value	df value	p value
Dhanmondi	678.8819	28.113	136.97	< 2.2e-16
Lalbagh	118.3297			≅ 0 <0.01

The box plot for Dhanmondi shown in Figure 4.22 (a) shows electricity consumption (kWh/hr per m²) for 69 mixed-use buildings in Dhanmondi, with Q1 at 35.817, median at 71.4439, and Q3 at 170.1888. The IQR is 134.3718, and the mean is 523.4256, indicating a right-skewed distribution (skewness 7.8044). Most buildings consume between 35.82 and 170.19 kWh/hr per m², but high outliers suggest some buildings have significantly higher usage, likely due to mixed-use facilities with high energy demands.

The box plot for Lalbagh shown in Figure 4.22(b) shows electricity consumption (kWh/hr per m²) for 5,581 mixed-use buildings in Lalbagh, with Q1 at 44.535, median at 100.3418, and Q3 at 286.3269. The IQR is 241.792, and the mean is 1,731.8714, indicating a highly right-skewed distribution (skewness 76.4742). Most buildings consume between 44.54 and 286.33 kWh/hr per m², but high outliers suggest some buildings have significantly higher usage, likely due to large or energy-intensive mixed-use facilities.

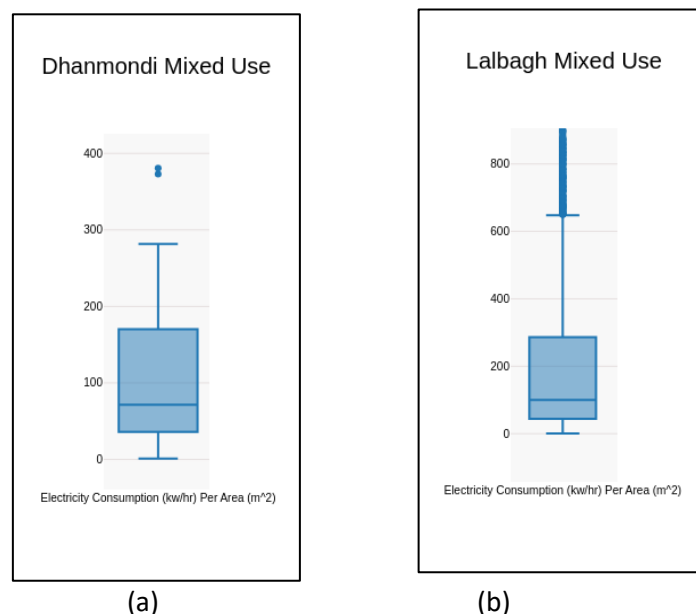


Figure 4.22: Boxplot of Mixed-Use Electricity Consumption Per Area in Dhanmondi and Lalbagh
Source: DPDC

This study found that electricity consumption per m² for mixed use in Dhanmondi is statistically significantly lower compared to Lalbagh

Location	Mean	t value	df value	p value
Dhanmondi	299.9382	23.435	221.58	< 2.2e-16
Lalbagh	635.582			$\cong 0$ <0.01

CHAPTER 5 : SPATIAL ANALYSIS OF ELECTRICITY CONSUMPTION TREND

This chapter provides a comprehensive overview of per-consumer electricity consumption trends across four distinct usage categories—Residential, Commercial, Industrial, and Institutional in the Lalbagh and Dhanmondi regions from 2017 to 2023. In addition, it tried to find out the trends in total electricity consumption across four distinct uses in Lalbagh and Dhanmondi for different uses from 2017 to 2023

5.1 Trend Analysis of Electricity Consumption Based on Consumption Per Consumer in Lalbagh

As mentioned in chapter four, five types of land use are prominent in Lalbagh. The per-consumer electricity consumption pattern shows considerable differences among them. Figure 5.1 illustrates the consumption per consumer trends in Lalbagh from 2017 to 2023.

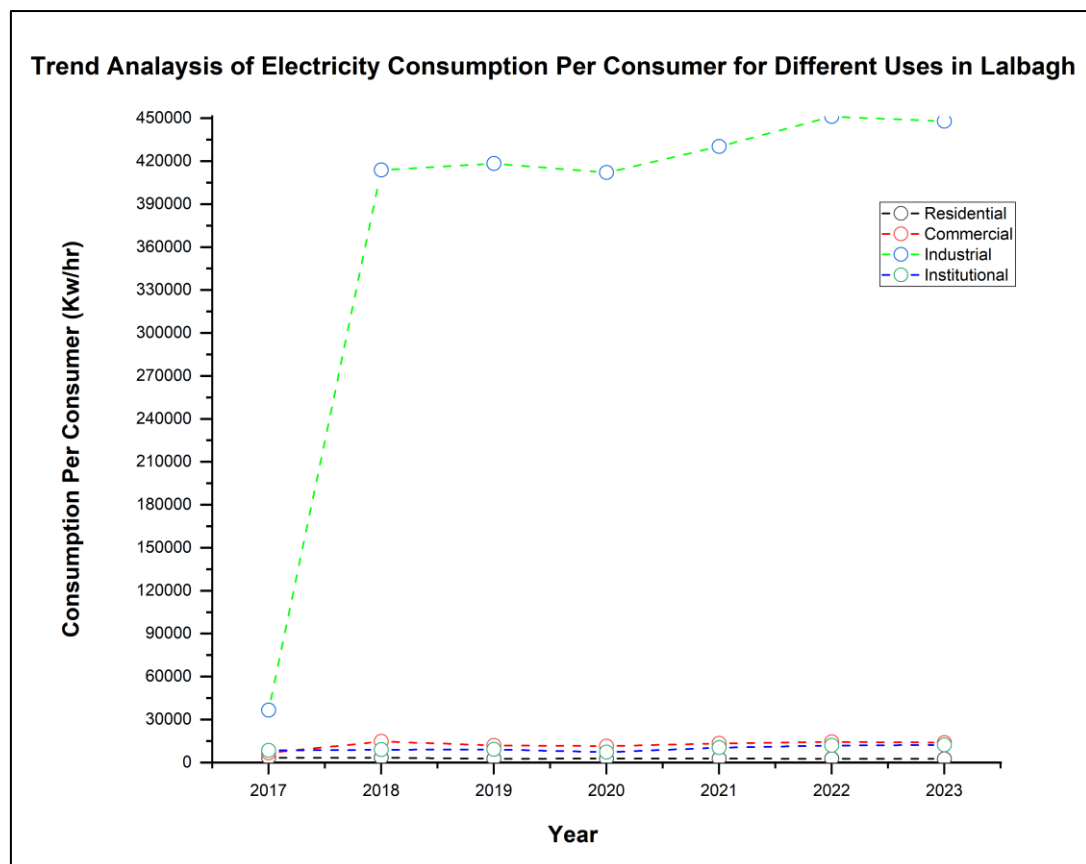


Figure 5.1: Trend Analysis of Electricity Consumption Per Consumer for Different Uses in Lalbagh
Source: DPDC, 2024

The per-consumer electricity consumption for residential use in Lalbagh exhibited a consistent downward trend over the seven-year period shown in Figure 5.1. Starting at 3,324.16kwh in 2017, consumption slightly decreased to 3,313.04kwh in 2018 before experiencing a more pronounced decline to 2,605.84kwh in 2019. Despite marginal increases in 2020 (2,678.58kwh) and 2021 (2,696.22kwh), the overall trajectory continued downward, reaching a low of 2,503.61kwh by 2023. This steady decline suggests a gradual reduction in residential electricity demand per consumer, demographic shifts, or changes in household consumption patterns. The COVID outbreak caused lockdowns in 2020 and 2021. So, people stayed at home and used more electricity. Besides, low-income people live in this region. Due to migration, especially low-income people come to Dhaka for work, and low-quality housing is increasing in the area. However, their consumption level is stable as before. So, the customer number is increasing, and but the consumption per Consumer is decreasing.

On the other hand, commercial electricity consumption per consumer in Lalbagh shows variation over the years. From 6,609.94kwh in 2017, consumption rose to 14,653.70kwh in 2018 and peaked at 19,449.48kwh in 2019. However, this was followed by a sharp decline to 14,174.38kwh in 2020 and further reductions in subsequent years because of COVID, with a brief fluctuation in 2022 (14,366.68kwh) before settling at 13,904.22kwh in 2023.

But the industrial per consumer consumption in Lalbagh experienced a rise from 36548.93kwh in 2017 to 413710.8kwh in 2018, with further increases to 418261.2kwh in 2019. A slight decrease occurred in 2020 (412038kwh). The electricity consumption per consumer then gradually increases in 2021 and 2022 (451116.7kwh) before a slight decrease to 447818.7kwh in 2023. The decrease between 2020 and 2021 might have occurred due to the COVID lockdown. Industrial customer numbers increased rapidly in 2018 and 2019.

Institutional electricity consumption per consumer in Lalbagh showed an overall upward trend despite some variability. Beginning at 8,365.04kwh in 2017, consumption increased to 9,080.74kwh in 2019 before dropping sharply to 7,221.34kwh in 2020. This was followed by a strong recovery, with consumption rising steadily to 12,185.61kwh by 2023. The 2020 decline may be attributed to temporary reductions in institutional activities, possibly due to the COVID lockdown, while the subsequent increase suggests a return to or expansion of institutional operations.

5.2 Trend Analysis of Electricity Consumption Based on Total Consumption in Lalbagh:

Figure 5.2 illustrates the total consumption trends in Lalbagh from 2017 to 2023

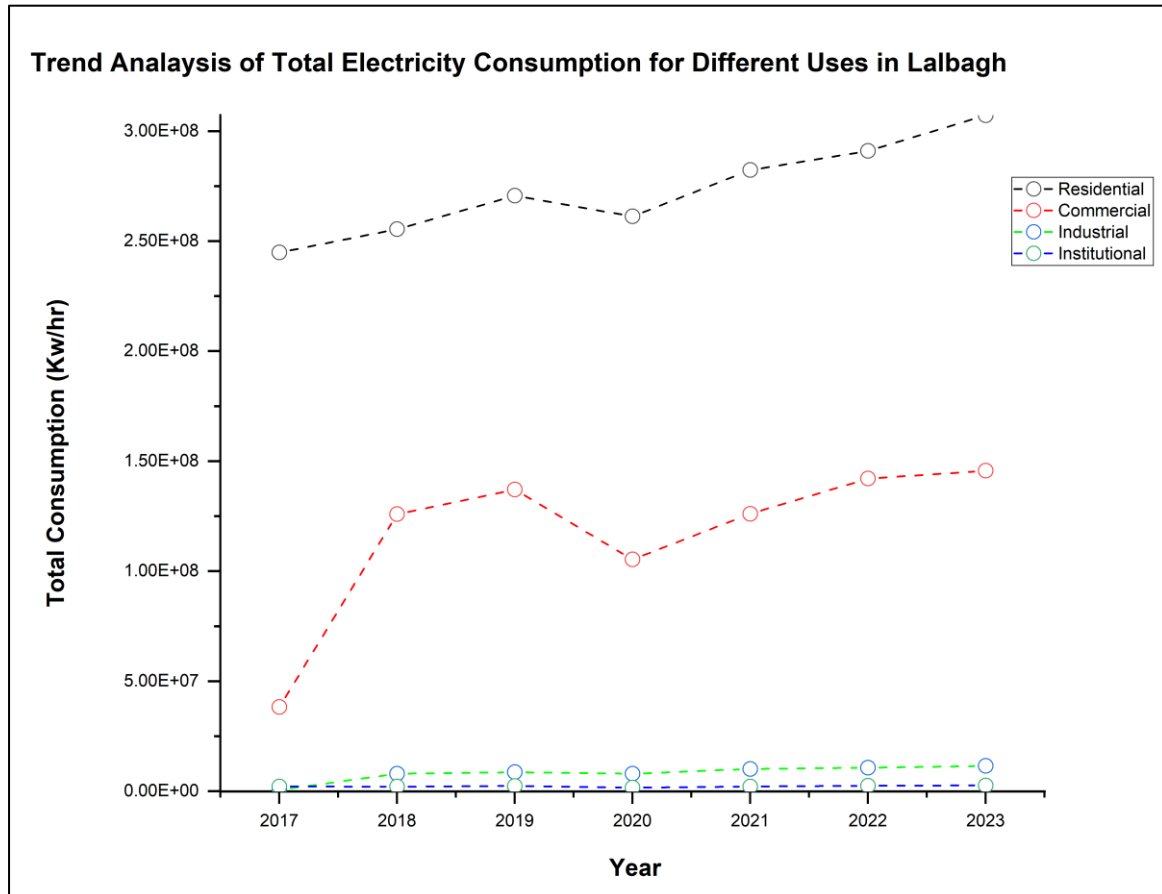


Figure 5.2: Trend Analysis of Total Electricity Consumption for Different Uses in Lalbagh
Source: DPDC, 2024

The residential total electricity consumption in Lalbagh, from Figure 5.2 showed an overall increasing trend from 2017 to 2023. The residential total electricity consumption in 2017 was 183,082,534 kW/hr. By 2018, it increased to 195,464,174 kW/hr, reflecting a rise in usage. In 2020, it decreased to 194,063,668 kW/hr, showing a slight decline from the previous year. By 2023, it peaked at 244,532,071 kW/hr, indicating a significant overall increase over the seven-year period.

But the commercial total electricity consumption, followed a pattern of growth with a decline in 2020. In 2017, the commercial total electricity consumption stood at 41,381,302 kW/hr. It rose to 47,566,181 kW/hr in 2018, marking an upward trend. By 2020, it dropped to 43,820,471

kW/hr, reflecting a reduction from the 2018 peak. In 2023, it reached its highest level at 63,613,951 kW/hr, showing a strong increase over the years.

On the other hand, the institutional total consumption of electricity demonstrates a consistent increase from 2017 to 2023. The institutional total electricity consumption in 2017 was 2,829,867 kW/hr. It grew slightly to 2,950,735 kW/hr in 2018, indicating a modest rise. By 2020, it fell to 2,928,994 kW/hr, a minor decrease from the previous year. In 2023, it reached 4,426,684 kW/hr, demonstrating a moderate increase over the period.

The industrial total consumption of electricity displays a steady and consistent increase from 2017 to 2023. In 2017, the industrial total electricity consumption was 159,353,335 kW/hr. It increased significantly to 197,340,054 kW/hr in 2018, showing substantial growth. By 2020, it rose further to 210,551,426 kW/hr, continuing the upward trend. In 2023, it reached 246,748,119 kW/hr, marking a considerable increase across the seven years.

5.3 Trend Analysis of Electricity Consumption Based on Consumption Per Consumer in Dhanmondi

Like Lalbagh, the consumption pattern varies according to land use in Dhanmondi. The per-consumer electricity consumption pattern shows considerable differences among them. Figure 5.3 illustrates the consumption per consumer trends in Dhanmondi from 2017 to 2023.

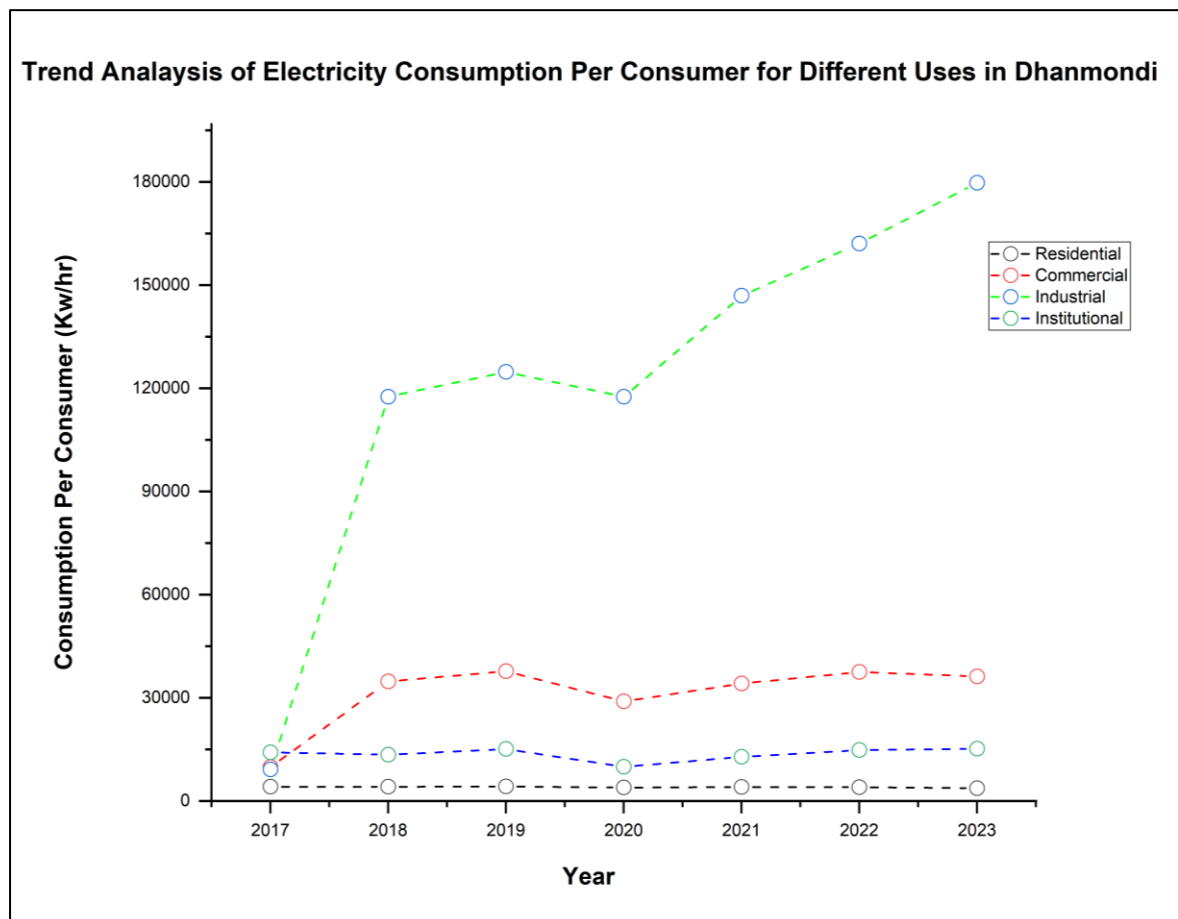


Figure 5.3: Trend Analysis of Electricity Consumption Per Consumer for Different Uses in Dhanmondi
Source: DPDC, 2024

From Figure 5.3, residential per-consumer electricity consumption began at 4,106.40kwh in 2017 and increased to 4,173.85kwh in 2019 in Dhanmondi. However, a decline followed, with consumption falling to 3,888.55kwh in 2020 before fluctuating in subsequent years and finally increasing to 4,026.85kwh in 2023. The post-2019 decline mirrors the trend observed in Lalbagh, though Dhanmondi consistently maintained higher residential consumption levels, suggesting a higher baseline of residential energy use in this region. Relatively higher-income

people live in Dhanmondi, and their demand for electricity appliances is higher than low-income people.

On the other hand, the commercial per-consumer consumption in Dhanmondi exhibited significant growth from 9,816.96kwh in 2017 to a peak of 37,675.73kwh in 2019. This was followed by a decline to 29,925.34kwh in 2020, after which consumption rose again to 37,447.27kwh in 2022 before slightly decreasing to 36,189.92kwh in 2023. The overall trend indicates a substantial increase in commercial electricity use over the period, with the 2020 dip potentially reflecting temporary economic disruptions. The COVID outbreak caused lockdowns in 2020 and 2021. So, some commercial institutions remained shut down during the period. The increased consumption after the lockdown reflected economic recovery.

The industrial consumption per consumer in Dhanmondi began at a high level of 9135.74kwh in 2017 but sharply rose to 117,512.53kwh in 2018. From 2018 onward, consumption fluctuated, with a general upward trend, reaching 179,721.06kwh by 2023. The subsequent increase from 2017 to 2023 suggests a gradual expansion of industrial activities.

Like Lalbagh, the institutional per-consumer consumption in Dhanmondi showed an overall upward trend, starting at 14,123.32kwh in 2017 and peaking at 15,088.61kwh in 2019. A sharp decline to 9,936.40kwh in 2020 was followed by a steady recovery, with consumption reaching 15,155.99kwh by 2023. Similar to Lalbagh, the 2020 drop likely reflects temporary reductions in institutional operations, with a subsequent return to growth.

5.4 Trend Analysis of Electricity Consumption Based on Total Consumption in Dhanmondi:

5.4 illustrates the total consumption trends in Dhanmondi from 2017 to 2023

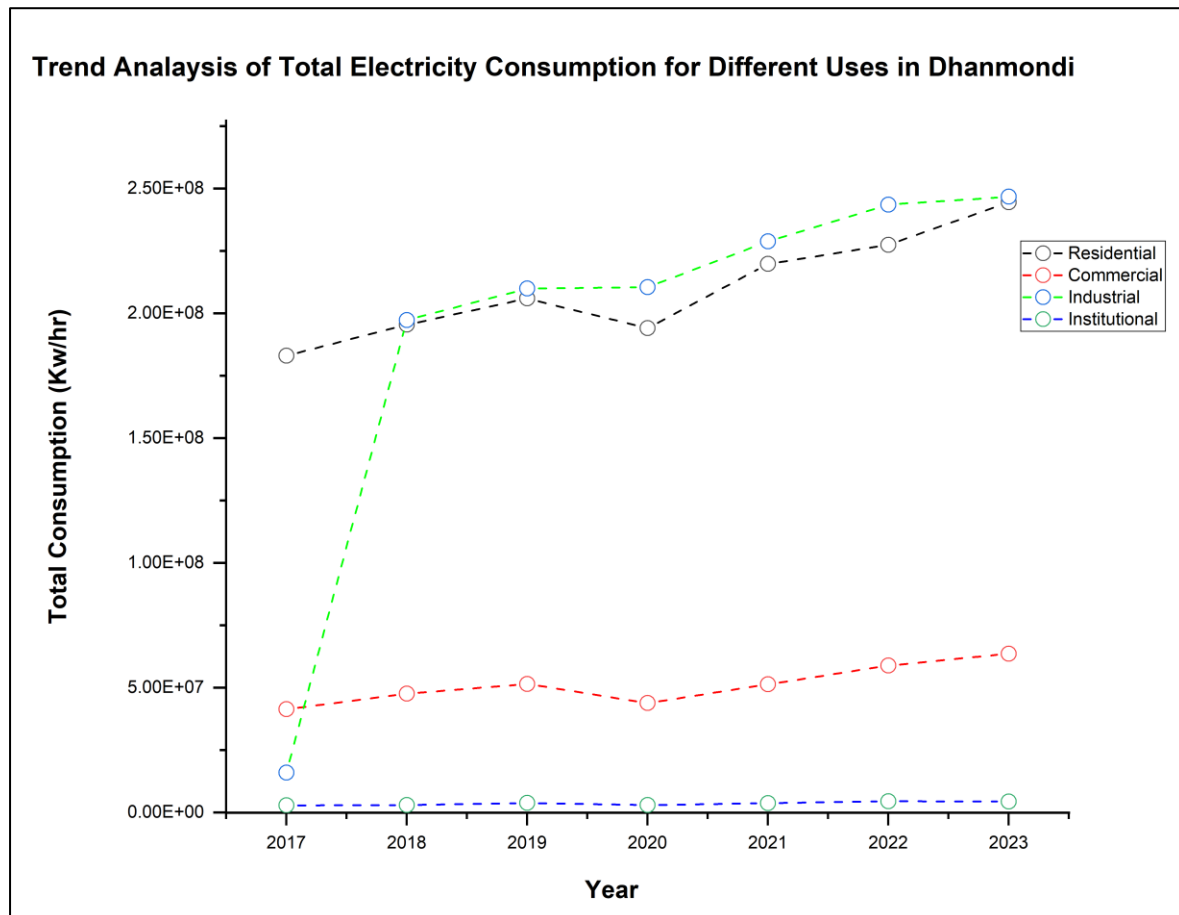


Figure 5.4: Trend Analysis of Total Electricity Consumption for Different Use in Dhanmondi
Source: DPDC, 2024

Figure 5.4 reveals that residential electricity consumption increased from 244,815,538 units in 2017 to 307,169,084 units in 2023, marking a 25.5% growth over the period. A temporary decline occurred in 2020 (261,201,717 units), followed by consistent annual increases thereafter. The sector maintained its position as the largest consumer of electricity across all years.

In line with that, commercial consumption rose from 38,246,867 units in 2017 to 145,628,245 units in 2023, with notable volatility. A sharp increase occurred between 2017 and 2018 (38.2 million to 125.9 million units), followed by fluctuations, including a dip in 2020 (105,288,239

units) and subsequent recovery. The sector exhibited the highest year-on-year variability among all categories.

But the institutional electricity usage, remained relatively stable, starting at 2,160,868 units in 2017 and reaching 2,637,143 units in 2023 shown in Figure 7. A decline was observed in 2020 (1,629,570 units), after which consumption gradually increased. This sector consistently represented the smallest proportion of total electricity demand.

On the other hand, industrial consumption demonstrated exponential growth, rising from 612,095 units in 2017 to 11,502,148 units in 2023—an 18-fold increase. Growth was uninterrupted, with annual increments even during 2020 (7,988,273 units). By 2023, industrial demand surpassed commercial consumption, positioning it as the second-largest sector after residential.

CHAPTER 6 : FACTORS OF ELECTRICITY CONSUMPTION

In this study, a total of eleven variables mentioned in chapter two have been used to determine the annual electricity usage of a building for different purposes. These variables define the geospatial, physical, and social characteristics of the building and its surrounding area. The study includes electricity consumption varies on different uses as dependent variable and the following independent variables:

Normalized Difference Vegetation Index (NDVI): The NDVI map of the study area is shown in Figures B7 and B8 of Appendix B. It is calculated from satellite imagery of Landsat 8 using ArcGIS. In Dhanmondi, NDVI values range from 0.02 to 0.20, with a mean of approximately 0.08, indicating a moderate to high vegetation presence due to its planned residential layout with parks and green spaces. In Lalbagh, NDVI ranges from 0.01 to 0.18, with a mean of about 0.06, suggesting slightly less vegetation cover. Dhanmondi generally shows higher NDVI values than Lalbagh, reflecting more green cover.

Average Modified Normalized Difference Water Index (MNDWI): The MNDWI (shown in Figures B3 and B4 of Appendix B) captures the spatial distribution of water bodies, also derived from remote sensing data (Landsat 8 is used for the study). MNDWI detects water bodies and moisture content. Dhanmondi's MNDWI values range from -0.20 to -0.05, with a mean of around -0.12, indicating a moderate presence of water or moisture. Lalbagh's values range from -0.22 to -0.04, with a mean of approximately -0.13, suggesting a slightly higher water presence, possibly due to its proximity to the Buriganga River. Lalbagh exhibits marginally lower (more negative) MNDWI values compared to Dhanmondi, indicating greater moisture or water body influence.

Distance from Nearby Permeable Areas (m): This variable measures how far buildings are from permeable areas like parks (shown in Figures B9 and B10 of Appendix B). Dhanmondi's distances range from 0 to 500 meters, with a mean of around 150 meters, showing good access to green spaces. Lalbagh's distances range from 0 to 600 meters, with a mean of about 200 meters, indicating buildings are farther from such areas. Dhanmondi has shorter average distances than Lalbagh, reflecting its planned layout.

Floor Number: It refers the total number of floors. Dhanmondi's buildings range from 1 to 15 floors, with a mean of about five floors, showing a mix of low to mid-rise structures typical

of modern residential areas. Lalbagh's buildings range from 1 to 12 floors, with a mean of approximately four floors, reflecting mostly low-rise, older constructions.

Building Density (250m): Building density means the number of buildings within a 250m buffer of each building (shown in Figures B1 and B2 of Appendix B). In Dhanmondi, densities range from 50 to 300 buildings (mean ~150), indicating a moderate concentration due to planned spacing. Lalbagh's densities range from 100 to 400 buildings (mean ~250), showing a higher concentration typical of its dense urban fabric. Lalbagh exhibits greater building density than Dhanmondi, reflecting its crowded historical development.

Building Area (sqm) and Volume (cubic meters): Building Area defines the ground area of each building, and building volume refers multiplication of building area and total building height. These metrics indicate building size and scale. Dhanmondi's building areas range from 10 to 1000 sqm (mean ~200 sqm) and volumes from 50 to 10,000 cubic meters (mean ~1500 cubic meters), suggesting larger, more spacious structures. Lalbagh's areas range from 5 to 800 sqm (mean ~150 sqm) and volumes from 20 to 8000 cubic meters (mean ~1000 cubic meters), indicating smaller buildings. Dhanmondi has larger building areas and volumes than Lalbagh, consistent with its affluent character.

Shape Factor : The perimeter-to-area ratio (shown in Figures B5 and B6 of Appendix B) indicates building shape complexity. Dhanmondi's ratios range from 0.1 to 1.5 (mean ~0.5), suggesting a mix of simple and complex shapes. Lalbagh's ratios range from 0.2 to 2.0 (mean ~0.8), indicating more irregularly shaped buildings due to constrained plots or incremental construction. Lalbagh shows higher average ratios than Dhanmondi, pointing to greater shape irregularity.

Customer No: Customer number represents the number of house units per building. In Dhanmondi, it ranges from 1 to 50, with a mean of about 10, indicating more units per building due to larger structures. In Lalbagh, it ranges from 1 to 30, with a mean of around 8, suggesting fewer units. Dhanmondi has a slightly higher customer number than Lalbagh.

Building age: Building Age is the duration of a building's sanction date to the electricity consumption year (2023) considered for the study. In Dhanmondi, it ranges from 1 to 25 years, averaging around 10 years, indicating a mix of older and newer buildings. In contrast, Lalbagh has a narrower range of 3 to 7 years, with a mean of 6, suggesting relatively recent development.

Structure Type: From Figure 4.4, Lalbagh shows pucca structures as the most prevalent, with a significant presence of semi-permanent construction and a smaller but notable portion of temporary buildings. On the other hand, in Dhanmondi, pucca structures are the most common and constitute over half of all buildings, reinforcing the interpretation of a stable, developed region. semi-pucca structures are also significant, while katcha structures are less prevalent, and the minimal presence of under-construction buildings suggests little current development activity (shown in Figure 4.5)

A comprehensive set of five distinct MLR models was systematically developed for the five categories of uses in two distinct areas following the methodology. All models show strong significance ($p < 0.0005$) and R^2 (0.07 – 0.80), demonstrating their potential effectiveness in predicting electricity consumption based on identified input parameters. In the beginning, the Pearson Coefficient Test was conducted to determine whether the input parameters have any multi-collinearity effect. According to Kim (2019) and Shrestha (2020), the input variables with Pearson correlation coefficients exceeding 0.8 should be excluded to build a statistically stable model. Thus, the input variables with Pearson correlation coefficient values up to 0.80 were considered strongly correlated and did not proceed in the stepwise MLR modelling process in this study. All the MLR models were formulated in OriginPro 2016 software. Firstly, the MLR model was prepared with all parameters to quantify electricity consumption. Finally, the best best-suited MLR model was chosen after several attempts.

Pearson multi-collinearity test

Figures 6.1 to 6.5 depicts the Pearson collinear analysis among the variables for five different uses in both Lalbagh and Dhanmondi areas, and the p-values are given in Figures C1 to C10 of Appendix C

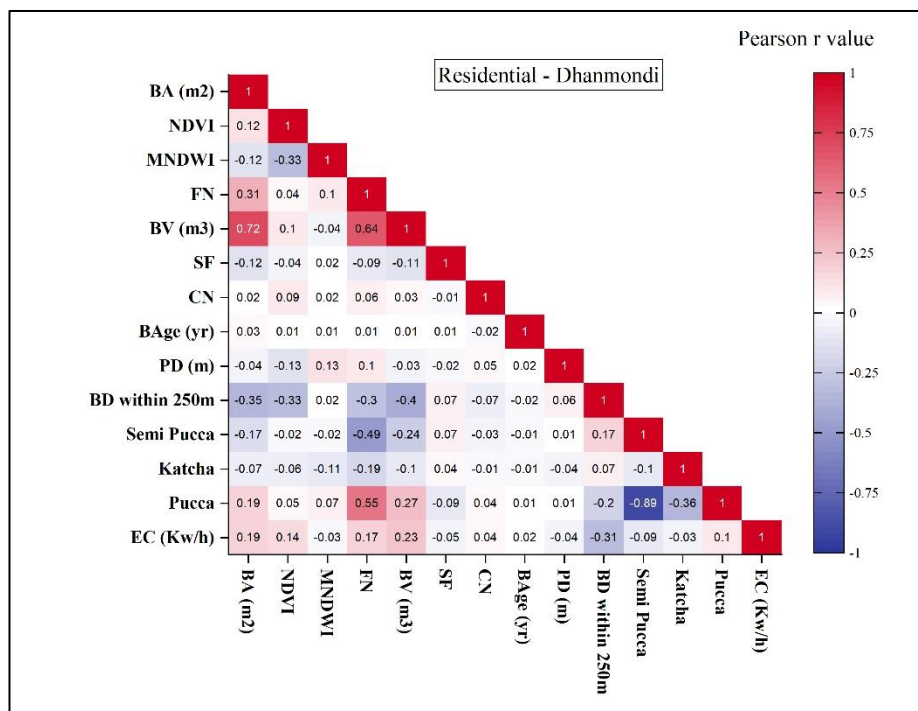
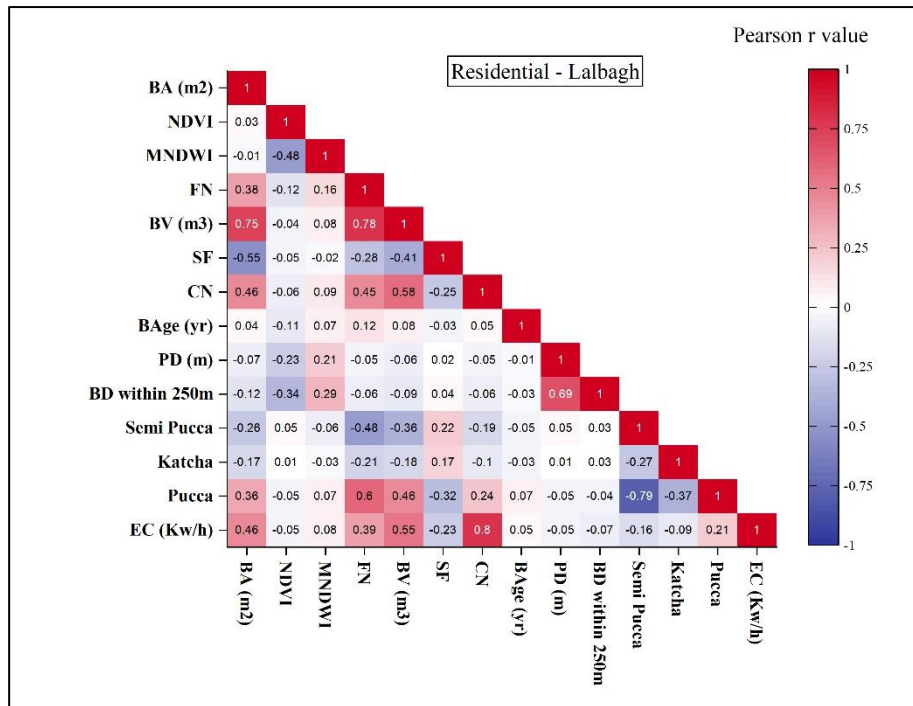


Figure 6.1: Pearson Collinearity Test Analysis for Lalbagh and Dhanmondi Areas for Residential Use

In Figure 6.1, the independent variables for residential use are more strongly correlated with electricity consumption (EC) in the Lalbagh area compared to Dhanmondi area. BV (r value 0.55) and customer no (r value 0.80) are highly correlated with EC in Lalbagh area whereas the BD within 250 m (-0.31) and BV (0.23) exhibit higher r values in the Dhanmondi area. In the Lalbagh area, BA is highly correlated with BV, shape factor and customer no (ranging between ± 0.40 to ± 0.75), indicating that larger areas generally correlate with larger buildings at high energy use. Contrastingly, the shape factor shows a negative correlation with building volume (-0.41), suggesting that more compact or regular-shaped buildings tend to have increased volume, possibly due to more efficient use of space. In Dhanmondi, where the correlation between building volume and floor number (0.64) stands out, reinforcing the idea that multi-storied structures contribute substantially to the total volume of buildings. Pucca shows a moderate positive correlation with FN in both Dhanmondi (0.55) and Lalbagh (0.6) possibly as taller structures require robust, permanent materials to ensure structural stability and compliance with urban building codes. Interestingly, electricity consumption here shows a negative correlation (-0.31) with building density within a 250m radius, which could indicate that higher-density areas benefit from more shared resources or infrastructure efficiencies, leading to lower per-building energy consumption.

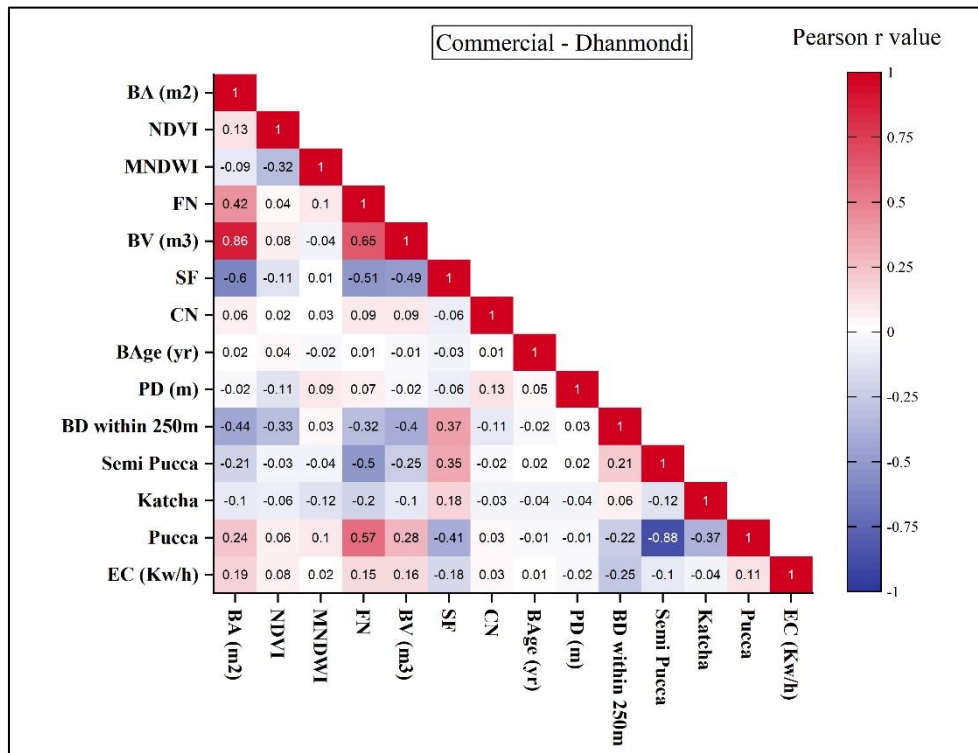
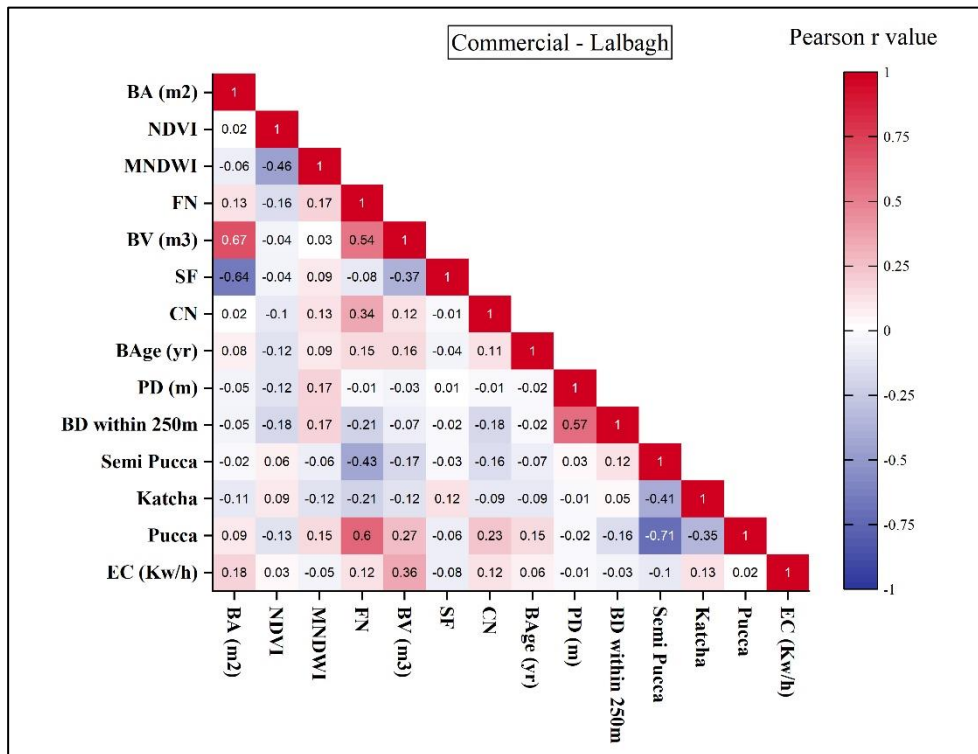


Figure 6.2: Pearson Collinearity Test Analysis for Lalbagh and Dhanmondi Areas for Commercial Use

In Figure 6.2, the relation with independent variables with electricity consumption depicts higher in Lalbagh area compared to Dhanmondi area. In Lalbagh, the correlation between building area (BA) and building volume (BV) is highly positive (0.67), indicating that larger commercial spaces typically possess greater volumes. A significant negative correlation is observed between shape factor and BV (-0.49), suggesting that more irregularly shaped buildings tend to have smaller volumes. Furthermore, electricity consumption (EC) is moderately correlated with floor number (0.36), hinting that higher buildings might consume more energy, possibly due to increased usage areas like elevators and lighting for additional floors. In Dhanmondi's commercial matrix, the correlation between BA and BV is remarkably high (0.86), which is higher than in Lalbagh, reflecting a consistent trend where larger floor areas in commercial buildings correlate with substantial volumetric increases. The shape factor's negative correlation with BV (-0.49) is also significant, aligning with the Lalbagh findings (-0.37). However, Dhanmondi and Lalbagh display a unique aspect where BD within 250m shows a moderate negative correlation with floor number (-0.32) and (-0.21) respectively, suggesting that taller commercial buildings often take up more space or are spread out to meet safety rules or city plans, so areas with many tall buildings tend to have fewer total buildings overall.

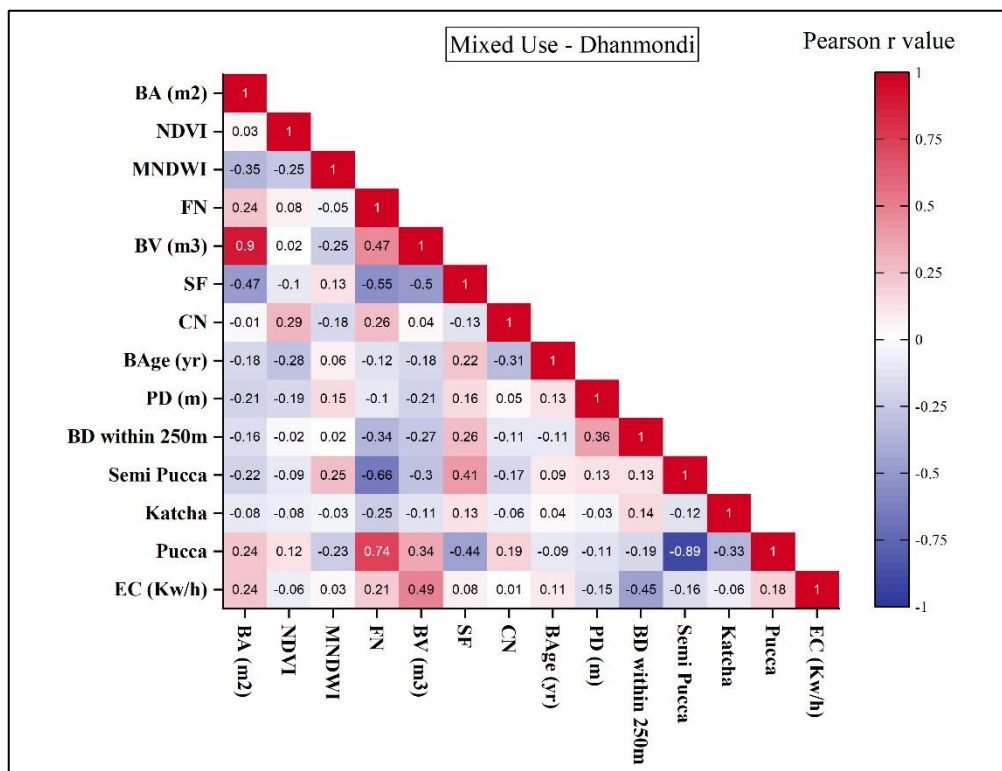
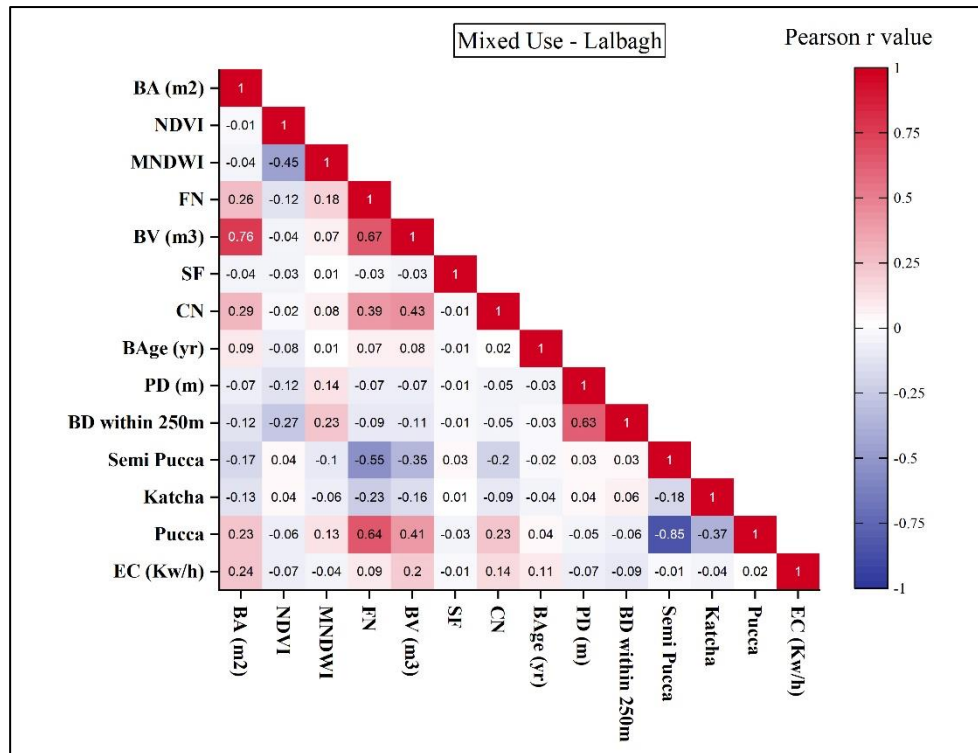


Figure 6.3: Pearson Collinearity Test Analysis for Lalbagh and Dhanmondi Areas for Mixed-Use

In Figure 6.3, the relation between independent variables with electricity consumption for mixed-use is depicted as higher in the Dhanmondi (-0.06 to 0.49) area compared to the Lalbagh area (-0.01 to 0.24). For Lalbagh, a notable strong positive correlation (0.76) exists between BA and BV, suggesting a direct relationship between the size of the area and the volume. The shape factor has a notable negative correlation (-0.55) with BV, indicating that as buildings become more geometrically compact, their volume decreases in Dhanmondi. Notably, the correlation between BA and BV is even stronger (0.9), suggesting a highly consistent trend across mixed-use environments where larger areas invariably mean larger building volumes. However, EC shows a positive correlation with building volume (0.49), its negative correlation with BD within 250m (-0.45) implies that denser areas might lead to somewhat reduced electricity usage per building. Pucca shows a moderate positive correlation with FN in both Dhanmondi (0.74) and Lalbagh (0.64), indicating that building structure material has a great impact on taller buildings.

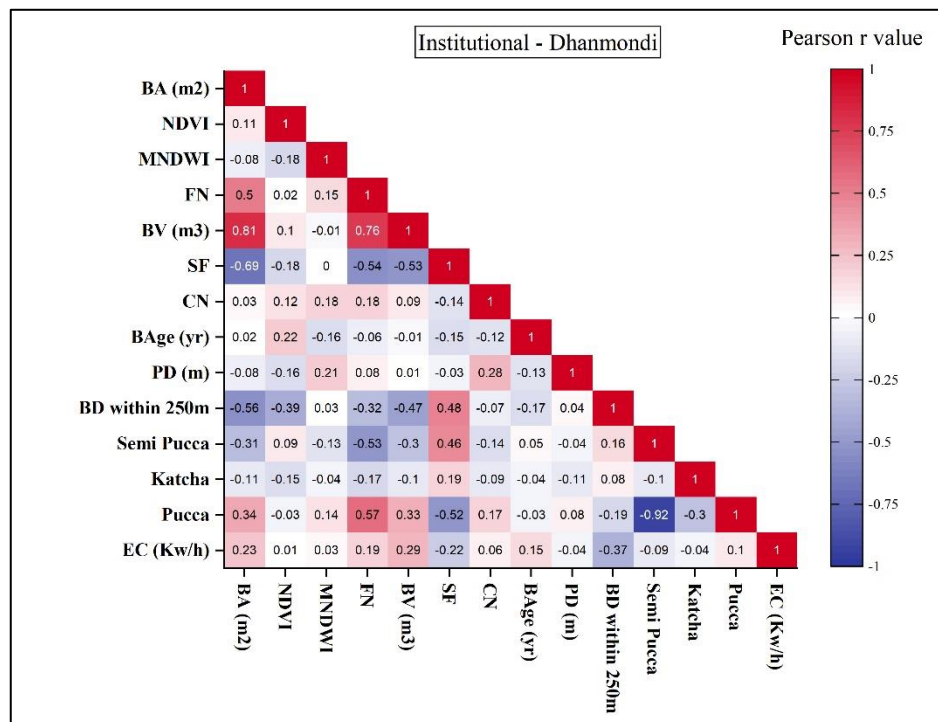
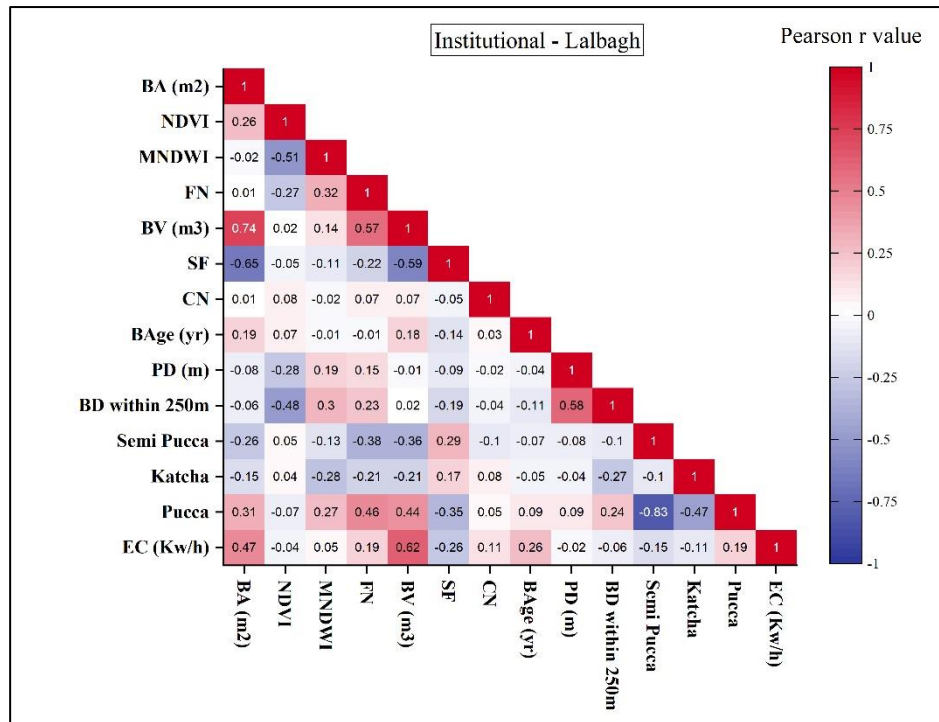


Figure 6.4: Pearson Collinearity Test Analysis for Lalbagh and Dhanmondi Areas for Institutional Use

In Figure 6.4, the relation with independent variables with EC for institutional depicted higher in the Lalbagh (-0.26 to 0.62) area compared to the Dhanmondi area (-0.36 to 0.29). In Lalbagh, there is a notably strong positive correlation between BA and BV of 0.74, suggesting a very direct proportionality between the size and volumetric capacity of institutional buildings. Another significant observation is the high correlation (0.58) between (BD within 250m) and PD, indicating that denser institutional zones tend to be clustered. The negative correlation of (-0.59) between the shape factor and BV implies that more irregularly shaped institutions may have larger volumes. Additionally, the EC shows a strong positive correlation with floor number (0.62) and BA (0.47), which could reflect higher utility use in larger, multi-storied institutional structures. Contrastingly, the matrix for Dhanmondi shows a similarly strong positive correlation between BA and BV (0.81) and between floor number and BV (0.74), reinforcing the trend observed in Lalbagh, where larger and taller buildings are common in institutional setups. However, the negative correlations are more pronounced in Dhanmondi, such as a significant (-0.56) between BD within 250m and BA, which suggests that more expansive institutions might be situated in less densely built environments. Moreover, the EC's negative correlation with BD within 250m (-0.37) highlights an interesting dynamic where institutions in denser areas might have more efficient energy usage or shared utilities.

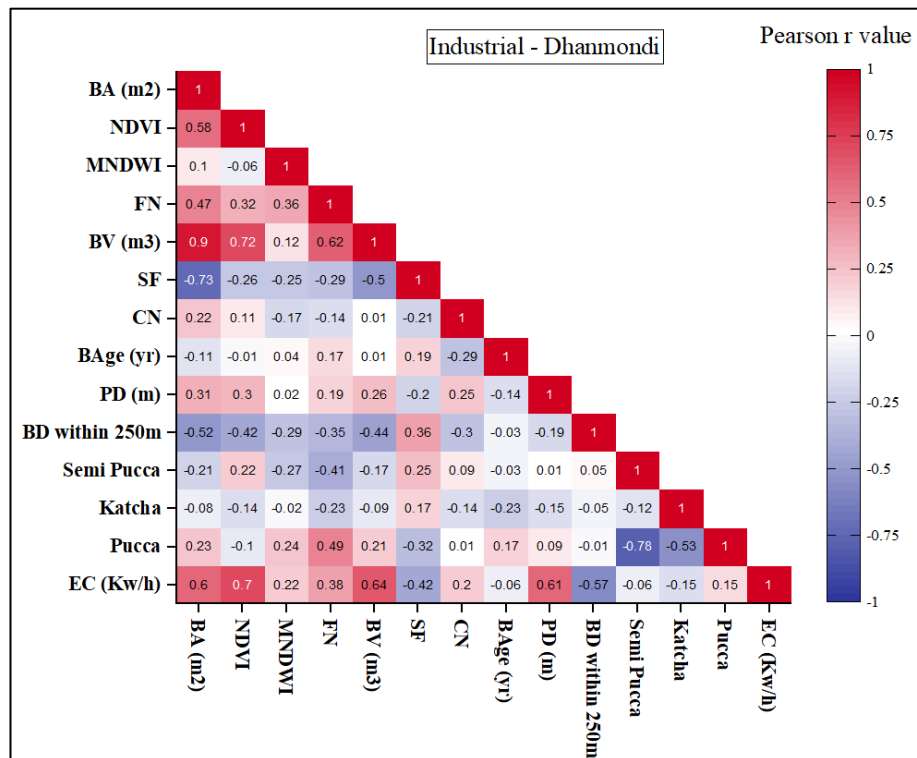
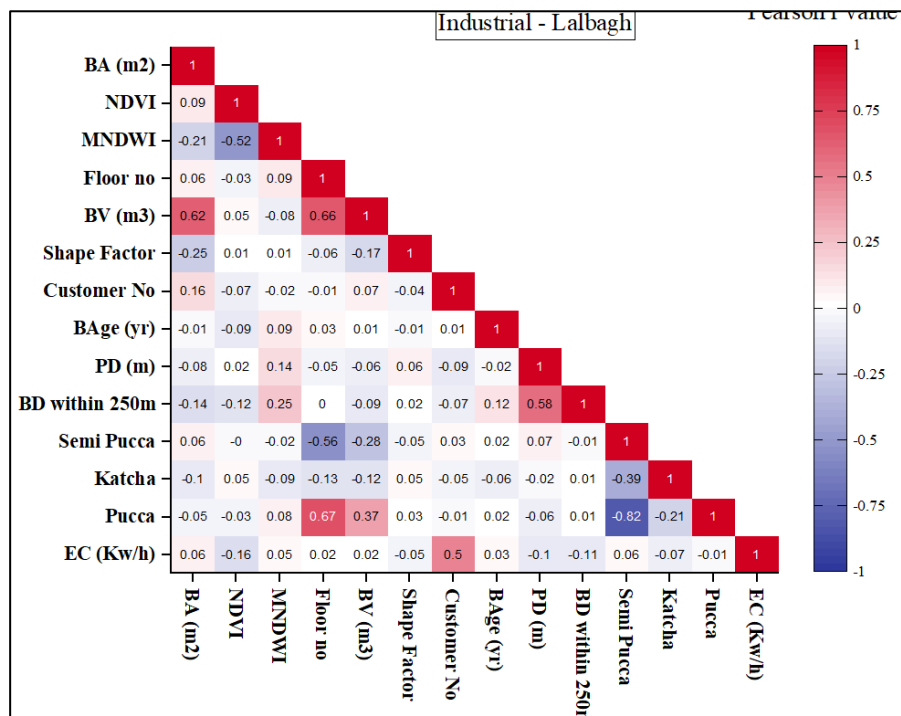


Figure 6.5: Pearson Collinearity Test Analysis for Lalbagh and Dhanmondi Areas for Industrial Use

In Figure 6.5, the relation with independent variables with EC for industrial use show robust and stronger in Dhanmondi (-0.57 to 0.70) area compared to Lalbagh area (-0.16 to 0.50). In Lalbagh, there is a strong positive correlation between BA and BV at 0.62, suggesting a close relationship where larger areas generally correspond to larger industrial volumes. Notably, the correlation between EC and BV is also significant at 0.5, indicating that larger industrial facilities tend to consume more electricity, which is likely due to increased machinery or operational demands. Additionally, there is a negative correlation of -0.25 between shape factor and BA, which might suggest that more irregularly shaped industrial buildings have smaller floor areas. In Dhanmondi's industrial sector, the BA and BV correlation is extremely high at 0.9, reinforcing the trend observed in Lalbagh but with greater intensity. EC is strongly correlated with BV at 0.64 and shows a considerable positive correlation with floor number (0.64), suggesting that taller and larger industrial buildings consume significantly more power, potentially reflecting higher operational or mechanical loads. A notable pattern is the strong negative correlation of -0.73 between shape factor and BV, indicating that more regular or efficiently shaped buildings in Dhanmondi tend to be larger, possibly due to the optimization of space for industrial activities.

6.1 Residential Use

Lalbagh: In Table 6.1, the observed negative correlation between electricity consumption for residential use in Lalbagh and factors such as the perimeter/area ratio of buildings, NDVI (Normalized Difference Vegetation Index), and building density within a 250m buffer can be attributed to several interrelated factors. Firstly, the perimeter/area ratio is a significant determinant of energy efficiency in buildings. A lower perimeter/area ratio typically indicates a more compact building design, which can reduce heat loss and improve thermal performance, thereby decreasing the need for heating and cooling, which are major components of residential electricity consumption (Antoniucci et al., 2021). This is particularly relevant in low-income areas where residents may prioritize energy efficiency due to financial constraints. Additionally, higher NDVI values, which indicate more vegetation and green spaces, are associated with lower urban heat island effects and can contribute to reduced cooling needs in residential buildings (Ko, 2013). A Study has shown that urban greenery can significantly lower surrounding temperatures, thus reducing the reliance on air conditioning systems (Ko, 2013). Moreover, building density within a 250m radius can influence energy consumption patterns. Higher building density often correlates with shared resources and infrastructure, which can lead to lower individual energy consumption per household (Antoniucci et al., 2021). In densely populated areas, the proximity of services and amenities can also reduce the need for transportation, further lowering overall energy use (Ko, 2013).

The positive correlation between electricity consumption and factors such as the number of floors in a building and the number of customers in Lalbagh can be explained through various socio-economic and structural dynamics. Firstly, multi-story buildings tend to accommodate more residents, which can lead to increased electricity consumption. Each additional floor can house more families or individuals, thus amplifying the cumulative demand for electricity for essential services such as lighting, heating, and cooling. This phenomenon is particularly pronounced in low-income neighborhoods where housing is often characterized by high-density living arrangements, such as tenement buildings, which can lead to significant electricity usage due to the sheer number of occupants. However, it is important to note that some studies have found a non-significant relationship between the size of a building and electricity consumption, while others have concluded a positive relationship (Ubani et al., 2024). Moreover, the number of customers in a building directly correlates with electricity demand. In low-income areas, where residents may rely heavily on electricity for basic needs, the aggregation of households in a single building can lead to increased consumption levels.

This is supported by findings that indicate higher electricity usage in densely populated environments, where the demand for energy services is amplified by the number of residents. Additionally, as electricity becomes more accessible through improved infrastructure and technology, low-income households may increase their consumption to meet their growing needs for modern conveniences and electronic devices (Saidi et al., 2015). Furthermore, the economic context of low-income households plays a crucial role in this positive correlation. As households experience improvements in their economic conditions, they are likely to increase their electricity consumption to enhance their quality of life. This is particularly relevant in contexts where government policies or subsidies facilitate access to electricity, thereby encouraging higher usage among low-income residents. The willingness to pay for improved housing conditions, as noted in studies, also reflects a broader trend where households invest in better living environments that often necessitate increased energy consumption (Addo, 2015).

The final equation of Electricity Consumption (Kw/hr) in year =

$$1547.80 - 5060.76 \times \text{NDVI} + 194.4 \times \text{FN} - 604.56 \times \text{SF} + 2172.95 \times \text{CN} - 0.72 \times \text{BD}$$

This model explains approximately 64.3% of the variance in electricity consumption (EC in kWh).

Dhanmondi: In Table 6.1, the observed relationships between electricity consumption, building density, building volume, and the structure type (pucca, katcha and semi pucca) in Dhanmondi residential areas can be attributed to several interrelated factors. Firstly, the negative correlation between electricity consumption and building volume may be explained by the efficiency of larger buildings. Larger residential structures often incorporate advanced energy-efficient technologies and designs that reduce overall energy consumption per unit area. For instance, Pickering et al. discuss how building design and technology influence energy consumption, suggesting that larger buildings may utilize energy more efficiently due to better insulation and HVAC systems (Pickering et al., 2017).

Conversely, the positive correlation between electricity consumption and building volume can be attributed to the fact that larger volumes typically indicate more extensive heating, cooling, and electrical needs. Larger buildings often require more energy for lighting, heating, and cooling due to their greater volume and surface area exposed to the external environment. This

aligns with findings from Wang et al., who note that built environment characteristics significantly influence energy consumption, particularly in larger structures where energy demands scale with volume (Wang et al., 2023). The relationship between building density and electricity consumption is more nuanced. Higher building density, particularly within a 250m buffer, may lead to increased energy consumption due to the urban heat island effect, where densely built areas experience higher temperatures, thus increasing cooling demands. Building structure types significantly impact residential electricity use. Katcha (temporary materials, poor insulation) increases consumption due to higher heating/cooling needs. Pucca (durable, well-insulated) reduces energy use through efficiency. Semi-pucca (mixed materials) shows variable consumption. Studies confirm pucca buildings mitigate electricity demand compared to katcha (Amber et al., 2019; Brounen et al., 2012; Wahlström & Hårsman, 2015).

The final equation of Electricity Consumption (Kw/hr) in year =

$$94335.54 + 5.52 \times BV - 106.94 \times BD - 2.82E17 \times \text{Semi pucca} - 2.82E17 \times \text{Katcha} + 2.82E17 \times \text{Pucca}$$

This model explains about 11.3 % of the variance in electricity consumption (EC in kWh). The factors can not explain electricity consumption effectively in Dhanmondi, unlike Lalbagh.

Table 6.1: Multi-linear Regression Model of Electricity Consumption for Residential Use

Location	Dependent Variable	Independent Variables	Coefficient Value	P Value	R ²	F Value
Lalbagh	Electricity Consumption (Kw/hr)	NDVI	-5060.76	2.24E-6	0.65	5105.29
		FN	194.39	1.22E-6		
		SF	-604.55	3.51E-4		
		CN	2172.95	1.12E-6		
		BD	-0.72	3.31E-5		
Dhanmondi	Electricity Consumption (Kw/hr)	Semi Pucca	-2.82E17	5.50E-4	0.11	326.25
		Katcha	-2.82E17	5.50E-4		
		Pucca	-2.82E17	5.50E-4		
		BV	5.53	4.03E-28		
		BD	-106.94	7.27E-85		

Here, NDVI means Normalized Difference Vegetation Index), FN = Floor No, SF = Shape Factor, CN = Customer No, BD = Building Density, BV= Building Volume

6.2 Commercial Use

Lalbagh: In Table 6.2, the observed relationships between electricity consumption for commercial use in low-income areas and various building characteristics, including the negative correlation with floor number and area, as well as the positive correlation with building volume and the Modified Normalized Difference Water Index (MNDWI), can be explained through several interrelated factors.

Negative Correlation with Floor Number and Area can be explained by building characteristics, socio-economic characteristics and occupancy pattern. In low-income areas, buildings are often designed with limited space and resources, leading to smaller floor areas and fewer floors. Research indicates that smaller buildings typically have less efficient energy systems, which can result in higher per-unit energy consumption (Buys et al., 2015). For instance, Buys et al. highlight that larger floor areas tend to be associated with higher electricity consumption, particularly among higher-income households, suggesting that low-income buildings may not benefit from economies of scale in energy efficiency (Buys et al., 2015). Low-income households often face financial limitations that restrict their ability to invest in energy-efficient technologies. This is compounded by the fact that lower-floor buildings may have higher occupancy rates, leading to increased energy demands for basic needs such as lighting and cooling (Woo & Cho, 2018). The lack of investment in energy-efficient appliances and systems in these buildings contributes to higher overall electricity consumption, particularly in smaller, low-rise structures (Mastrucci et al., 2014). The demographic characteristics of low-income areas, including larger family sizes and higher occupancy rates in smaller units, can lead to increased energy consumption per unit area. As noted by Woo and Cho, the energy consumption patterns in mixed-use buildings can vary significantly based on the proportion of residential versus commercial use, with higher residential use often leading to increased overall energy consumption (Woo & Cho, 2018). This suggests that as the floor number decreases, the energy consumption per unit area may increase due to the higher number of occupants sharing limited space.

The positive correlation between electricity consumption and building volume can be attributed to the increased energy demands associated with larger structures. Larger buildings typically require more energy for heating, cooling, and lighting, as they encompass more space and often house more occupants or commercial activities (Clement & Schultz, 2011). The relationship between building volume and energy consumption is supported by findings that indicate larger buildings tend to have higher energy use intensity (Szabó et al., 2016). The Modified

Normalized Difference Water Index (MNDWI) is often used to assess the presence of water bodies and vegetation in urban areas, which can influence local microclimates and energy consumption patterns. Higher MNDWI values may indicate better access to green spaces and water bodies, which can mitigate urban heat effects and reduce cooling demands in larger buildings (Pandey et al., 2022). This relationship suggests that buildings with higher volumes located in areas with better environmental conditions may experience lower energy consumption due to reduced cooling needs. The interplay between building volume and environmental factors, such as local climate, can also affect energy consumption. Research indicates that larger buildings may benefit from better thermal performance and energy efficiency due to their design and orientation, which can lead to lower overall energy use (Sena et al., 2018). Additionally, the presence of vegetation and water bodies, as indicated by MNDWI, can further enhance energy efficiency by moderating temperature extremes (Pandey et al., 2022).

The final equation of Electricity Consumption (Kw/hr) in year =

$$796.18 - 37013.65 \times \text{MNDWI} + 194.50 \times \text{FN} + 8.26 \times \text{BV} + 2319.45 \times \text{CN}$$

This model explains approximately 15.09 % of the variance in electricity consumption (EC in kWh).

Dhanmondi: From Table 6.2, the observed relationships between electricity consumption, perimeter/area ratio, building density, and building area in high-income commercial areas can be explained through various interrelated factors.

Firstly, the negative correlation between electricity consumption and the perimeter/area ratio of buildings can be attributed to the efficiency of energy use in building design. A lower perimeter/area ratio often indicates a more compact building shape, which can lead to reduced heat loss and better thermal performance. This is supported by findings that suggest buildings with a smaller surface area relative to their volume tend to consume less energy for heating and cooling due to lower exposure to external temperature fluctuations (Chen et al., 2017; Ko, 2013). Additionally, compact building designs may facilitate more efficient HVAC systems, further reducing energy consumption (Chen & Hong, 2018). The negative correlation between electricity consumption and building density within a 250m buffer can be explained by the benefits of shared resources and infrastructure in densely built environments. Higher building

density often leads to shared energy resources, such as district heating or cooling systems, which can be more efficient than individual systems (Antoniucci et al., 2021). Additionally, densely populated areas may promote energy-efficient behaviors among occupants, such as reduced reliance on personal vehicles, which can indirectly lower overall electricity consumption (Sena et al., 2021).

Conversely, the positive correlation between electricity consumption and building area is likely due to the increased demand for energy-intensive services in larger buildings. Larger commercial spaces typically house more equipment, lighting, and HVAC systems, which collectively contribute to higher electricity use (Lee et al., 2019; Khan et al., 2021). Furthermore, larger buildings may also accommodate more occupants or activities, leading to increased energy consumption for lighting, heating, and cooling (Amoodi & Azar, 2018). This aligns with research indicating that larger buildings inherently require more energy to operate due to their scale and the complexity of their systems (Woo & Cho, 2018; Boukarta & Berezowska, 2017).

The final equation of Electricity Consumption (Kw/hr) in year =

$$162059.44 - 49892.26 \times SF + 47.66 \times BA - 87.28 \times BD$$

This model explains approximately 7.38 % of the variance in electricity consumption. These factors can not explain the commercial electricity consumption effectively in both areas.

Table 6.2: Multi-linear Regression Model of Electricity Consumption for Commercial Use

Location	Dependent Variable	Independent Variables	Coefficient Value	P Value	R ²	F Value
Lalbagh	Electricity Consumption (Kw/hr)	MNDWI	-37013.65	0.03	0.15	70.13
		FN	-1768.50	5.01E-6		
		BV	8.26	3.76E-48		
		CN	2319.45	2.70E-7		
Dhanmondi	Electricity Consumption (Kw/hr)	BA	47.66	0.04E-7	0.07	326.25
		SF	- 49892.26	0.004		
		BD within 250m	-87.28	1.13E-17		

Here, MNDWI means Modified Normalized Difference Water Index, FN = Floor No, SF = Shape Factor, CN = Customer No, BD = Building Density, BV= Building Volume, BA = Building Area.

6.3 Mixed-Use

Lalbagh: In Table 6.3, the observed relationships between electricity consumption, Normalized Difference Vegetation Index (NDVI), building density, the number of customers in a building, and the area of a building in Lalbagh can be explained through various socio-economic and environmental factors.

Firstly, the negative correlation between electricity consumption and NDVI can be attributed to the presence of vegetation, which typically provides shade and cooling effects, thereby reducing the need for air conditioning and other energy-intensive cooling systems. Studies have shown that urban greenery can significantly lower ambient temperatures, leading to decreased electricity demand for cooling purposes (Jang et al., 2021). Additionally, areas with higher NDVI often indicate better environmental quality and potentially lower population density, which can also correlate with reduced energy consumption as fewer buildings and residents may lead to lower overall demand (Xiao-ping & Reiner, 2016). Building density, measured as the number of buildings within a 250m buffer, also exhibits a negative correlation with electricity consumption. This can be explained by the phenomenon of energy efficiency that often accompanies higher-density living. In densely populated areas, shared resources and infrastructure can lead to lower per capita energy consumption due to economies of scale in energy use (Kawka & Cetin, 2021). Furthermore, densely built environments may promote the use of public transportation and reduce reliance on personal vehicles, which can further decrease overall energy consumption (Lee et al., 2020).

Conversely, the positive correlation between electricity consumption and the number of customers in a building, as well as the area of the building, can be understood through the lens of increased demand for electrical appliances and services. Larger buildings typically house more occupants, leading to higher cumulative energy needs for lighting, heating, cooling, and powering appliances (Kusumaningrum, 2018). Additionally, as income levels rise, even within low-income groups, there is an observed increase in electricity consumption as households acquire more electrical appliances to improve their quality of life (Daioglou et al., 2012). This is particularly relevant in mixed-income communities where social dynamics may encourage higher consumption patterns among low-income residents due to the influence of their higher-income neighbours (Zu et al., 2019).

The final equation of Electricity Consumption (Kw/hr) in year =

$$65249.29 - 397777.15 \times \text{NDVI} + 244.70 \times \text{BA} - 604.56 \times \text{SF} + 1563.15 \times \text{CN} - 29.49 \times \text{BD}$$

This model explains approximately 7 % of the variance in electricity consumption (EC in kWh).

Dhanmondi: From Table 6.3, in relatively high-income areas like Dhanmondi, the relationship between electricity consumption and various building characteristics reveals complex dynamics.

The negative correlation between electricity Consumption and building density within a 250m buffer and electricity consumption is indicative of urban planning dynamics. In high-density areas, the proximity of buildings can lead to shared resources and energy systems, which may reduce individual building electricity consumption. However, the overall demand in such areas can still be high due to the cumulative effect of multiple buildings operating simultaneously (Shan & Cao, 2017). This interplay between density and consumption highlights the importance of considering urban design in energy management strategies.

Conversely, the positive correlation between electricity consumption and building volume suggests that as the volume of a building increases, so does its energy demand. This can be explained by the fact that larger volumes often require more extensive heating, ventilation, and air conditioning (HVAC) systems, which are significant contributors to energy consumption (Ghadi & Baniyounes, 2018). Additionally, buildings with greater volume may also have more complex electrical systems and amenities that further increase their energy needs (Ghadi & Baniyounes, 2018). The perimeter-to-area ratio also plays a crucial role in energy consumption patterns. A higher perimeter-to-area ratio indicates a more complex building shape, which can lead to increased heat loss or gain, depending on the climate (Zhao & Magoulès, 2012). Buildings with a high perimeter-to-area ratio typically have more exterior surface area relative to their volume, which can result in higher energy consumption for heating and cooling. This aligns with the findings that suggest buildings with intricate designs or those that are less compact tend to consume more energy due to inefficiencies in thermal management (Zhao & Magoulès, 2012).

The final equation of Electricity Consumption (Kw/hr) in year =

$$65428.67 - 256223.25 \times SF + 22.89 \times BV - 167.59 \times BD$$

This model explains about 54 % of the variance in electricity consumption (EC in kWh). These factors can explain the mixed-use consumption effectively in Dhanmondi, unlike Lalbagh

Table 6.3: Multi-linear Regression Model of Electricity Consumption for Mixed Use

Location	Dependent Variable	Independent Variables	Coefficient Value	P Value	R ²	F Value
Lalbagh	Electricity Consumption (Kw/hr)	NDVI	-397777.15	7.55E-11	0.07	110.57
		BA	244.70	1.37E-51		
		CN	1563.15	9.04E-9		
		BD	-29.49	9.25E-10		
Dhanmondi	Electricity Consumption (Kw/hr)	SF	256223.25	3.27E-5	0.54	24.54
		BV	22.89	3.35E-8		
		BD	-167.59	1.48E-5		

Here, NDVI means Normalized Difference Vegetation Index), SF = Shape Factor, CN = Customer No, BD = Building Density, BV= Building Volume, BA = Building Area

6.4 Institutional Use

Lalbagh: In Table 6.4, the observed relationships between electricity consumption and various building characteristics in Lalbagh can be explained through a combination of factors related to building design, environmental context, and socio-economic conditions.

Firstly, the negative correlation between electricity consumption and the floor number of a building can be attributed to the energy efficiency of higher buildings. Taller buildings often incorporate more advanced energy management systems and design features aimed at reducing energy consumption per unit area. For instance, Mashhoodi and Timmeren Mashhoodi & Timmeren (2018) suggest that higher buildings may benefit from economies of scale in energy efficiency measures, which can lead to lower overall energy consumption. Furthermore, the negative correlation with NDVI (Normalized Difference Vegetation Index) indicates that buildings surrounded by more vegetation may experience reduced energy needs due to natural cooling effects provided by trees and green spaces, which can mitigate the urban heat island effect.

In contrast, the positive correlation between electricity consumption and building density within a 250m radius can be explained by the increased demand for energy services in densely populated areas. Higher building density often leads to greater energy consumption due to the concentration of activities and services that require electricity, such as lighting, heating, and cooling. Additionally, the volume of the building and its age can also contribute to higher energy consumption. Older buildings typically have less efficient insulation and outdated heating and cooling systems, leading to increased energy use (Ghadi & Baniyounes, 2018). The volume of a building can influence its energy consumption as larger structures may require more energy for heating, cooling, and lighting, particularly if they are not designed with energy efficiency in mind (Asfour & Alshawaf, 2015). Moreover, the building age factor is significant because older buildings often lack modern energy-efficient technologies and materials, which can lead to higher electricity consumption compared to newer constructions that are designed with sustainability in mind (Jo & Kim, 2022). This aligns with findings from Jo and Kim (Jo & Kim, 2022), who emphasize that construction age significantly impacts energy consumption patterns.

The final equation of Electricity Consumption (Kw/hr) in year =

$$3027.68 - 62529.78 \times \text{NDVI} - 2831.39 \times \text{FN} - 14019.78 \times \text{SF} + 10.33 \times \text{BV} + 299.67 \times \text{BAge}$$

This model explains approximately 48 % of the variance in electricity consumption (EC in kWh).

Dhanmondi: In Table 6.4, the observed negative correlation between electricity consumption and building density, alongside the positive correlation with building volume in institutional settings within high-income areas, can be attributed to multiple interrelated factors.

Firstly, the negative correlation with building density can be explained through the concept of energy efficiency associated with urban design. In high-density areas, buildings are often designed to maximize space efficiency, which can lead to shared resources and infrastructure, such as heating and cooling systems, that reduce overall energy consumption per unit area. Studies have shown that higher urban density can lead to reduced energy consumption due to the proximity of services and amenities, which diminishes the need for extensive transportation and the associated energy costs (Boukarta & Berezowska, 2017; Morikawa, 2012).

Furthermore, densely populated areas may benefit from economies of scale in energy distribution and management, leading to lower per capita energy use (Antoniucci et al., 2021).

Conversely, the positive correlation between electricity consumption and building volume can be attributed to the increased energy demands associated with larger buildings. Larger institutional buildings typically house more occupants and require more extensive heating, ventilation, and air conditioning (HVAC) systems, which significantly contribute to electricity consumption (Ghadi & Baniyounes, 2018). Research indicates that larger non-residential buildings often exhibit higher energy consumption due to their operational requirements, including lighting and climate control systems, which are more pronounced in expansive spaces (Cho et al., 2019). Additionally, the complexity of systems in larger buildings often leads to higher energy use, as they may not operate as efficiently as smaller, more compact structures (Wang et al., 2023). Moreover, the operational patterns of institutional buildings, particularly in high-income areas, can further elucidate these correlations. High-income institutions may have more advanced technological systems and amenities that, while enhancing comfort and functionality, also lead to increased energy consumption (Amoodi & Azar, 2018). For instance, the integration of sophisticated HVAC systems and extensive lighting can elevate electricity use, particularly in larger buildings that require more extensive infrastructure to maintain operational efficiency (Bastida et al., 2024).

The final equation of Electricity Consumption (Kw/hr) in year =

$$107484 + 4.42 \times BV - 67.16 \times BD$$

This model explains approximately 15 % of the variance in electricity consumption (EC in kWh). These factors can explain the institutional use of electricity consumption moderately in Lalbagh but cannot explain the consumption in Dhanmondi.

Table 6.4: Multi-linear Regression Model of Electricity Consumption for Institutional Use

Location	Dependent Variable	Independent Variables	Coefficient Value	P Value	R ²	F Value
Lalbagh	Electricity Consumption (Kw/hr)	NDVI	-62529.78	0.05	0.48	19.96
		FN	-2831.39	9.42E-4		
		SF	14019.78	0.009		
		BV	10.33	2.01E-13		
		BAge	299.67	0.07		
Dhanmondi	Electricity Consumption (Kw/hr)	BD	-67.16	0.002	0.15	11.22
		BV	4.42	0.007		

Here, NDVI means Normalized Difference Vegetation Index), SF = Shape Factor, CN = Customer No, BD = Building Density, BV= Building Volume, BAage= Building Age, FN = Floor No

6.5 Industrial Use

Lalbagh: In Table 6.5, the observed negative correlation between electricity consumption and building area, NDVI (Normalized Difference Vegetation Index), and building density within a 250m buffer can be attributed to several interrelated factors. These factors include the influence of urban density, the characteristics of the built environment, and the socio-economic dynamics of low-income areas.

Firstly, urban density plays a significant role in electricity consumption patterns. Research indicates that higher building density often leads to reduced per capita electricity consumption due to shared resources and infrastructure, such as communal heating and cooling systems, which are more efficient in densely populated areas (Antoniucci et al., 2021). For instance, the study by Sajadian et al. highlights that horizontal building density significantly affects energy consumption, suggesting that as buildings become more densely packed, the overall energy efficiency improves, leading to lower individual consumption rates (Sajadian et al., 2022). Additionally, the findings of Ghafoor et al. support this notion, indicating that larger residential areas tend to consume more energy due to increased demand for heating and cooling in larger spaces (Ghafoor et al., 2020). Furthermore, the NDVI, which reflects the amount of vegetation in an area, can also impact electricity consumption. Areas with higher vegetation typically experience lower temperatures, which can reduce the need for air conditioning and thus lower

electricity consumption (Wen et al., 2022). This is particularly relevant in low-income areas where residents may rely on less energy-intensive cooling methods, such as natural ventilation, due to economic constraints (Amber et al., 2021). The presence of vegetation can also enhance the microclimate, further reducing the reliance on electrical cooling systems (Wen et al., 2022). Moreover, the area of the building itself is a critical factor. Larger buildings often have higher overall energy demands due to the need for lighting, heating, and cooling across more extensive spaces. However, when considering the electricity consumption per unit area, larger buildings may exhibit lower efficiency, leading to a negative correlation with electricity consumption (Antoniucci et al., 2021). This phenomenon is supported by the work of Morikawa, which found that energy efficiency tends to be higher in densely populated urban settings, suggesting that as building area increases, the efficiency of energy use may decrease (Morikawa, 2012).

In contrast, the positive correlation between electricity consumption and the number of customers in a building can be explained by the increased demand for electrical appliances and services as occupancy rises. More occupants typically lead to higher usage of electrical devices, such as lighting, heating, and cooling systems, which directly correlates with increased electricity consumption (Cheung et al., 2013). This relationship is particularly pronounced in commercial and mixed-use buildings where customer footfall translates into higher energy demands for operational purposes (Woo & Cho, 2018).

The final equation of Electricity Consumption (Kw/hr) in year =

$$8593.44 - 570195.67 \times \text{NDVI} + 194.4 \times \text{FN} - 205.79 \times \text{BA} + 101898.49 \times \text{CN} - 36.67 \times \text{BD}$$

This model explains approximately 27 % of the variance in electricity consumption (EC in kWh).

Dhanmondi: In Table 6.5, the relationship between electricity consumption and various building characteristics in high-income industrial areas can be understood through several interrelated factors. The negative correlation between electricity consumption and the perimeter/area ratio of buildings, as well as building density, can be attributed to several reasons.

Firstly, a lower perimeter/area ratio typically indicates a more compact building design, which can lead to reduced surface area exposed to external temperatures. This compactness can reduce the need for heating and cooling systems that consume significant electricity (Antoniucci et al.,

2021). Moreover, higher building density may lead to shared infrastructure and resources, such as heating systems and cooling towers, which can further decrease individual electricity consumption through economies of scale (Liddle, 2013).

In contrast, the positive correlation between electricity consumption and building volume suggests that larger buildings inherently require more energy for lighting, heating, cooling, and powering equipment. This is particularly evident in industrial settings where larger facilities often house more machinery and technology, leading to increased energy demands (Liu et al., 2023). Furthermore, the distance of a building from its nearest permeable land can impact electricity consumption due to the urban heat island effect, where buildings located further from green spaces may experience higher ambient temperatures, necessitating greater use of air conditioning systems (Yana et al., 2024). The findings also resonate with the economic context of high-income areas, where the demand for comfort and convenience often drives higher electricity consumption. High-income households and industries are more likely to invest in energy-intensive appliances and systems, thus increasing overall consumption (Williams et al., 2021). The presence of advanced technologies and larger electrical appliances in these settings further exacerbates this trend, as these systems are designed to operate efficiently but still consume substantial amounts of electricity (Sena et al., 2021).

The final equation of Electricity Consumption (Kw/hr) in year =

$$266209.61 - 16.45 \times BV + 491.66 \times PD - 175327.37 \times SF - 94.07 \times BD$$

This model explains approximately 80 % of the variance in electricity consumption in kWh. These factors can explain the industrial consumption effectively in Dhanmondi and moderately in Lalbagh.

Table 6.5: Multi-linear Regression Model of Electricity Consumption for Industrial Use

Location	Dependent Variable	Independent Variables	Coefficient Value	P Value	R ²	F Value
Lalbagh	Electricity Consumption (Kw/hr)	NDVI	-570195.67	8.04E-7	0.27	69.34
		BA	-205.79	0.008		
		CN	101898.49	3.00E-55		
		BD within 250m	-36.67	0.001		
Dhanmondi	Electricity Consumption (Kw/hr)	NDVI	290174.57	3.27E-5	0.80	27.40
		BV	16.45	2.87E-6		
		BD within 250m	-94.07	2.34E-4		
		PD (m)	491.66	1.23E-6		
		SF	-175327.37	0.005		

Here, NDVI means Normalized Difference Vegetation Index), SF = Shape Factor, CN = Customer No, BD = Building Density, BV= Building Volume, BA = Building Area, PD = Distance from Permeable Area

Overall, the factors analyzed demonstrate varying effectiveness in explaining electricity consumption across different land-use categories in Dhanmondi and Lalbagh. While industrial electricity consumption is effectively explained in Dhanmondi and moderately in Lalbagh, institutional use shows moderate explanatory power only in Lalbagh, with no significant correlation in Dhanmondi. Mixed-use consumption is effectively captured in Dhanmondi but not in Lalbagh, whereas commercial electricity usage remains poorly explained in both areas. Overall, these factors fail to adequately explain electricity consumption patterns in Dhanmondi compared to Lalbagh, highlighting disparities in how urban form, socio-economic variables, and infrastructure interact across the two regions. This inconsistency underscores the need for localized models that account for contextual differences in land use, building typologies, and energy-use behaviors.

CHAPTER 7 : MAJOR FINDINGS AND RECOMMENDATIONS

7.1 Major Findings

This section shows the findings from trend analysis, spatial patterns, land use diversity, building typology, regression models, hypothesis testing, and statistical interactions to present a cohesive understanding of how socioeconomic status, urban structure, and environmental factors shape energy use. The findings are listed below.

- Total electricity consumption increased in both Lalbagh and Dhanmondi across residential, commercial, industrial, and institutional usages from 2017 to 2023. Dhanmondi consistently showed higher consumption, particularly in residential and commercial uses, reflecting its affluent population and bustling commercial activity. Lalbagh exhibited lower, more stable usage, consistent with its residential-industrial character.
- In Lalbagh, residential consumption per customer declined slightly, possibly due to a growing customer base diluting individual usage amid urban migration, while total consumption grew. Dhanmondi maintained higher and more stable per-customer usage in residential areas, indicative of sustained demand for energy-intensive appliances. Commercial and industrial per-customer usage rose sharply in Dhanmondi, whereas Lalbagh saw fluctuations in commercial usage and steady industrial growth.
- Both neighbourhoods experienced a sharp decline in consumption in 2020, likely due to the COVID-19 pandemic, affecting residential, commercial, and institutional sectors. Post-2020 recovery was robust, with Dhanmondi's commercial and industrial sectors rebounding more strongly, maybe highlighting its economic dynamism.
- In Lalbagh, lower consumption prevails in northern and central areas, with higher usage concentrated in the southeast, likely linked to industrial or commercial hubs. In contrast, higher consumption clusters in northern and central zones correspond to commercial and mixed-use areas. Dhanmondi's extensive high-consumption zones stem from its diverse land use and planned development, while Lalbagh's concentrated usage reflects its industrial focus, illustrating how urban layout shapes spatial energy demands.
- Dominated by residential and mixed-use areas, Lalbagh has moderate land use diversity (Shannon Diversity Index: 1.79). This residential focus supports consistent but lower

electricity consumption. With a mix of residential, commercial, and institutional zones, Dhanmondi exhibits higher land use diversity (Shannon Diversity Index: 2.03)

- Lalbagh features a mix of permanent (pucca), semi-permanent (semi-pucca), and temporary (katcha) structures, reflecting its varied development stages and socio-economic conditions. In contrast, Dhanmondi is predominantly permanent (pucca) structures dominate, consistent with its planned, affluent character and modern infrastructure.
- In Lalbagh, electricity consumption across various uses is primarily driven by environmental factors and occupancy levels, with additional influences from building characteristics such as floor number in residential settings, building size in commercial contexts, and building age in institutional uses, while industrial use aligns with the broader emphasis on environmental aspects and occupancy. In contrast, Dhanmondi's consumption is predominantly shaped by structural factors: building volume and density are key for residential, institutional, and industrial uses, with building area and shape driving commercial and mixed uses, and an additional factor of proximity to permeable areas, specifically impacting industrial consumption.
- The R^2 values for Dhanmondi are low, while those for Lalbagh are relatively higher. This suggests that Lalbagh follows a moderate linear trend, whereas Dhanmondi is likely to exhibit a non-linear pattern.
- Lalbagh shows higher consumption per m^2 in industrial and mixed-use buildings and lower consumption in residential, commercial, and institutional buildings than Dhanmondi. Dhanmondi's elevated per m^2 consumption in residential and commercial sectors reflects its affluence and energy-intensive amenities, while Lalbagh's focus on industrial and mixed-use consumption highlights its dense, activity-driven energy use.
- Building structure type, location, and use significantly influence electricity consumption, with notable interactions among these factors. The combined effect of these variables results in distinct consumption patterns that vary by area's unique character and building type. Dhanmondi's consumption dynamics differ from Lalbagh's due to its structural and locational characteristics, underscoring the need for area-specific energy management approaches.
- The low value of R for residential and institutional indicates that the built environment is a minor factor in determining and forecasting electricity consumption patterns in Dhanmondi. On the other hand, built environmental factors could sufficiently explain

this usage institutional usage in Lalbagh. As socio-economic indicators like income, education level is considerable higher in Dhanmondi than Lalbagh may be socio-economic factors have larger role to play in determining and forecasting electricity consumption pattern in Dhanmondi.

- The regression also indicates that built environmental factors have little impact on energy consumption for commercial usage, while they play important role in industrial use.

7.2 Recommendation

To address the disparities in electricity consumption patterns and enhance energy efficiency in Dhanmondi and Lalbagh, the following strategies are given:

- Promoting energy-efficient building designs, such as compact building designs with lower perimeter/area ratios, to enhance thermal efficiency by reducing heat loss and cooling demands, particularly in residential (Lalbagh) and commercial (Dhanmondi) sectors. This can lower overall electricity demand.
- Increasing vegetation (higher NDVI) in both areas to mitigate urban heat island effects and reduce cooling-related electricity use (Ming et al., 2020; Lotfabadi & Hançer, 2019). In Lalbagh, preserve open spaces near the river; in Dhanmondi, integrate green roofs or parks amidst commercial zones.
- Designing context-specific policies as three-way ANOVA shows significant interactions between structure type, location, and use ($p < 0.001$), meaning uniform policies won't suffice. Implementing area-specific energy policies. Therefore, generic policies may not be effective (Zhang et al., 2024; Mechri et al., 2010).
- Conducting targeted interventions in high-consumption hotspots. In Dhanmondi, targeting commercial and mixed-use areas with energy-saving measures like efficient lighting and HVAC systems ((Ghahramani et al., 2016; Stojanović et al., 2014).
- A comprehensive energy efficiency strategy should incorporate retrofitting initiatives alongside the mandatory installation of solar systems in new and existing buildings, fostering long-term sustainability and reduced energy usage (Kotov et al., 2024)..

This study investigates the impact of socioeconomic, urban, and environmental factors on electricity consumption in Dhaka's Dhanmondi and Lalbagh neighborhoods from 2017 to 2023. Employing trend analysis, spatial distribution, and regression models, it demonstrates how urban configurations and socio-economic conditions shape energy usage across various sectors. Throughout the period, both neighborhoods experienced an increase in electricity usage across residential, commercial, industrial, and institutional sectors, with Dhanmondi experiencing higher consumption, attributed to its affluent population and commercial vibrancy. In contrast, Lalbagh exhibited more stable, lower consumption, reflecting its industrial-residential character. The COVID-19 pandemic caused a significant dip in 2020, with a notable recovery in Dhanmondi, particularly in its commercial and industrial sectors, indicating economic resilience. Spatial analysis revealed lower consumption in Lalbagh's northern and central areas, with higher figures near southeastern industrial hubs, while Dhanmondi displayed consistently high usage in its central and northern zones, correlating with its diverse and planned development. Lalbagh's land use diversity and mix of building structures suggest varied development stages and socio-economic conditions, in contrast to Dhanmondi's predominantly permanent, modern structures. The study explores the interdependence of all three factors (use, location and structure type) creates specific scenarios where consumption is influenced in ways that cannot be predicted by looking at each factor individually. Finally, the study underscores the need for localized energy management strategies to cater to the distinct characteristics of each neighborhood, with Dhanmondi's energy consumption largely driven by structural factors like building volume and density, whereas Lalbagh's consumption is influenced by environmental and occupancy factors.

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APPENDIX A

নং	গ্রাহক শ্রেণী এবং জাতি	এনার্জি বোর্ড/সার্ভ (টাকা/কি.ও.ঘ.)	ডিম্বাক বোর্ড/সার্ভ [টাকা/কি.ও./মাস]
১	এনার্জি-১: আবাসিক		
	স্ট্যান্ডার্ড	১০.০৫	
	অন্য পিক সময়	৯.৫০	৯০.০০
	নিরপেক্ষ সময়	১০.২৯	
২	এনার্জি-২: বণিক/কৃষিকার্মে ব্যবহৃত পাম্প		
	স্ট্যান্ডার্ড	১১.৬০	
	অন্য পিক সময়	১০.৪৮	৯০.০০
	নিরপেক্ষ সময়	১৪.৫৭	
৩	এনার্জি-৩: শিল্প		
	স্ট্যান্ডার্ড	১০.৮৮	
	অন্য পিক সময়	৯.৭৫	৯০.০০
	নিরপেক্ষ সময়	১০.৬২	
৪	এনার্জি-৪: নির্মাণ		
	স্ট্যান্ডার্ড	১৪.০৮	
	অন্য পিক সময়	১২.৯৯	১২০.০০
	নিরপেক্ষ সময়	১৮.০৬	
৫	এনার্জি-৫: সাধারণ*		
	স্ট্যান্ডার্ড	১০.০১	
	অন্য পিক সময়	৯.৫৬	৯০.০০
	নিরপেক্ষ সময়	১০.৪২	
৬	এনার্জি- ৬: অস্থায়ী	১৯.০২	১২০.০০
৭	এনার্জি- ৭: বাটারি চার্জিং স্টেশন		
	স্ট্যান্ডার্ড	৯.৪৯	
	অন্য পিক সময়	৮.৬০	৯০.০০
	সুপার অন্যান্য পিক সময়	৭.৭১	
	নিরপেক্ষ সময়	১২.১৪	
৮	এনার্জি- ৮: সোল/বৃষ্টিজলে ব্যবহৃত পাম্প		

নং	গ্রাহক শ্রেণী এবং জাতি	এনার্জি বোর্ড/সার্ভ (টাকা/কি.ও.ঘ.)	ডিম্বাক বোর্ড/সার্ভ [টাকা/কি.ও./মাস]
১	এনার্জি- এ: আবাসিক		
	সাইক্ল লাইন: ০০-৫০ ইউনিট	৪.৬০	
	প্রথম ধাপ: ০০ থেকে ৭৫ ইউনিট	৫.২৬	
	দ্বিতীয় ধাপ: ৭৬ থেকে ২০০ ইউনিট	৭.২০	
	তৃতীয় ধাপ: ২০১ থেকে ৫০০ ইউনিট	৭.৫৯	৪২.০০
	চতুর্থ ধাপ: ৫০১ থেকে ৪০০ ইউনিট	৮.০২	
	পঞ্চম ধাপ: ৪০১ থেকে ৬০০ ইউনিট	১২.৬৭	
	ষষ্ঠ ধাপ: ৬০১ ইউনিট থেকে উপরে	১৪.৬১	
২	এনার্জি- বি: সোল/বৃষ্টিজলে ব্যবহৃত পাম্প	৫.২৫	৪২.০০
৩	এনার্জি- সি ১: ক্ষুদ্র শিল্প		
	স্ট্যান্ডার্ড বোর্ড	১০.৭৬	
	অন্য পিক সময়	৯.৬৮	৪৮.০০
	নিরপেক্ষ সময়	১২.৯৫	
৪	এনার্জি- সি ২: নির্মাণ	১৫.১৫	১২০.০০
৫	এনার্জি- ডি ১: শিক্ষা, স্বাস্থ্য ও সার্বজনীন প্রয়োজন এবং হাসপাতাল	৭.৫৫	৬০.০০
৬	এনার্জি- ডি ২: রাজস্ব বাতি ও পানির পাম্প	৯.৭১	৯০.০০
৭	এনার্জি- ডি ৩: বাটারি চার্জিং স্টেশন		
	স্ট্যান্ডার্ড	৯.৬২	
	অন্য পিক সময়	৮.৬৬	৯০.০০
	সুপার অন্যান্য পিক সময়	৭.৬৮	
	নিরপেক্ষ সময়	১২.১৪	
৮	এনার্জি- ই: বণিক/কৃষিকার্মে ব্যবহৃত পাম্প		

Figure A1: Tariff Categorization by DPDC

Source: DPDC, 2024

Objective	Variable	Level of Measurement	Data Collection
1. To identify trends in electricity consumption over the years in selected area of Dhaka	Electricity consumption	Ratio	DPDC
2. To examine the relationship of built environment factors with electricity consumption	Normalized Difference Vegetation Index (NDVI)	Ratio	Landsat 8 (USGS)
	Average Modified Normalized Difference Water Index (MNDWI)	Ratio	Landsat 8 (USGS)
	Distance from nearby permeable area	Ratio	DAP
	Floor No	Ratio	DPDC
	Building Density(250m)	Ratio	DPDC
	Building Area(sqm)	Ratio	DPDC
	Building Volume (cubic meter)	Ratio	DPDC
	Perimeter/Area ratio	Ratio	DPDC
	Customer No	Ratio	DPDC
	Building Age	Ratio	DPDC
	Structure Type	Nominal	DAP

Figure A2: Variables, Level of Measurement and Sources

Source: Author, 2024

APPENDIX B

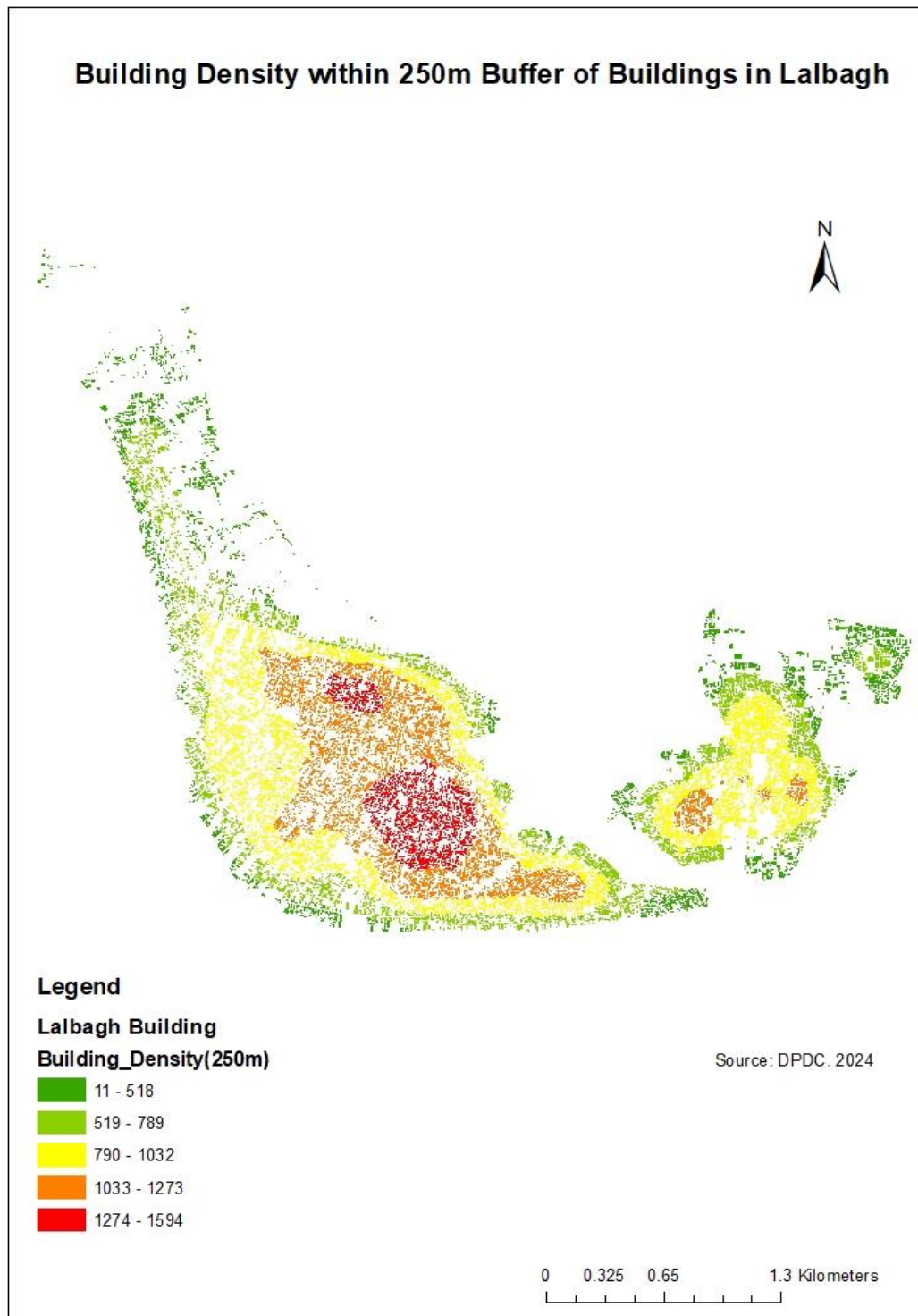


Figure B1: Building Density of Buildings in Lalbagh

Source: DPDC



Figure B2: Building Density of Buildings in Dhanmondi

Source: DPDC

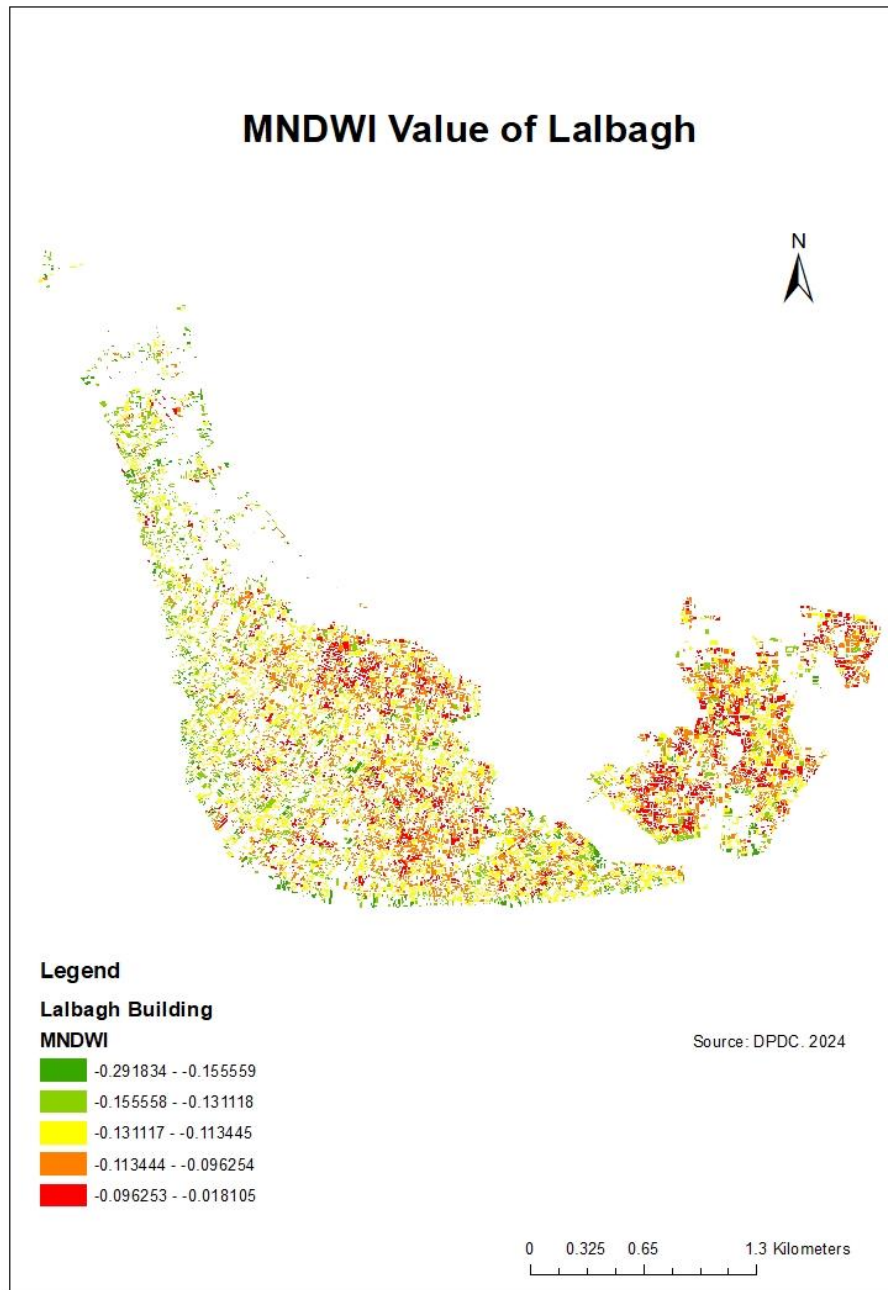


Figure B3: MNDWI Value of Lalbagh

Source: USGS

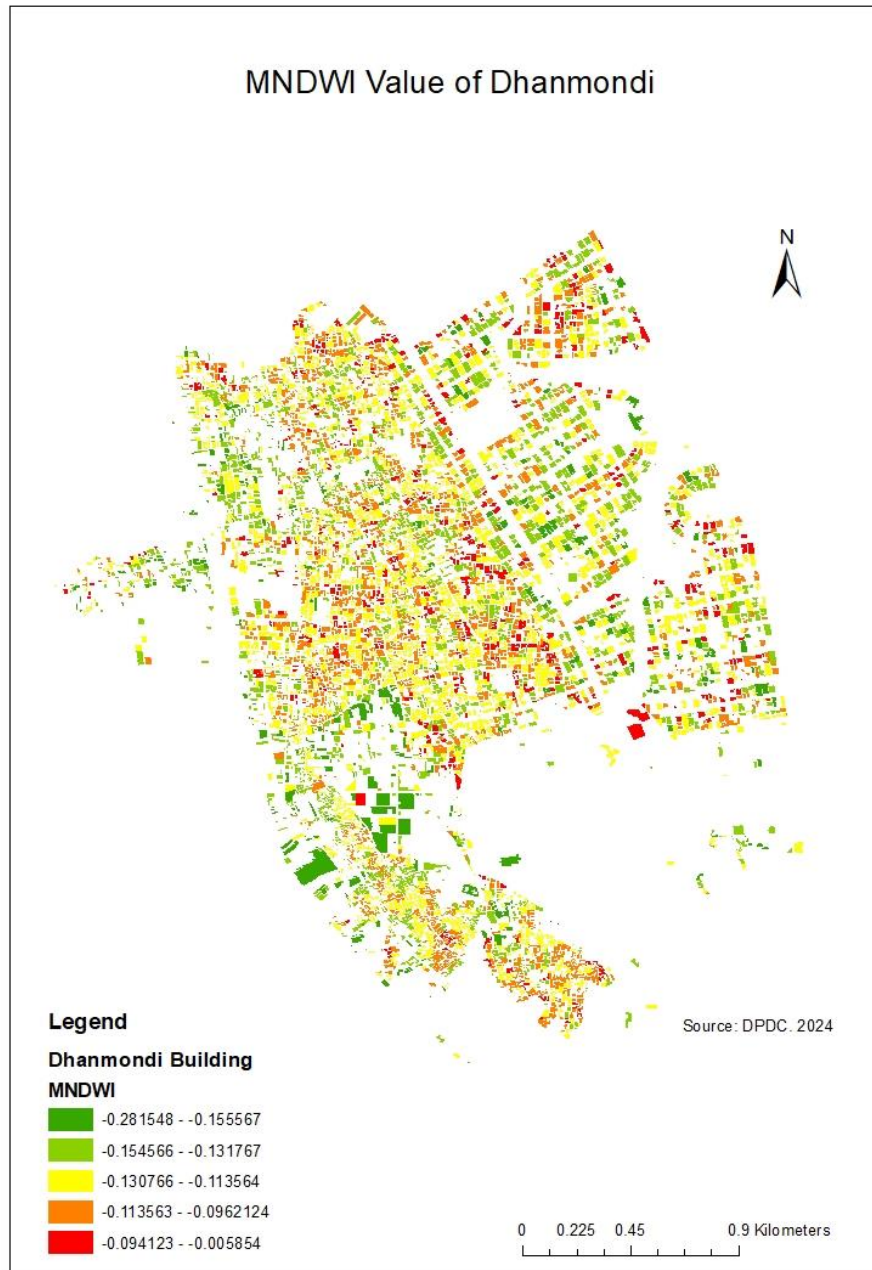


Figure B4: MNDWI Value of Lalbagh

Source: USGS

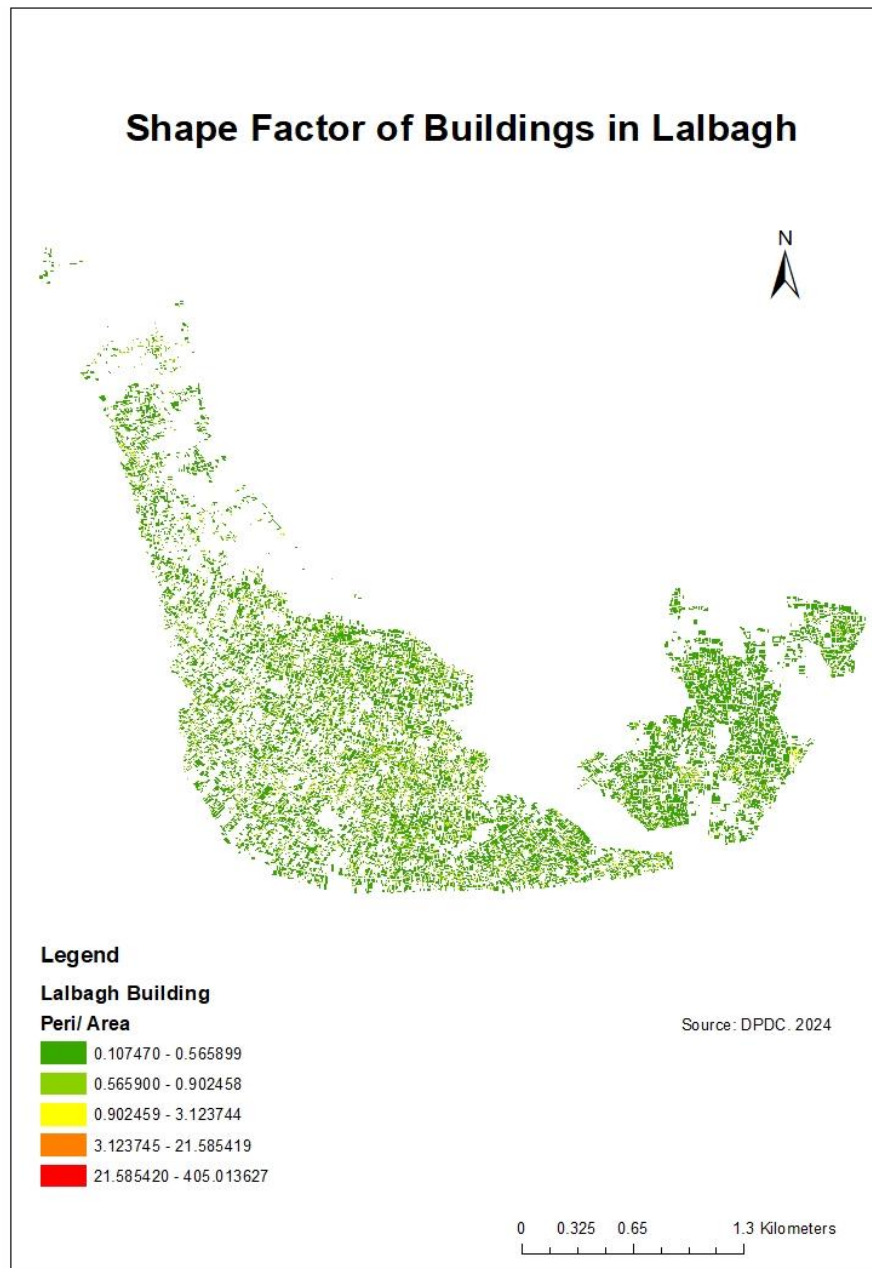


Figure B5: Shape Factor of Buildings in Lalbagh

Source: DPDC



Figure B6: Shape Factor of Buildings in Dhanmondi

Source: DPDC

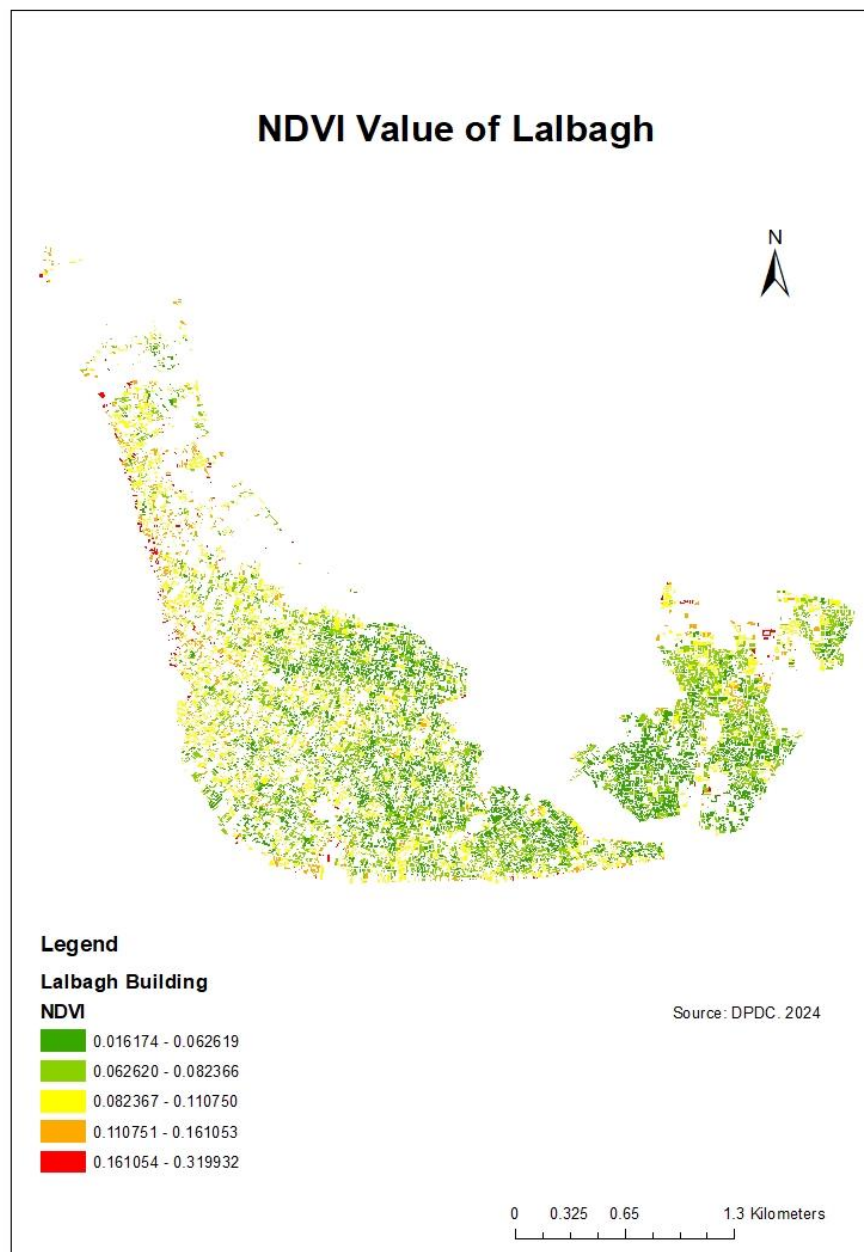


Figure B7: NDVI Value of Lalbagh

Source: USGS

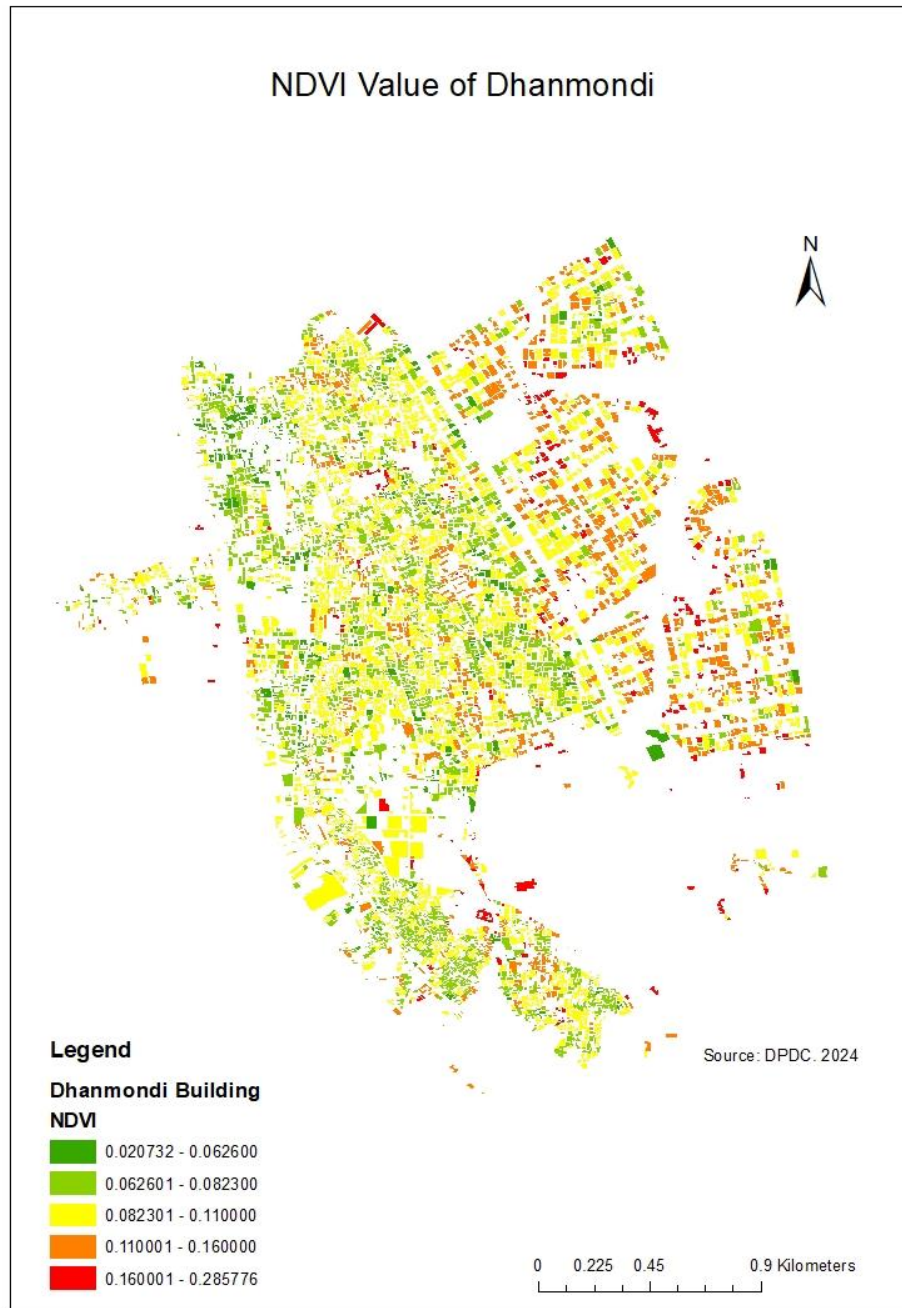


Figure B8: NDVI Value of Dhanmondi

Source: USGS

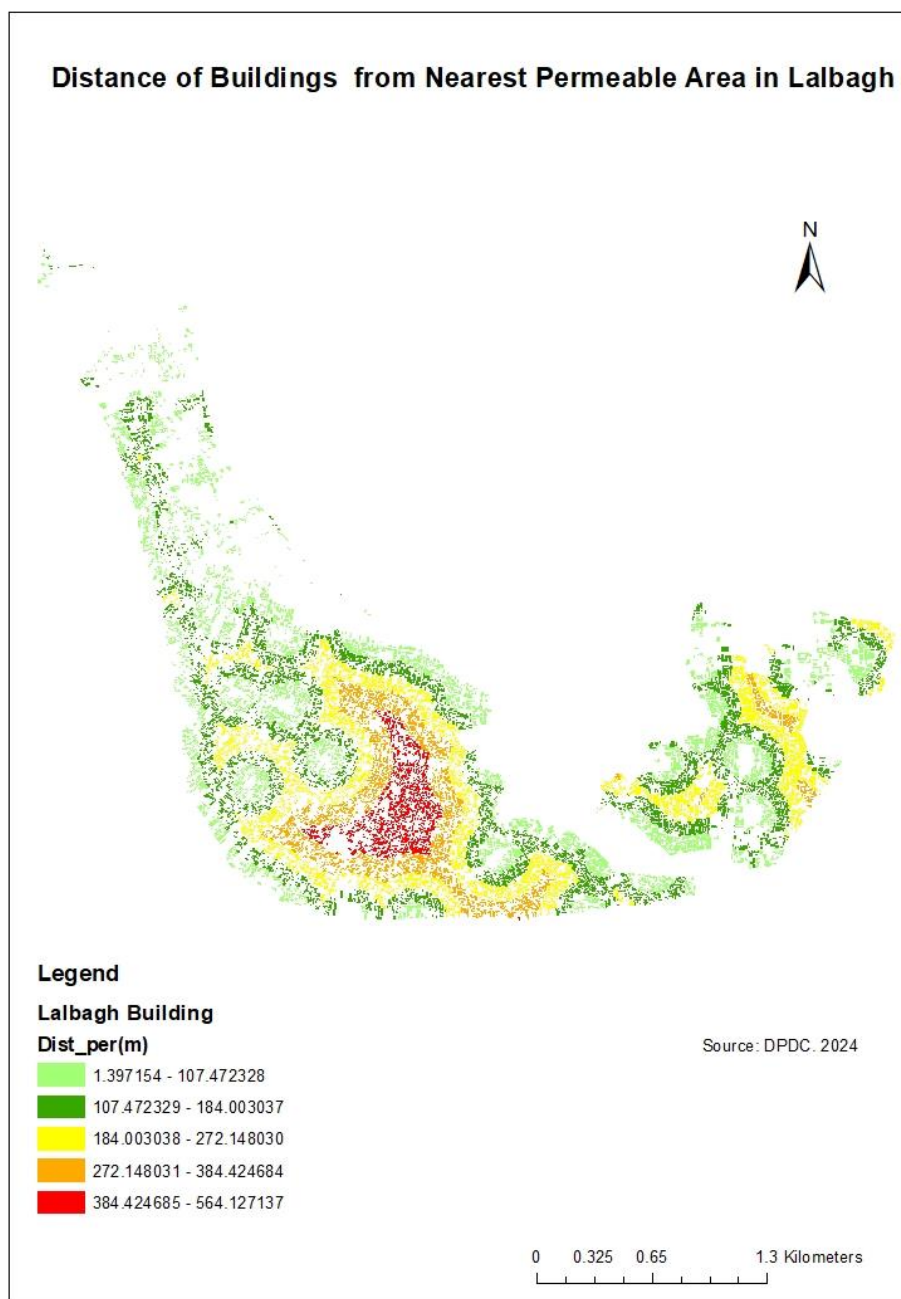


Figure B9: Distance of Buildings from Nearest Permeable Area in Lalbagh

Source: DPDC

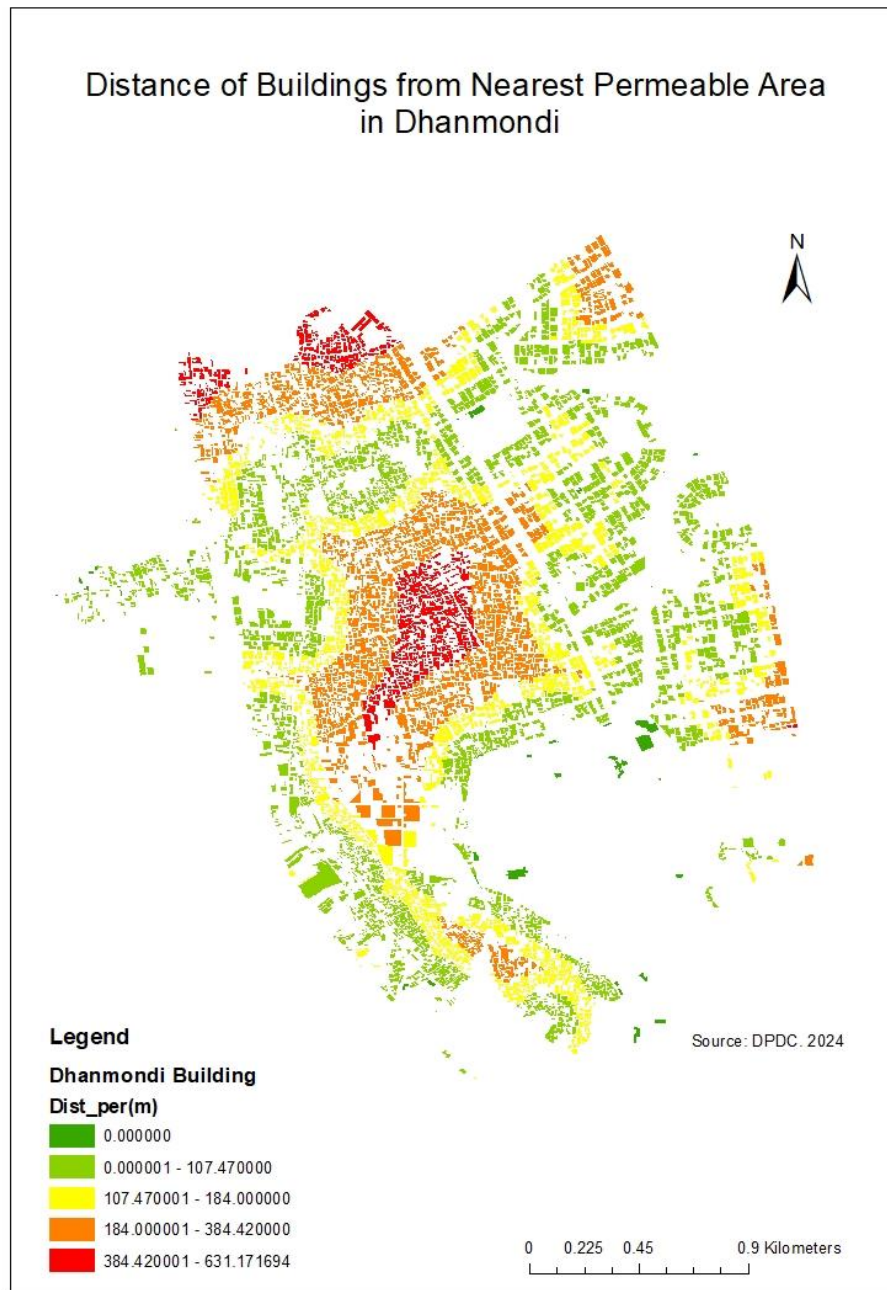


Figure B10: Distance of Buildings from Nearest Permeable Area in Dhanmondi

Source: DPDC

APPENDIX C

Table C1: Shannon Diversity Index Calculation of Lalbagh

Source: DAP

LANDUSE	Total Area(sqm)	Proportion(Pi)	Percentage(%)	(-PiLnPi)
Mixed Use	1769812	0.283961924	28.39619239	0.357484
Residential	2338381	0.375187403	37.51874027	0.367807
Vacant Land	335664	0.053856452	5.385645211	0.157338
Open Space	47784	0.007666824	0.766682369	0.037344
Institutional	12981	0.002082769	0.208276909	0.012859
Education and Research	160309	0.025721179	2.572117946	0.094151
Water body	420935	0.067537972	6.753797151	0.182019
Community Facilities	50575	0.008114633	0.811463268	0.039065
Restricted	94152	0.015106454	1.510645371	0.063336
Commercial	229386	0.036804412	3.680441192	0.121533
Industrial	304463	0.048850329	4.885032943	0.147479
Health Facilities	20575	0.003301207	0.330120746	0.018861
Agriculture	1691	0.000271317	0.027131673	0.002228
Transport and Commu	445756	0.071520439	7.152043909	0.188655
Administrative	104	1.66865E-05	0.001668654	0.000184
Total Area of All Use	6232568		Shannon Diversity Index (H')	1.790343

Table C2: Shannon Diversity Index Calculation of Dhanmondi

LANDUSE	Total Area(sqm)	Proportion(Pi)	Percentage(%)	(-PiLnPi)
Commercial	313418	0.04807043	4.807042972	0.14589799
Restricted	687804	0.105491816	10.54918155	0.23726395
Mixed Use	682715	0.104711291	10.47112911	0.23628609
Transport and Communication	735410	0.112793377	11.27933773	0.24613744
Residential	2414649	0.370346359	37.03463587	0.36787119
Institutional	4900	0.000751537	0.075153662	0.0054061
Education and Research	491965	0.075455044	7.545504392	0.1949923
Community Facilities	167416	0.025677399	2.567739907	0.09403434
Industrial	336427	0.051599431	5.159943098	0.15295334
Open Space	173134	0.026554396	2.65543963	0.09635422
Vacant Land	192438	0.029515144	2.951514385	0.10397748
Water body	319699	0.049033777	4.903377697	0.14784889
Total Area of All Uses	6519975		Shannon Diversity Index (H')	2.02902332

Pearson Correlations														
	BA (m2)	NDVI	MNDWI	Floor no	BV (m3)	Shape Factor	Customer No	BAge (yr)	PD (m)	BD within 250m	Semi Pucca	Katcha	Pucca	EC (Kwh)
BA (m2)	Pearson Corr.	1	0.03233*	-0.00918	0.37691*	0.7472*	-0.55447*	0.4604*	0.03993*	-0.06793*	-0.11884*	-0.25821*	-0.17309*	0.35931*
	p-value	--	1.1838E-4	0.27454	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
NDVI	Pearson Corr.	0.03233*	1	-0.48002*	-0.12009*	-0.04384*	-0.00489	-0.05673*	-0.10604*	-0.22573*	-0.34072*	0.04979*	0.00491	-0.0512*
	p-value	1.1838E-4	--	<0.0001	<0.0001	<0.0001	0.56048	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.55873	<0.0001
MNDWI	Pearson Corr.	-0.00918	-0.48002*	1	0.15618*	0.08195*	0.01554	0.08689*	0.06963*	0.21217*	0.29276*	-0.05878*	-0.02742*	0.07419*
	p-value	0.27454	<0.0001	--	<0.0001	<0.0001	0.06427	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.00109	<0.0001
Floor no	Pearson Corr.	0.37691*	-0.12009*	0.15618*	1	0.78135*	-0.27856*	0.44868*	0.11571*	-0.05332*	-0.06421*	-0.48482*	-0.21289*	0.60343*
	p-value	<0.0001	<0.0001	<0.0001	--	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
BV (m3)	Pearson Corr.	0.7472*	-0.04384*	0.08195*	0.78135*	1	-0.40762*	0.57584*	0.07928*	-0.0565*	-0.09362*	-0.36424*	-0.17644*	0.46383*
	p-value	<0.0001	<0.0001	<0.0001	<0.0001	--	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
Shape Factor	Pearson Corr.	-0.55447*	-0.00489	-0.01554	-0.27856*	-0.40762*	1	-0.25401*	-0.03055*	0.02151*	0.04067*	0.21951*	0.16959*	-0.31972*
	p-value	<0.0001	0.56048	0.06427	<0.0001	<0.0001	--	<0.0001	2.74351E-4	0.01042	<0.0001	<0.0001	<0.0001	<0.0001
Customer No	Pearson Corr.	0.4604*	-0.05673*	0.08689*	0.44868*	0.57584*	-0.25401*	1	0.05043*	-0.04755*	-0.05944*	-0.1857*	-0.10196*	0.2441*
	p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	--	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
BAge (yr)	Pearson Corr.	0.03993*	-0.10604*	0.06963*	0.11571*	0.07928*	-0.03055*	0.05043*	1	-0.01076	-0.02555*	-0.05073*	-0.03314*	0.07004*
	p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	--	0.2003	0.00235	<0.0001	<0.0001	<0.0001
PD (m)	Pearson Corr.	-0.06793*	-0.22573*	0.21217*	-0.05332*	-0.0565*	0.02151*	-0.04755*	-0.01076	1	0.69078*	0.04949*	0.00947	-0.0538*
	p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.01042	<0.0001	0.2003	--	<0.0001	<0.0001	0.25975	<0.0001
BD within 250m	Pearson Corr.	-0.11884*	-0.34072*	0.29276*	-0.06421*	-0.09362*	0.04067*	-0.05944*	-0.02555*	0.69078*	1	0.02662*	0.0277*	-0.0433*
	p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	--	0.00152	9.74872E-4	<0.0001
Semi Pucca	Pearson Corr.	-0.25821*	0.04979*	-0.05878*	-0.48482*	-0.36424*	0.21951*	-0.1857*	-0.05073*	0.04949*	0.02662*	1	-0.27394*	-0.79162*
	p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	--	<0.0001	<0.0001
Katcha	Pearson Corr.	-0.17309*	0.00491	-0.02742*	-0.21289*	-0.17644*	0.16959*	-0.10196*	-0.03314*	0.00947	0.0277*	-0.27394*	1	-0.37079*
	p-value	<0.0001	0.55873	0.00109	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.25975	9.74872E-4	<0.0001	--	<0.0001
Pucca	Pearson Corr.	0.35931*	-0.0512*	0.07419*	0.60343*	0.46383*	-0.31972*	0.2441*	0.07004*	-0.0538*	-0.0433*	-0.79162*	-0.37079*	1
	p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
EC (Kwh)	Pearson Corr.	0.45908*	-0.05324*	0.07812*	0.38688*	0.54717*	-0.22602*	0.80081*	0.05294*	-0.05399*	-0.06688*	-0.16098*	-0.08521*	0.20947*
	p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	--

2-tailed test of significance is used

Figure C1: Pearson Correlation Test of Consumption for Residential Use in Lalbagh

Source: DPDC, 2023

Pearson Correlations														
	BA (m2)	NDVI	MNDWI	Floor no	BV (m3)	Shape Factor	Customer No	BAge (yr)	PD (m)	BD within 250m	Semi Pucca	Katcha	Pucca	EC (Kwh)
BA (m2)	Pearson Corr.	1	0.01633	-0.06317*	0.12654*	0.66798*	-0.63985*	0.01778	0.08022*	-0.04645	-0.05061*	-0.00193	-0.10704*	
	p-value	--	0.5163	0.01194	<0.0001	<0.0001	<0.0001	0.47973	0.0014	0.06464	0.04406	0.9388	<0.0001	
NDVI	Pearson Corr.	0.01633	1	-0.46127*	-0.15648*	-0.03628	-0.03887	-0.10473*	-0.12204*	-0.12387*	-0.18191*	0.05767*	0.0949*	
	p-value	0.5163	--	<0.0001	<0.0001	0.14906	0.12209	<0.0001	<0.0001	<0.0001	<0.0001	0.02176	1.55965E-4	
MNDWI	Pearson Corr.	-0.06317*	-0.46127*	1	0.17178*	0.02711	0.08842*	0.1259*	0.08839*	0.16747*	0.17284*	-0.0568*	-0.12143*	
	p-value	0.01194	<0.0001	--	<0.0001	0.28099	4.28031E-4	<0.0001	4.30167E-4	<0.0001	<0.0001	0.02382	<0.0001	
Floor no	Pearson Corr.	0.12654*	-0.15648*	0.17178*	1	0.536*	-0.08385*	0.34262*	0.15253*	-0.01004	-0.21326*	-0.42775*	-0.20817*	
	p-value	<0.0001	<0.0001	<0.0001	--	<0.0001	8.3963E-4	<0.0001	<0.0001	0.68981	<0.0001	<0.0001	<0.0001	
BV (m3)	Pearson Corr.	0.66798*	-0.03628	0.02711	0.536*	1	-0.37077*	0.11583*	0.15776*	-0.02918	-0.07472*	-0.17394*	-0.11613*	
	p-value	<0.0001	0.14906	0.28099	<0.0001	--	<0.0001	<0.0001	<0.0001	0.24589	0.00293	<0.0001	<0.0001	
Shape Factor	Pearson Corr.	-0.63985*	-0.03887	0.08842*	-0.08385*	-0.37077*	1	-0.00683	-0.04304	0.00641	-0.01596	-0.0267	0.11647*	
	p-value	<0.0001	0.14906	4.28031E-4	8.3963E-4	<0.0001	--	0.78603	0.0869	0.79883	0.52569	0.28842	<0.0001	
Customer No	Pearson Corr.	0.01778	-0.10473*	0.1259*	0.34262*	0.11583*	-0.00683	1	0.11137*	-0.01055	-0.18107*	-0.15874*	-0.09113*	
	p-value	0.47973	<0.0001	<0.0001	<0.0001	<0.0001	0.78603	--	<0.0001	0.67483	<0.0001	2.83044E-4	<0.0001	
BAge (yr)	Pearson Corr.	0.08022*	-0.12204*	0.08839*	0.15253*	0.15776*	-0.04304	0.11137*	1	-0.01566	-0.02332	-0.07103*	-0.09437*	
	p-value	0.0014	<0.0001	4.30167E-4	<0.0001	<0.0001	0.0869	<0.0001	--	0.53352	0.35376	0.00469	1.69753E-4	
PD (m)	Pearson Corr.	-0.04645	-0.12387*	0.16747*	-0.01004	-0.02918	0.00641	-0.01055	-0.01566	1	0.56587*	0.03054	-0.01082	
	p-value	0.06464	<0.0001	<0.0001	0.68981	0.24589	0.79883	0.67483	0.53352	--	<0.0001	0.22464	0.6672	
BD within 250m	Pearson Corr.	-0.05061*	-0.18191*	0.17284*	-0.21326*	-0.07472*	-0.01596	-0.18107*	-0.02332	0.56587*	1	0.11755*	0.04787	
	p-value	0.04406	<0.0001	<0.0001	<0.0001	0.00293	0.52569	<0.0001	0.35376	<0.0001	<0.0001	<0.0001	0.05687	
Semi Pucca	Pearson Corr.	-0.00193	0.05767*	-0.0568*	-0.42775*	-0.17394*	-0.0267	-0.15874*	-0.07103*	0.03054	0.11755*	1	-0.41328*	
	p-value	0.9388	0.02176	0.02382	<0.0001	<0.0001	0.28842	<0.0001	0.00469	0.22464	<0.0001	--	<0.0001	
Katcha	Pearson Corr.	-0.10704*	0.0949*	-0.12143*	-0.20817*	-0.11613*	0.11647*	-0.09113*	-0.09437*	-0.01082	0.04787	-0.41328*	1	
	p-value	<0.0001	1.55965E-4	<0.0001	<0.0001	<0.0001	<0.0001	2.83044E-4	1.69753E-4	0.6672	0.05687	<0.0001	--	
Pucca	Pearson Corr.	0.08502*	-0.13292*	0.15261*	0.60139*	0.26897*	-0.06289*	0.23394*	0.14625*	-0.02302	-0.15802*	-0.70787*	-0.35065*	
	p-value	7.09298E-4	<0.0001	<0.0001	<0.0001	<0.0001	0.01233	<0.0001	<0.0001	0.36012	<0.0001	<0.0001	<0.0001	
EC (Kwh)	Pearson Corr.	0.17818*	0.03205	-0.04818	0.12372*	0.35821*	-0.07684*	0.12394*	0.05782*	-0.01157	-0.03144	-0.09805*	0.12972*	
	p-value	<0.0001	0.20255	0.06625	<0.0001	<0.0001	0.00222	<0.0001	0.02141	0.64545	0.21127	<0.0001	<0.0001	

2-tailed test of significance is used

*. Correlation is significant at the 0.05 level

Figure C2: Pearson Correlation Test of Consumption for Commercial Use in Lalbagh

Source: DPDC, 2023

Pearson Correlations															
		BA (m2)	NDVI	MNDWI	Floor no	BV (m3)	Shape Factor	Customer No	BAge (yr)	PD (m)	BD within 250m	Semi Pucca	Katcha	Pucca	EC (Kwh)
BA (m2)	Pearson Corr.	1	0.26491*	-0.02027	0.00542	0.74033*	-0.64889*	0.01173	0.18597*	-0.08214	-0.05597	-0.25636*	-0.14766	0.30972*	0.47036*
	p-value	--	0.00439	0.83048	0.95437	<0.0001	<0.0001	0.90146	0.04759	0.38497	0.55418	0.0059	0.11892	7.98632E-4	<0.0001
NDVI	Pearson Corr.	0.26491*	1	-0.51271*	-0.27329*	0.02035	-0.05277	0.0791	0.07109	-0.27686*	-0.48216*	0.05363	0.03974	-0.06971	-0.04411
	p-value	0.00439	--	<0.0001	0.00326	0.8298	0.57708	0.40283	0.45227	0.00286	<0.0001	0.5709	0.67467	0.4611	0.64119
MNDWI	Pearson Corr.	-0.02027	-0.51271*	1	0.32205*	0.13648	-0.10799	-0.02114	-0.01189	0.19141*	0.30447*	-0.12874	-0.28222*	0.27131*	0.05519
	p-value	0.83048	<0.0001	--	4.75464E-4	0.14765	0.25275	0.8233	0.90008	0.04135	9.89344E-4	0.17222	0.00235	0.0035	0.5906
Floor no	Pearson Corr.	0.00542	-0.27329*	0.32205*	1	0.57071*	-0.22047*	0.07092	-0.0089	0.14845	0.22808*	-0.3811*	-0.21457*	0.45767*	0.19258
	p-value	0.95437	0.00326	4.75464E-4	--	<0.0001	0.01842	0.45335	0.92509	0.11496	0.01466	<0.0001	0.02188	<0.0001	0.04008
BV (m3)	Pearson Corr.	0.74033*	0.02035	0.13648	0.57071*	1	-0.58999*	0.07214	0.18433*	-0.09051	0.01875	-0.36496*	-0.20676*	0.43899*	0.61706
	p-value	<0.0001	0.8298	0.14765	<0.0001	--	<0.0001	0.44561	0.04961	0.14961	0.84305	<0.0001	0.0273	<0.0001	<0.0001
Shape Factor	Pearson Corr.	-0.64889*	-0.05277	-0.10799	-0.22047*	-0.58999*	1	-0.0517	-0.13541	-0.09081	-0.18979*	0.29318*	0.16864	-0.35408*	-0.25804*
	p-value	<0.0001	0.57708	0.25275	0.01842	<0.0001	--	0.58486	0.15008	0.03728	0.03974	0.00155	0.07288	0.11247E-4	0.00633
Customer No	Pearson Corr.	0.01173	0.0791	-0.02114	0.07092	0.07214	-0.0517	1	0.03019	-0.02035	0.0059	0.07694	0.04685	0.11189	0.00449
	p-value	0.90146	0.40283	0.8233	0.45335	0.44561	0.58486	--	0.74983	0.8011	0.69429	0.27973	0.40319	0.22355	0.23585
BAge (yr)	Pearson Corr.	0.18597*	0.07109	-0.01189	-0.0089	0.18433*	-0.13541	0.03019	1	-0.03537	-0.10695	-0.06553	-0.05277	0.08753	0.25832
	p-value	0.04759	0.45227	0.90008	0.92509	0.04961	0.15088	0.74983	--	1	0.70869	0.26189	0.48852	0.57709	0.33544
PD (m)	Pearson Corr.	-0.08214	-0.27686*	0.19141*	-0.14845	-0.09051	-0.09061	-0.02385	-0.03537	1	0.58461*	-0.07536	-0.03769	0.08786	-0.0184
	p-value	0.38497	0.00286	0.04135	0.11496	0.91998	0.33766	0.8011	0.70869	--	<0.0001	0.42549	0.89055	0.35258	0.8453
BD within 250m	Pearson Corr.	-0.05597	-0.48216*	0.30447*	0.22808*	0.01875	-0.18979*	-0.03744	-0.10595	0.58461*	1	-0.10329	-0.27116*	0.24256*	-0.063
	p-value	0.55418	<0.0001	9.89344E-4	0.01466	0.84305	0.04313	0.69249	0.26189	<0.0001	--	0.10329	0.00352	0.00932	0.5034
Semi Pucca	Pearson Corr.	-0.25636*	0.05363	-0.12874	-0.3811*	-0.36496*	0.29318*	-0.1021	-0.06553	0.07536	-0.10329	1	-0.09867	-0.83271*	-0.1461
	p-value	0.0059	0.5709	0.17222	<0.0001	<0.0001	0.00155	0.27973	0.48852	0.42549	0.27414	--	0.29626	<0.0001	0.12008
Katcha	Pearson Corr.	-0.14766	0.03974	-0.28222*	-0.21457*	-0.20676*	0.16864	0.07904	-0.05277	-0.03769	-0.27116*	-0.09867	1	-0.46884*	-0.1050
	p-value	0.11892	0.67467	0.00325	0.02188	0.0273	0.07288	0.40319	0.57709	0.69055	0.00352	0.29626	--	<0.0001	0.2660
Pucca	Pearson Corr.	0.30972*	-0.06971	0.27131*	0.45767*	0.43899*	-0.35406*	0.04665	0.08753	0.08786	0.24256*	-0.83271*	-0.46884*	1	0.18815
	p-value	7.98632E-4	0.4611	0.0035	<0.0001	<0.0001	0.11247E-4	0.62214	0.35444	0.35258	0.00932	<0.0001	<0.0001	--	0.04499
EC (Kwh)	Pearson Corr.	0.47036*	-0.04411	0.05091	0.19258*	0.61706*	-0.25804*	0.11199	0.25932*	-0.01847	-0.0633	-0.14613	-0.10505	0.18815*	1
	p-value	<0.0001	0.64119	0.59065	0.04009	<0.0001	0.00539	0.23552	0.00534	0.84539	0.50346	0.12082	0.26801	0.04499	--
2-tailed test of significance is used															
* Correlation is significant at the 0.05 level															

Pearson Correlations		BA (m2)	NDVI	MNDWI	Floor no	BV (m3)	Shape Factor	Customer No	BAge (yr)	PD (m)	BD within 250m	Semi Pucca	Katcha	Pucca	EC (Kwh)
BA (m2)	Pearson Corr.	1	-0.00527	-0.03082*	0.26202*	0.75093*	-0.04143*	0.28668*	0.08559*	-0.0065*	-0.11607*	-0.17169*	-0.12563*	0.22961*	0.23561*
	p-value	--	0.68966	0.00523	<0.0001	<0.0001	0.00168	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
NDVI	Pearson Corr.	-0.00527	1	-0.45491*	-0.11852*	-0.04465*	-3.31474E-4	-0.01671	-0.08128*	-0.11682*	-0.27019*	0.0391*	0.03836*	-0.05751*	-0.0665*
	p-value	0.68966	--	<0.0001	<0.0001	0.70786E-4	0.97995	0.20507	<0.0001	<0.0001	<0.0001	0.00302	0.00362	<0.0001	<0.0001
MNDWI	Pearson Corr.	-0.03682*	-0.45491*	1	0.17512*	0.07447*	0.00987	0.07595*	0.01476	0.13744*	0.23199*	-0.10052*	-0.05914*	0.12672*	-0.03683*
	p-value	0.00523	<0.0001	--	<0.0001	<0.0001	0.45413	<0.0001	0.2631	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.00547
Floor no	Pearson Corr.	0.26202*	-0.11856*	0.17512*	1	0.66582*	-0.02701*	0.39296*	0.07134*	-0.07496*	-0.09304*	-0.54692*	-0.2277*	0.63912*	0.08801*
	p-value	<0.0001	<0.0001	<0.0001	--	<0.0001	0.04059	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
BV (m3)	Pearson Corr.	0.75093*	-0.04465*	0.07447*	0.66582*	1	-0.03159*	0.42774*	0.0799*	-0.06987*	-0.10736*	-0.34641*	-0.15703*	0.41166*	0.19783*
	p-value	<0.0001	0.70786E-4	<0.0001	<0.0001	--	0.01658	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
Shape Factor	Pearson Corr.	-0.04143*	-3.31474E-4	0.00987	-0.02701*	-0.03159*	1	-0.01058	-0.00845	-0.00993	-0.00824	0.03296*	0.00518	-0.03394*	-0.00983
	p-value	0.00168	0.97995	0.45413	0.04059	0.01658	--	0.42232	0.52175	0.45141	0.53234	0.01244	0.69469	0.01005	0.45621
Customer No	Pearson Corr.	0.28668*	-0.01671	0.07595*	0.39296*	0.42774*	-0.01058	1	0.022	-0.04524*	-0.05148*	-0.19555*	-0.09024*	0.23324*	0.14048*
	p-value	<0.0001	0.20507	<0.0001	<0.0001	<0.0001	0.42232	--	0.0953	5.99646E-4	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
BAge (yr)	Pearson Corr.	0.08559*	-0.08128*	0.01476	0.07134*	0.0799*	-0.00845	0.022	1	-0.03412*	-0.02529	-0.01866	-0.04354*	0.04095*	0.11018*
	p-value	<0.0001	<0.0001	0.2631	<0.0001	<0.0001	0.52175	0.0953	--	0.00967	0.05514	0.15709	9.59618E-4	0.0019	<0.0001
PD (m)	Pearson Corr.	-0.0665*	-0.11682*	0.13744*	-0.07496*	-0.06987*	-0.00993	-0.04524*	-0.03412*	1	0.63229*	0.0318*	0.04322*	-0.0532*	-0.0652*
	p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.45141	5.99646E-4	0.00967	--	<0.0001	0.01591	0.00105	<0.0001	<0.0001
BD within 250m	Pearson Corr.	-0.11607*	-0.27019*	0.23199*	-0.09304*	-0.10736*	-0.00824	-0.05148*	-0.02529	0.63229*	1	0.03332*	0.05912*	-0.06316*	-0.0859*
	p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.53234	<0.0001	0.05514	<0.0001	--	0.01151	<0.0001	<0.0001	<0.0001
Semi Pucca	Pearson Corr.	-0.17169*	0.0391*	-0.10052*	-0.54692*	-0.34641*	0.03296*	-0.19555*	-0.01866	0.0318*	0.03332*	1	-0.17871*	-0.85011*	-0.00142
	p-value	<0.0001	0.00302	<0.0001	<0.0001	<0.0001	0.01244	<0.0001	0.15709	0.01591	0.01151	--	<0.0001	<0.0001	0.81445
Katcha	Pearson Corr.	-0.12563*	0.03836*	0.05914*	-0.2277*	-0.15703*	0.00518	-0.09024*	-0.04354*	0.04322*	0.05912*	-0.17871*	1	-0.3662*	-0.0422*
	p-value	<0.0001	0.00362	<0.0001	<0.0001	<0.0001	0.69469	<0.0001	9.59618E-4	0.00105	<0.0001	<0.0001	--	<0.0001	0.00137
Pucca	Pearson Corr.	0.22961*	-0.05751*	0.12672*	0.63912*	0.41166*	-0.03394*	0.23324*	0.04095*	-0.0532*	-0.06316*	-0.85011*	-0.3662*	1	0.02392
	p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.01005	<0.0001	0.0019	<0.0001	<0.0001	<0.0001	<0.0001	--	0.06968
EC (Kwh)	Pearson Corr.	0.23561*	-0.0665*	-0.03683*	0.08801*	0.19783*	-0.00983	0.14048*	0.11018*	-0.0652*	-0.0859*	-0.00142	-0.0422*	0.02392	1
	p-value	<0.0001	<0.0001	0.00547	<0.0001	<0.0001	0.45621	<0.0001	<0.0001	<0.0001	<0.0001	0.91445	0.00137	0.06968	--

2-tailed test of significance is used
* Correlation is significant at the 0.05 level

Figure C4: Pearson Correlation Test of Consumption for Mixed Use in Lalbagh
Source: DPDC, 2023

Pearson Correlations		BA (m2)	NDVI	MNDWI	Floor no	BV (m3)	Shape Factor	Customer No	BAge (yr)	PD (m)	BD within 250m	Semi Pucca	Katcha	Pucca	EC (Kwh)
BA (m2)	Pearson Corr.	1	0.09413*	-0.20949*	0.05949	0.61983*	-0.24686*	0.16425*	-0.01123	-0.07763*	-0.14369*	0.06227	-0.09917*	-0.00454	0.05843
	p-value	--	0.00435	<0.0001	0.07192	<0.0001	<0.0001	0.73427	0.01878	<0.0001	0.05959	0.00266	0.89091	0.07712	
NDVI	Pearson Corr.	0.09413*	1	-0.51681*	-0.0302	0.04947	0.00813	-0.06579*	-0.08896*	0.02213	-0.11614*	-0.00138	0.05315	-0.03157	-0.16373*
	p-value	0.00435	--	<0.0001	0.36131	0.13459	0.80588	0.04652	0.00706	0.50355	4.28093E-4	0.96671	0.10796	0.33992	<0.0001
MNDWI	Pearson Corr.	-0.20949*	-0.51681*	1	0.08638*	-0.07618*	0.00974	-0.02034	0.0872*	0.14376*	0.2472*	-0.02211	-0.09351*	0.08163*	0.00488
	p-value	<0.0001	<0.0001	--	0.00891	0.02112	0.7684	0.53857	0.00827	<0.0001	<0.0001	0.50384	0.00462	0.01347	0.88282
Floor no	Pearson Corr.	0.05949	-0.0302	0.08638*	1	0.66063*	-0.05778	-0.00965	0.03052	-0.04771	0.00269	-0.55509*	-0.12792*	0.6695*	0.0022
	p-value	0.07192	0.36131	0.00891	--	<0.0001	0.0805	0.77043	0.35619	0.1491	0.93514	<0.0001	1.03522E-4	<0.0001	0.94699
BV (m3)	Pearson Corr.	0.61983*	0.04947	-0.07618*	0.66063*	1	-0.17367*	0.07125*	0.0064	-0.0593	-0.08893*	-0.29031*	-0.11595*	0.37*	0.02136
	p-value	<0.0001	0.13459	0.02112	<0.0001	--	<0.0001	0.03106	0.84651	0.07283	0.00708	<0.0001	4.37685E-4	<0.0001	0.51859
Shape Factor	Pearson Corr.	-0.24686*	0.00813	0.00974	-0.05778	-0.17367*	1	-0.04181	-0.00589	0.06163	0.01987	-0.05291	0.04945	0.0255	-0.04931
	p-value	<0.0001	0.80588	0.7684	0.0805	<0.0001	--	0.20617	0.8586	0.06226	0.54802	0.10953	0.1348	0.44076	0.13591
Customer No	Pearson Corr.	0.16425*	-0.06579*	-0.02034	-0.00965	0.07125*	-0.04181	1	0.01009	-0.08604*	-0.06771*	0.03316	-0.04588	-0.00673	0.49599*
	p-value	<0.0001	0.04652	0.53857	0.77043	0.03106	0.20617	--	0.76031	0.00918	0.04047	0.31609	0.16532	0.83887	<0.0001
BAge (yr)	Pearson Corr.	-0.01123	-0.08896*	0.0872*	0.03052	0.0064	-0.00589	0.01009	1	-0.01879	0.12084*	0.01755	-0.05966	0.01843	0.02937
	p-value	0.73427	0.00706	0.00827	0.35619	0.84651	0.8586	0.76031	--	0.57012	2.46703E-4	0.59576	0.07113	0.57755	0.37463
PD (m)	Pearson Corr.	-0.07763*	0.02213	0.14376*	-0.04771	-0.0593	0.06163	-0.08604*	-0.01879	1	0.57588*	0.06655*	-0.02083	-0.05779	-0.10499*
	p-value	0.01878	0.50355	<0.0001	0.1491	0.07283	0.06226	0.00918	0.57012	--	<0.0001	0.04405	0.52894	0.08046	0.00146
BD within 250m	Pearson Corr.	-0.14369*	-0.11614*	0.2472*	0.00269	-0.08893*	0.01987	-0.06771*	0.12084*	0.57588*	1	-0.01244	0.00976	0.00716	-0.10735*
	p-value	<0.0001	4.28093E-4	<0.0001	0.93514	0.00708	0.54802	0.04047	2.46703E-4	<0.0001	--	0.70688	0.7681	0.82867	0.00114
Semi Pucca	Pearson Corr.	0.06227	-0.00138	-0.02211	-0.55509*	-0.29031*	-0.05291	0.03316	0.01755	0.06655*	-0.01244	1	-0.39057*	-0.82011*	0.05543
	p-value	0.05959	0.96671	0.50384	<0.0001	<0.0001	0.10953	0.31609	0.59576	0.04405	0.70688	--	<0.0001	<0.0001	0.09361
Katcha	Pearson Corr.	-0.09917*	0.05315	-0.09351*	-0.12792*	-0.11595*	0.04945	-0.04588	-0.05966	-0.02083	0.00976	-0.39057*	1	-0.20645*	-0.07082*
	p-value	0.00266	0.10796	0.00462	1.03522E-4	4.37685E-4	0.1348	0.16532	0.07113	0.52894	0.7681	<0.0001	--	<0.0001	0.03209
Pucca	Pearson Corr.	-0.00454	-0.03157	0.08163*	0.6695*	0.37*	0.0255	-0.00673	0.01843	-0.05779	0.00716	-0.82011*	-0.20645*	1	-0.01489
	p-value	0.89091	0.33992	0.01347	<0.0001	<0.0001	0.44076	0.83887	0.57755	0.08046	0.82867	<0.0001	<0.0001	--	0.65257
EC (Kwh)	Pearson Corr.	0.05843	-0.16373*	0.00488	0.0022	0.02136	-0.04931	0.49599*	0.02937	-0.10499*	-0.10735*	0.05543	-0.07082*	-0.01489	1
	p-value	0.07712	<0.0001	0.88282	0.94699	0.51859	0.13591	<0.0001	0.37463	0.00146	0.00114	0.09361	0.03209	0.65257	--

Figure C5: Pearson Correlation Test of Consumption for Industrial Use in Lalbagh
Source: DPDC, 2023

Pearson Correlations		BA (m2)	NDVI	MNDWI	Floor no	BV (m3)	Shape Factor	Customer No	BAge (yr)	P (m)	PD (m)	BD within 250m	Semi Pucca	Katcha	Pucca	EC (Kwh)
BA (m2)	Pearson Corr.	1	0.03131	-0.34932*	0.23547	0.89995*	-0.47395*	-0.00836	-0.18497	0.91589*	-0.20957	-0.15977	-0.2156	-0.08401	0.24253*	0.239
	p-value	--	<0.7984	0.00326	0.05145	<0.0001	<0.0001	0.94562	0.12811	<0.0001	0.08393	0.18974	0.07521	0.49251	0.04465	0.04
NDVI	Pearson Corr.	0.03131	1	-0.25376*	0.07989	0.01818	-0.10441	0.29446*	-0.27791*	0.021	-0.18698	-0.02165	-0.08688	-0.07842	0.11776	-0.06
	p-value	0.7984	--	0.03538	0.51405	0.88212	0.39324	0.01405	0.02077	0.864	0.12396	0.85986	0.47781	0.52185	0.33519	0.621
MNDWI	Pearson Corr.	-0.34932*	-0.25376*	1	-0.04734	-0.24842*	0.12973	-0.17612	0.06492	-0.28754*	0.153	0.02254	0.25403*	-0.02919	-0.22815	0.03
	p-value	0.00326	0.03538	--	0.69928	0.03957	0.28804	0.14773	0.59612	0.01659	0.20944	0.85412	0.03518	0.8118	0.05937	0.78
Floor no	Pearson Corr.	0.23547	0.07989	-0.04734	1	0.46939*	-0.55154*	0.26346*	-0.12104	0.41712*	-0.10393	-0.34344*	-0.66319*	-0.24729*	0.74102*	0.210
	p-value	0.05145	0.51405	0.69928	--	<0.0001	<0.0001	0.02872	0.32185	0.36307E-4	0.39542	0.00386	<0.0001	0.04051	<0.0001	0.082
BV (m3)	Pearson Corr.	0.89995*	0.01818	-0.24842*	0.46939*	1	-0.49579*	0.04018	-0.1783	0.96802*	-0.21412	-0.2659*	-0.30405*	-0.11423	0.34012*	0.485
	p-value	<0.0001	0.88212	0.03957	<0.0001	--	<0.0001	0.04018	0.1427	<0.0001	0.07729	0.02723	0.01109	0.34998	0.00424	<0.00
Shape Factor	Pearson Corr.	-0.47395*	-0.10441	0.12973	-0.55154*	-0.49579*	1	-0.12705	0.22429	-0.62482*	0.16177	0.26038*	0.40564*	0.13245	-0.44479*	0.081
	p-value	<0.0001	0.39324	0.28804	<0.0001	<0.0001	--	0.2982	0.06392	<0.0001	0.18418	0.03071	5.44442E-4	0.27798	1.28594E-4	0.505
Customer No	Pearson Corr.	-0.00836	0.29446*	-0.17612	0.26346*	0.04018	-0.12705	1	-0.30671*	0.02245	0.05364	-0.10625	-0.17383	-0.05871	0.19149	0.007
	p-value	0.94562	0.01405	0.14773	0.02872	0.74306	0.2982	--	0.01037	0.85472	0.66159	0.38491	0.15315	0.63181	0.11498	0.955
BAge (yr)	Pearson Corr.	-0.18497	-0.27791*	0.06492	-0.12104	-0.1783	0.22429	-0.30671*	1	-0.16028	0.13386	-0.10869	0.09458	0.00429	-0.09176	0.107
	p-value	0.12811	0.02077	0.59612	0.32185	0.1427	0.06392	0.01037	--	0.18832	0.27285	0.37399	0.43951	0.97208	0.45334	0.376
P (m)	Pearson Corr.	0.91589*	0.021	-0.28754*	0.41712*	0.96802*	-0.62482*	0.02245	-0.16028	1	-0.21073	-0.253*	-0.29764*	-0.11568	0.33468*	0.379
	p-value	<0.0001	0.864	0.01659	0.36307E-4	<0.0001	<0.0001	0.85472	0.18832	--	0.0822	0.03595	0.013	0.34386	0.00494	0.001
PD (m)	Pearson Corr.	-0.20957	-0.18698	0.153	-0.10393	-0.21412	0.16177	0.05364	0.13386	-0.21073	1	0.36306*	0.13202	-0.03034	-0.11175	-0.146
	p-value	0.08393	0.12396	0.20944	0.39542	0.07729	0.18418	0.66159	0.27285	0.0822	--	0.00217	0.27955	0.80457	0.36063	0.225
BD within 250m	Pearson Corr.	-0.15977	-0.02165	0.02254	-0.34344*	-0.2659*	0.26038*	-0.10625	-0.10869	-0.253*	0.36306*	1	0.13485	0.14322	-0.19245	-0.452
	p-value	0.18974	0.85986	0.85412	0.00386	0.02723	0.03071	0.38491	0.37399	0.03595	0.00217	--	0.26927	0.24041	0.11314	-0.00
Semi Pucca	Pearson Corr.	-0.2156	-0.08688	0.25403*	-0.60319*	-0.30405*	-0.17383	0.09458	-0.29764*	0.13202	0.13485	0.13485	1	-0.1219	-0.89498*	-0.155
	p-value	0.07521	0.47781	0.03518	<0.0001	0.01109	5.44442E-4	0.15315	0.43951	0.013	0.27955	0.26927	--	0.31836	<0.0001	0.1
Katcha	Pearson Corr.	-0.08401	-0.07842	-0.02919	-0.24729*	-0.11423	0.13245	-0.05871	0.00429	-0.11568	-0.03034	0.14322	-0.1219	1	-0.33371*	-0.055
	p-value	0.49251	0.52185	0.8118	0.04051	0.34998	0.27798	0.63181	0.97208	0.34386	0.80457	0.24041	0.31836	--	0.00508	0.621
Pucca	Pearson Corr.	0.24253*	0.11776	-0.22815	0.74102*	0.34012*	-0.44479*	0.19149	-0.09176	0.33468*	-0.11175	-0.19245	-0.89498*	-0.33371*	1	0.177
	p-value	0.04465	0.33519	0.05937	<0.0001	0.00424	1.28594E-4	0.11498	0.45334	0.00494	0.36063	0.11314	<0.0001	0.00508	<0.0001	--
EC (Kwh)	Pearson Corr.	0.23926*	-0.0594	0.0329	0.21045	0.48516*	0.08158	0.00764	0.10766	0.37901*	-0.14668	-0.45293*	-0.05941	0.05941	0.17735	--
	p-value	0.0477	0.62778	0.78844	0.08262	<0.0001	0.50514	0.95035	0.37857	0.00132	0.22912	<0.0001	0.193	0.62777	0.14488	--

Figure C6: Pearson Correlation Test of Consumption for Commercial Use in Dhanmondi
Source: DPDC, 2023

Pearson Correlations		BA (m2)	NDVI	MNDWI	Floor no	BV (m3)	Shape Factor	Customer No	BAge (yr)	P (m)	PD (m)	BD within 250m	Semi Pucca	Katcha	Pucca
BA (m2)	Pearson Corr.	1	0.12148*	-0.12068*	0.30834*	0.71606*	-0.11664*	0.01759	0.00323	0.85318*	-0.04019*	-0.35221*	-0.16616*	-0.07265*	0.18908*
	p-value	--	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.12369	0.77765	<0.0001	4.31985E-4	<0.0001	<0.0001	<0.0001	<0.0001
NDVI	Pearson Corr.	0.12148*	1	-0.3255*	0.04209*	0.09649*	-0.04246*	0.00925	0.01016	0.17093*	-0.12713*	-0.32935*	-0.02376*	-0.06135*	0.05031*
	p-value	<0.0001	--	<0.0001	2.28021E-4	<0.0001	2.00444E-4	0.41795	0.37393	<0.0001	<0.0001	<0.0001	0.03754	<0.0001	<0.0001
MNDWI	Pearson Corr.	-0.12068*	-0.3255*	1	0.10015*	-0.03611*	0.00193	0.02154	5.30566E-4	-0.1179*	0.12867*	0.02137	-0.02407*	-0.11023*	0.07294*
	p-value	<0.0001	<0.0001	--	<0.0001	0.00157	0.8656	0.05934	0.96296	<0.0001	<0.0001	0.06134	0.03508	<0.0001	<0.0001
Floor no	Pearson Corr.	0.30834*	0.04209*	0.10015*	1	0.64458*	-0.08945*	0.06186*	0.01025	0.42582*	0.09625*	-0.30238*	-0.49173*	-0.18963*	0.54798*
	p-value	<0.0001	2.28021E-4	<0.0001	--	<0.0001	<0.0001	<0.0001	0.36973	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
BV (m3)	Pearson Corr.	0.71606*	0.09649*	-0.03611*	0.64458*	1	-0.11228*	0.03358*	0.00639	0.73992*	-0.0327*	-0.40166*	-0.23679*	-0.09501*	0.26556*
	p-value	<0.0001	<0.0001	0.00157	<0.0001	--	<0.0001	0.00328	0.57577	<0.0001	0.0042	-0.06737*	-0.0001	<0.0001	<0.0001
Shape Factor	Pearson Corr.	-0.11664*	-0.04246*	0.00193	-0.08945*	-0.11228*	1	-0.00624	0.00106	-0.17084*	-0.01512	0.06737*	0.07293*	0.04003*	-0.08671*
	p-value	<0.0001	2.00444E-4	0.8656	<0.0001	<0.0001	--	0.58462	0.92579	<0.0001	0.18566	<0.0001	<0.0001	4.563E-4	<0.0001
Customer No	Pearson Corr.	0.01759	0.00925	0.02154	0.06186*	0.03358*	-0.00624	1	-0.01599	0.03288*	0.05339*	-0.07438*	-0.03263*	-0.01482	0.03738*
	p-value	0.12369	0.41795	0.05934	0.0001	0.00328	0.58462	--	0.16168	0.00399	<0.0001	<0.0001	0.00428	0.19465	0.00106
BAge (yr)	Pearson Corr.	0.00323	0.01016	5.30566E-4	0.01025	0.00639	0.00106	-0.01599	1	0.00252	0.02433*	-0.01624	-0.00865	-0.00731	0.01145
	p-value	0.77765	0.37393	0.96296	0.36973	0.57577	0.92579	0.16168	--	0.8255	0.03318	0.15503	0.44909	0.52223	0.3161
P (m)	Pearson Corr.	0.85318*	0.17093*	-0.1179*	0.42582*	0.73992*	-0.17084*	0.03288*	0.00252	1	-0.02107	-0.46897*	-0.25273*	-0.08123*	0.27422*
	p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.00399	0.8255	--	0.06508	<0.0001	<0.0001	<0.0001	<0.0001
PD (m)	Pearson Corr.	-0.04019*	-0.12713*	0.12867*	0.09625*	-0.0327*	-0.01512	0.05339*	0.02433*	-0.02107	1	0.06314*	0.01082	-0.03807*	0.00723
	p-value	4.31985E-4	<0.0001	<0.0001	<0.0001	0.0042	0.18566	<0.0001	0.03318	0.06508	--	<0.0001	0.34336	8.57547E-4	0.52664
BD within 250m	Pearson Corr.	-0.35221*	-0.32935*	0.02137	-0.30238*	-0.40166*	0.06737*	-0.07438*	-0.01624	-0.46897*	0.06314*	1	0.17167*	0.07482*	-0.19524*
	p-value	<0.0001	<0.0001	0.06134	<0.0001	<0.0001	<0.0001	<0.0001	0.15503	<0.0001	<0.0001	--	<0.0001	<0.0001	<0.0001
Semi Pucca	Pearson Corr.	-0.16616*	-0.02376*	-0.02407*	-0.49173*	-0.23679*	0.07293*	-0.03263*	-0.00865	-0.25273*	0.01082	0.17167*	1	-0.10376*	0.89804*
	p-value	<0.0001	0.03754	0.03508	<0.0001	<0.0001	<0.0001	0.00428	0.44909	<0.0001	0.34336	<0.0001	--	<0.0001	<0.0001
Katcha	Pearson Corr.	-0.07265*	-0.06135*	-0.11023*	-0.18963*	-0.09501*	0.04003*	-0.01482	-0.00731	-0.08123*	-0.03807*	0.07482*	-0.10376*	1	-0.35943*
	p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.563E-4	0.19465	0.52223	<0.0001	8.57547E-4	<0.0001	<0.0001	--	<0.0001
Pucca	Pearson Corr.	0.18908*	0.05031*	0.07294*	0.54798*	0.26556*	-0.08671*	0.03738*	0.01145	0.27422*	0.00723	-0.19524*	-0.89804*	-0.35943*	1
	p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.00106	0.3161	<0.0001	0.52664	<0.0001	<0.0001	<0.0001	--
EC (Kwh)	Pearson Corr.	0.19138*	0.14096*	-0.02712*	0.17487*	0.23232*	-0.04632*	0.03574*	0.00174	0.25168*	-0.04185*	-0.31282*	-0.08566*	-0.03222*	0.09527*
	p-value	<0.0001	<0.0001	0.01758	<0.0001	<0.0001	<0.0001	0.00175	0.87887	<0.0001	2.47776E-4	<0.0001	<0.0001	0.00479	<0.0001

2-tailed test of significance is used
*. Correlation is significant at the 0.05 level

Figure C7: Pearson Correlation Test of Consumption for Residential Use in Dhanmondi
Source: DPDC, 2023

Pearson Correlations		BA (m2)	NDVI	MNDWI	Floor no	BV (m3)
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Pearson Correlations		BA (m2)	NDVI	MNDWI	Floor no	BV (m3)	Shape Factor	Customer No	BAge (yr)	PD (m)	BD within 250m	Semi Pucca	Katcha
BA (m2)	Pearson Corr.	1	0.01633	-0.06317*	0.12654*	0.66798*	-0.63985*	0.01778	0.08022*	-0.04645	-0.05061*	-0.00193	-0.10704*
	p-value	--	0.5163	0.01194	<0.0001	<0.0001	<0.0001	0.47973	0.0014	0.06464	0.04406	0.9388	<0.0001
NDVI	Pearson Corr.	0.01633	1	-0.46127*	-0.15648*	-0.03628	-0.03887	-0.10473*	-0.12204*	-0.12387*	-0.18191*	0.05767*	0.0949*
	p-value	0.5163	--	<0.0001	<0.0001	0.14906	0.12209	<0.0001	<0.0001	<0.0001	<0.0001	0.02176	1.55965E-4
MNDWI	Pearson Corr.	-0.06317*	-0.46127*	1	0.17178*	0.02711	0.08842*	0.1259*	0.08839*	0.16747*	0.17284*	-0.0568*	-0.12143*
	p-value	0.01194	<0.0001	--	<0.0001	0.28099	4.28031E-4	<0.0001	4.30167E-4	<0.0001	<0.0001	0.02382	<0.0001
Floor no	Pearson Corr.	0.12654*	-0.15648*	0.17178*	1	0.536*	-0.08385*	0.34262*	0.15253*	-0.01004	-0.21326*	-0.42775*	-0.20817*
	p-value	<0.0001	<0.0001	<0.0001	--	<0.0001	8.3963E-4	<0.0001	<0.0001	0.68981	<0.0001	<0.0001	<0.0001
BV (m3)	Pearson Corr.	0.66798*	-0.03628	0.02711	0.536*	1	-0.37077*	0.11583*	0.15776*	-0.02918	-0.07472*	-0.17394*	-0.11613*
	p-value	<0.0001	0.14906	0.28099	<0.0001	--	<0.0001	<0.0001	<0.0001	0.24589	0.00293	<0.0001	<0.0001
Shape Factor	Pearson Corr.	-0.63985*	-0.03887	0.08842*	-0.08385*	-0.37077*	1	-0.00683	-0.04304	0.00641	-0.01596	-0.0267	0.11647*
	p-value	<0.0001	0.12209	4.28031E-4	8.3963E-4	<0.0001	--	0.78603	0.0869	0.79883	0.52569	0.28842	<0.0001
Customer No	Pearson Corr.	0.01778	-0.10473*	0.1259*	0.34262*	0.11583*	-0.00683	1	0.11137*	-0.01055	-0.18107*	-0.15674*	-0.09113*
	p-value	0.47973	<0.0001	<0.0001	<0.0001	<0.0001	0.78603	--	<0.0001	0.67483	<0.0001	<0.0001	2.83044E-4
BAge (yr)	Pearson Corr.	0.08022*	-0.12204*	0.08839*	0.15253*	0.15776*	-0.04304	0.11137*	1	-0.01566	-0.02332	-0.07103*	0.09437*
	p-value	0.0014	<0.0001	4.30167E-4	<0.0001	<0.0001	0.0869	<0.0001	--	0.53352	0.35376	0.00469	1.69753E-4
PD (m)	Pearson Corr.	-0.04645	-0.12387*	0.16747*	-0.01004	-0.02918	0.00641	-0.01055	-0.01566	1	0.56587*	0.03054	-0.01082
	p-value	0.06464	<0.0001	<0.0001	0.68981	0.24589	0.79883	0.67483	0.53352	--	<0.0001	0.22464	0.6672
BD within 250m	Pearson Corr.	-0.05061*	-0.18191*	0.17284*	-0.21326*	-0.07472*	-0.01596	-0.18107*	-0.02332	0.56587*	1	0.11755*	0.04787
	p-value	0.04406	<0.0001	<0.0001	<0.0001	0.00293	0.52569	<0.0001	0.35376	<0.0001	--	<0.0001	0.05687
Semi Pucca	Pearson Corr.	-0.00193	0.05767*	-0.0568*	-0.42775*	-0.17394*	-0.0267	-0.15874*	-0.07103*	0.03054	0.11755*	1	-0.41328*
	p-value	0.9388	0.02176	0.02382	<0.0001	<0.0001	0.28842	<0.0001	0.00469	0.22464	<0.0001	--	<0.0001
Katcha	Pearson Corr.	-0.10704*	0.0949*	-0.12143*	-0.20817*	-0.11613*	0.11647*	-0.09113*	-0.09437*	-0.01082	0.04787	-0.41328*	1
	p-value	<0.0001	1.55965E-4	<0.0001	<0.0001	<0.0001	<0.0001	2.83044E-4	1.69753E-4	0.6672	0.05687	<0.0001	--
Pucca	Pearson Corr.	0.08502*	-0.13292*	0.15261*	0.60139*	0.26897*	-0.06289*	0.23394*	0.14625*	-0.02302	-0.15802*	-0.70787*	-0.35065*
	p-value	7.09298E-4	<0.0001	<0.0001	<0.0001	<0.0001	0.01233	<0.0001	<0.0001	0.36012	<0.0001	<0.0001	<0.0001
EC (Kwh)	Pearson Corr.	0.17818*	0.03205	-0.04618	0.12372*	0.35821*	-0.07684*	0.12394*	0.05782*	-0.01157	-0.03144	-0.09805*	0.12972*
	p-value	<0.0001	0.20255	0.06625	<0.0001	<0.0001	0.00222	<0.0001	0.02141	0.64545	0.21127	<0.0001	<0.0001

2-tailed test of significance is used
* Correlation is significant at the 0.05 level

Figure C9: Pearson Correlation Test of Consumption for Industrial Use in Lalbagh
Source: DPDC, 2023

Pearson Correlations		BA (m2)	NDVI	MNDWI	Floor no	BV (m3)	Shape Factor	Customer No	BAge (yr)	PD (m)	BD within 250m	Semi Pucca	Katcha	Pucca	EC (Kwh)
BA (m2)	Pearson Corr.	1	-0.00527	-0.03682*	0.26202*	0.75693*	-0.04143*	0.28658*	0.08559*	-0.0665*	-0.11607*	-0.17189*	-0.12563*	0.22981*	0.23561*
	p-value	--	0.68966	0.00523	<0.0001	<0.0001	0.00168	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
NDVI	Pearson Corr.	-0.00527	1	-0.45491*	-0.11856*	-0.04465*	-3.31474E-4	-0.01671	-0.08128*	-0.11682*	-0.27019*	0.0391*	0.03836*	-0.05751*	-0.0665*
	p-value	0.68966	--	<0.0001	<0.0001	0.0768E-4	9.7995	0.20507	<0.0001	<0.0001	<0.0001	0.00302	0.00362	<0.0001	<0.0001
MNDWI	Pearson Corr.	-0.03682*	-0.45491*	1	0.17512*	0.07447*	0.00987	0.07595*	0.01476	0.13744*	0.23199*	-0.10052*	-0.05914*	0.12672*	-0.03663*
	p-value	0.00523	<0.0001	--	<0.0001	<0.0001	0.45413	<0.0001	0.2631	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.00547
Floor no	Pearson Corr.	0.26202*	-0.11856*	0.17512*	1	0.66582*	-0.02701*	0.39286*	0.07134*	-0.07486*	-0.09304*	-0.54692*	-0.2277*	0.63912*	0.08801*
	p-value	<0.0001	<0.0001	<0.0001	--	<0.0001	0.04059	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
BV (m3)	Pearson Corr.	0.75693*	-0.04465*	0.07447*	0.66582*	1	-0.03159*	0.42774*	0.0799*	-0.06987*	-0.10736*	-0.34641*	-0.15703*	0.41166*	0.19783*
	p-value	<0.0001	0.0768E-4	<0.0001	<0.0001	--	0.01658	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
Shape Factor	Pearson Corr.	-0.04143*	-3.31474E-4	0.00987	-0.02701*	-0.03159*	1	-0.01058	-0.00845	-0.00993	-0.00824	0.03296*	0.00518	-0.03394*	-0.00983
	p-value	0.00168	0.97995	0.45413	0.04059	0.01658	--	0.42232	0.52175	0.45141	0.53234	0.01244	0.69469	0.01005	0.45621
Customer No	Pearson Corr.	0.28658*	-0.01671	0.07595*	0.39286*	0.42774*	-0.01058	1	0.022	-0.04524*	-0.05148*	-0.19555*	-0.09024*	0.23324*	0.14048*
	p-value	<0.0001	0.20507	<0.0001	<0.0001	<0.0001	0.42232	--	0.0953	5.99646E-4	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
BAge (yr)	Pearson Corr.	0.08559*	-0.08128*	0.01476	0.07134*	0.0799*	-0.00845	0.022	1	-0.03412*	-0.02529	-0.01866	-0.04354*	0.04095*	0.11018*
	p-value	<0.0001	<0.0001	0.2631	<0.0001	<0.0001	0.52175	0.0953	--	0.00967	0.05514	0.15709	9.59618E-4	0.0019	<0.0001
PD (m)	Pearson Corr.	-0.0665*	-0.11682*	0.13744*	-0.07486*	-0.06987*	-0.00993	-0.04524*	-0.03412*	1	0.63229*	0.0318*	0.04322*	-0.0532*	-0.0652*
	p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.45141	5.99646E-4	0.00967	--	<0.0001	0.01591	0.00105	<0.0001	<0.0001
BD within 250m	Pearson Corr.	-0.11607*	-0.27019*	0.23199*	-0.09304*	-0.10736*	-0.00824	-0.05148*	-0.02529	0.63229*	1	0.03332*	0.05912*	-0.06316*	-0.0859*
	p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.53234	<0.0001	0.05514	<0.0001	--	0.01151	<0.0001	<0.0001	<0.0001
Semi Pucca	Pearson Corr.	-0.17189*	0.0391*	-0.10052*	-0.54692*	-0.34641*	0.03296*	-0.19555*	-0.01866	0.0318*	0.03332*	1	-0.17871*	-0.85011*	-0.00142
	p-value	<0.0001	0.00302	<0.0001	<0.0001	<0.0001	0.01244	<0.0001	0.15709	0.01591	0.01151	--	<0.0001	<0.0001	0.91445
Katcha	Pearson Corr.	-0.12563*	0.03836*	-0.05914*	-0.2277*	-0.15703*	0.00518	-0.09024*	-0.04354*	0.04322*	0.05912*	-0.17871*	1	-0.3862*	-0.0422*
	p-value	0.0001	0.00362	<0.0001	<0.0001	<0.0001	0.69469	<0.0001	9.59618E-4	0.00105	<0.0001	<0.0001	--	<0.0001	0.00137
Pucca	Pearson Corr.	0.22981*	-0.05751*	0.12672*	0.63912*	0.41166*	-0.03394*	0.23324*	0.04095*	-0.0532*	-0.06316*	-0.85011*	-0.3862*	1	0.02392
	p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.01005	<0.0001	0.0019	<0.0001	<0.0001	<0.0001	<0.0001	--	0.06968
EC (Kwh)	Pearson Corr.	0.23561*	-0.0665*	-0.03663*	0.08801*	0.19783*	-0.00983	0.14048*	0.11018*	-0.0652*	-0.0859*	-0.00142	-0.0422*	0.02392	1
	p-value	<0.0001	<0.0001	0.00547	<0.0001	<0.0001	0.45621	<0.0001	<0.0001	<0.0001	<0.0001	0.91445	0.00137	0.06968	--

2-tailed test of significance is used
* Correlation is significant at the 0.05 level

Figure C10: Pearson Correlation Test of Consumption for Institutional Use in Lalbagh
Source: DPDC, 2023